



Presents

Product Management Casebook

2025-2026



PRODSOC



COPYRIGHTS

© 2024, ProdSoc – Product and Operations Society, FMS Delhi. All rights reserved. This book or parts thereof may not be reproduced in any form, stored in any retrieval system, or transmitted in any form by any means – electronic, mechanical, photocopy, recording, or otherwise – without prior written permission of ProdSoc – Product and Operations Society, FMS Delhi. For permission requests, write to ProdSoc at prodsoc@fms.edu

The Product Manager role is dynamic and ever-evolving. It requires a unique blend of business acumen, user empathy, and technical know-how. This book aims to equip you with the fundamental knowledge and practical tools to confidently showcase your product management potential to interviewers.

Part 1: Building Your Foundation

We begin by demystifying the essence of product management – its core responsibilities, various roles within the field, and the skills that set successful product managers apart. You'll learn about the challenges you might encounter and a comprehensive interview preparation strategy to tackle them head-on.

Part 2: Demystifying Tech Concepts

The modern product landscape is intertwined with technology. This section will equip you with a foundational understanding of key tech concepts relevant to product management, including A/B testing, database management systems, APIs, Big Data, AI, ML, and more. This knowledge will empower you to converse confidently with technical teams and make informed product decisions.

Part 3: Mastering Product Management Metrics

Knowing how to prioritize problems and measure success is essential for any product. This section will introduce you to key product management metrics, equipping you to track progress, analyze user behavior, and demonstrate the impact of your product decisions.

Part 4: Case Studies – Putting Your Skills to the Test

The heart of this book lies in the diverse case studies presented. You'll encounter practical case scenarios across different domains. By working through these cases, you'll gain valuable practice in applying your product management knowledge and interview techniques. Remember, excelling in product management interviews requires not just technical knowledge but also strong communication, problem-solving, and analytical abilities. This casebook provides you with the tools and practice to refine these skills. Utilize it not just to prepare for interviews, but also to spark your curiosity and ignite your passion for the world of product management.

Product and Operations Society 2025-26



Tajesh Nagpal
President



Abhinav Bende



Adithya Sailesh



Aman Yadav



Anvi Jain



Medha Bhattacharya



Saransh Jaiswal

Product and Operations Society 2024-25



Shivam Upadhyaya
President



Kovarthan S



Mahek Sengar



Reshav Shekhar



Rohit Paliwal



Samyak Jain



Srajal Gupta

Index

1. Introduction to Product Management

About Product Management
Types of Product Managers
Profile Guide
Preparation Strategy

2. Core Concepts

Innovation Pipeline
Product Life Cycle
Product Development Methodologies
Design
Users

3. Technical Concepts

How Internet Works
Tech Stack
API and Webhooks
Cloud Computing
DBMS
Big Data
AI, ML and LLMs
E-Commerce and Quick Commerce
IOT
Block Chain
Revenue Model
Basic Definitions

4. Frameworks

Business Model Canvas
Value Proposition Canvas
Product Market Fit
Problem Solution Fit
CIRCLES Framework
HEART Framework
AARRR Pirate Framework
Solution Prioritization Frameworks

5. Cases

Product Design

Design an app that helps individuals cook
Design a social media platform for Children
Design a Chair
Design an ATM for the Blind
Design a Turf Booking App
Design a Product for Software Developers
Design an Airport
Design a product for Homestay owners
Design a Mode of Transport for Long Distance
Design an App for FMS MBA 1st Year Students

Product Improvement

Improve Paytm
Improve YouTube
Improve Operational Efficiency of Airport
Improve WhatsApp

Index

Improve Spotify
Improve a Dating App
Improve Teaching in Tier-2 Cities

Root Cause Analysis (RCA)

Decrease in order fulfillment
Decrease in Weekly Active Users
Drop in Transacting Users
Decline in Add to Cart actions
Drop in Sales

Guesstimates

Estimating TAM and Martek Size
Tennis Balls in a Boeing 747
How many elevators ?
iPhones sold in India
Number of ATMs
Number of Flights

Category Launch

Pharmacy in Amazon
Alcohol Category in E-Commerce
Grocery in Flipkart
Furniture in Q-Comm
Maid on Urban Company

Puzzles

6. Interview Experiences

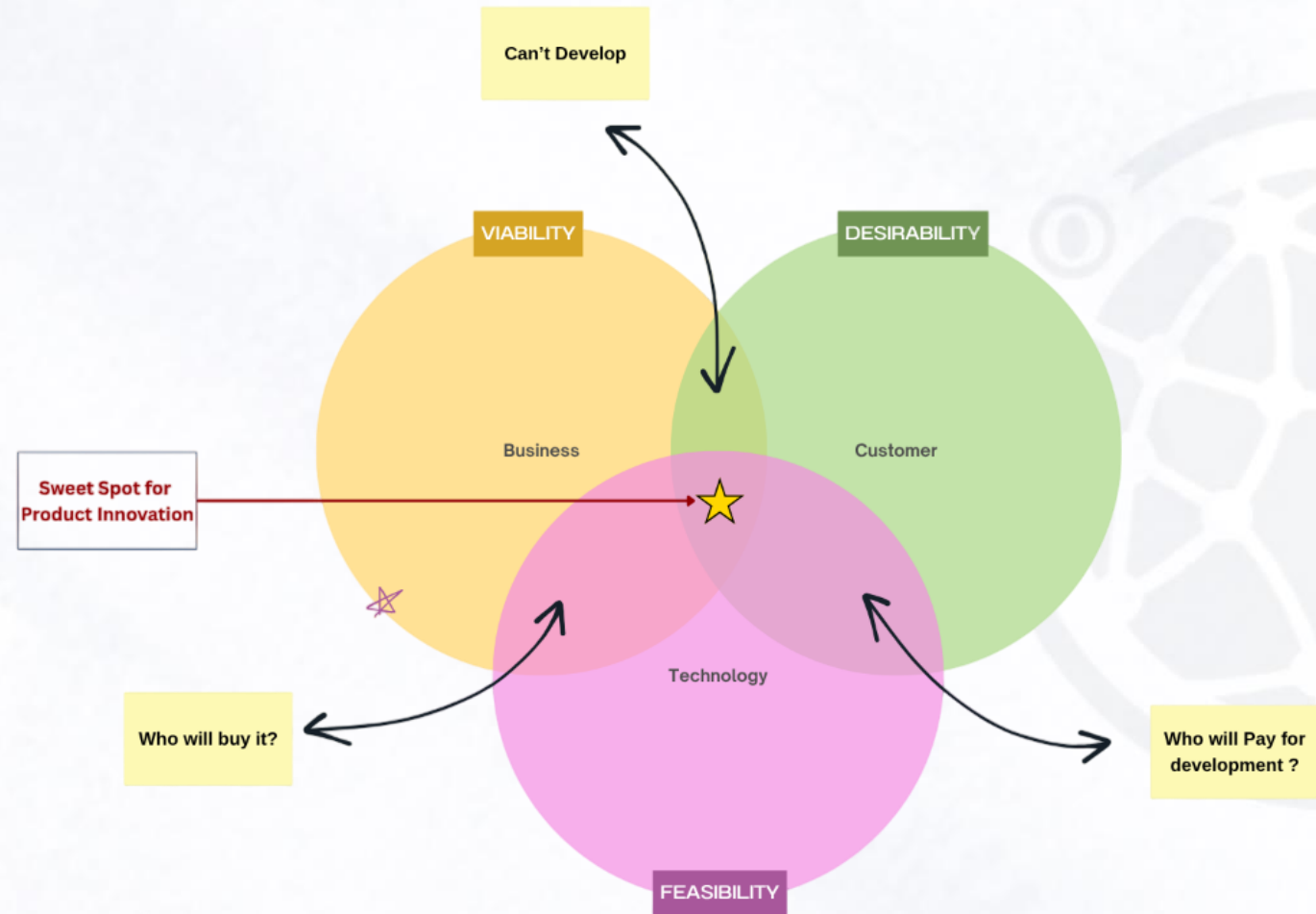
Introduction To Product Management

Product - A product is any item, service, software, or experience created and delivered to meet customer needs or solve a problem. It provides value to users and aligns with business objectives.

A recent article by Ken, titled "Do We Even Need Product Managers?" raises the question: what exactly does a product manager do, and are they truly necessary?

To better understand the role of a product manager, we can liken them to a mini CEO of a product. This comparison invites us to consider what a CEO does. A more accurate description of a product manager is that they empathize with users to identify the right problems to solve. They collaborate with engineering and design teams to gather resources and build effective solutions. Additionally, they work alongside sales, marketing, and growth teams to develop a compelling business proposition and communicate effectively with customers. Finally, they monitor feedback and oversee the product lifecycle to ensure ongoing customer value.

Product managers work to integrate user experience, technology, and business domains to achieve desirability, feasibility, and viability.



Types of Product Managers

Product manager roles vary from strategic to operational focus



Profile Overview

When hiring for product management roles from MBA colleges, recruiters generally look for two main criteria in candidates' profiles or CVs:

1. **Technical Background:** This includes your undergraduate degree and your experience during internships and previous jobs.

2. **Work Experience:** Many companies usually set a minimum work experience requirement. Having prior experience as an Associate Product Manager (APM) or Software Development Engineer (SDE) is also valued.

However, many companies, like Microsoft, seek holistic profiles and may give preference to candidates with non-technical backgrounds or those without prior work experience.

Additionally, some companies look for domain-specific expertise like Finance or Marketing.

CV Format

In addition to following common sense CV etiquette, it is crucial to strategically structure your CV to appeal to top product companies.

Your CV should **emphasize work experience and relevant projects** that demonstrate your capability to work in cross-functional, product-oriented roles. Preference should be given to:

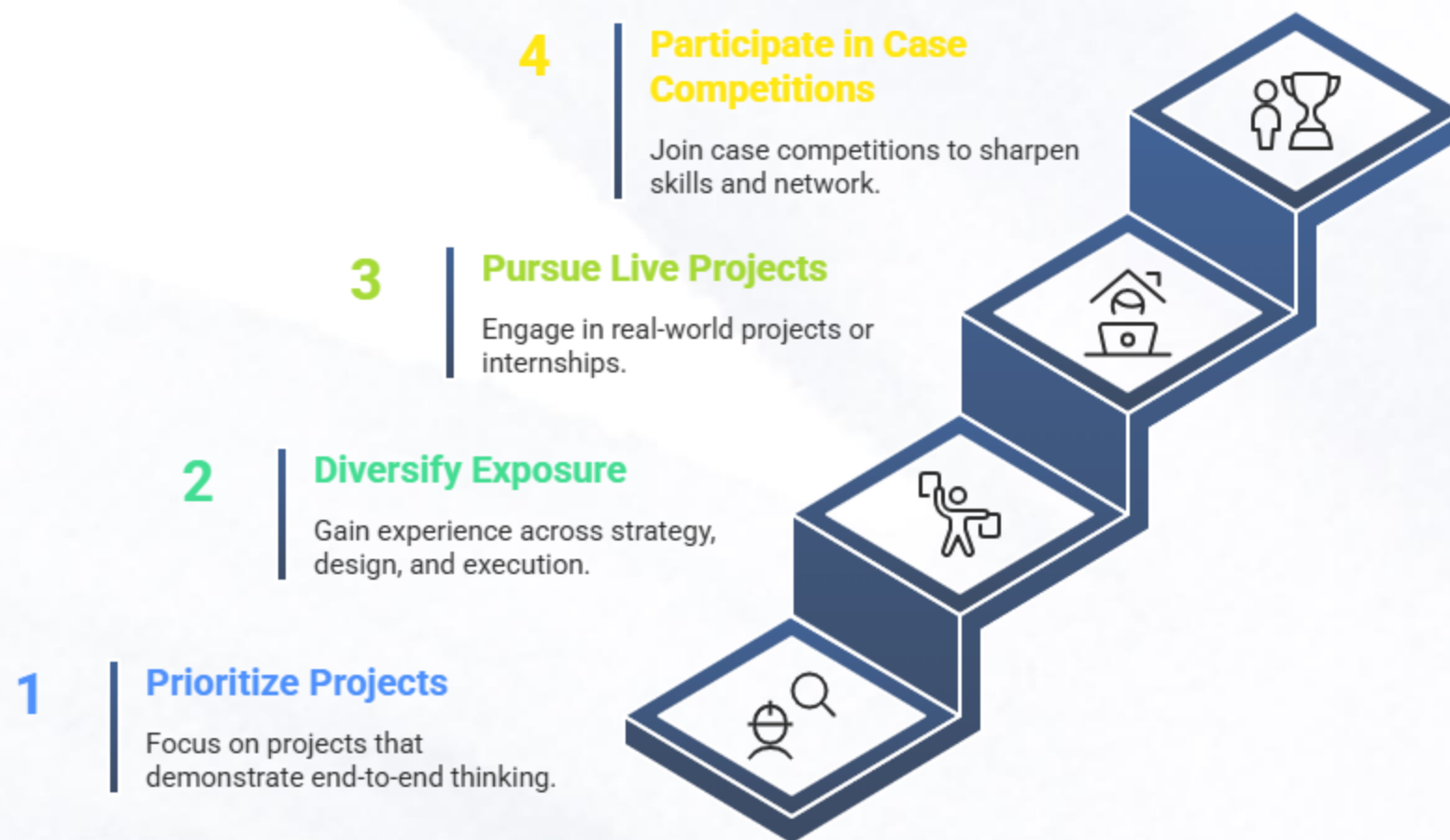
**Technical Projects**
that showcase your ability to build or contribute to a product, feature, or system.

**Product Management or Operations Projects**
that demonstrate skills such as problem-solving, stakeholder management, execution, and delivery.

**Analytics Projects**
where you have used data to drive decisions, measure success, or uncover user insights

Additionally, ensure that you **highlight key academic achievements**, especially if they demonstrate analytical rigor, innovation, or leadership (e.g., high CGPA, relevant coursework, awards, or scholarships). Certifications related to product, design thinking, analytics, or technology can also add value.

How to Build a Product focused Profile



Important Tips

- ✓ **Quantify impact** wherever possible e.g., “Improved onboarding completion rate by 15% through redesign of user flow.”
- ✓ **Tailor your CV** for product company’s focus area consumer tech, SaaS, fintech, etc. emphasizing the most relevant parts of your profile.
- ✓ **Get your CV reviewed** by seniors or mentors who have secured product roles; they can provide perspective on how your CV will be perceived

To ace product management interviews, prepare holistically combining technical awareness, structured thinking, and product sense.

Master the Basics

- ✓ Know every detail of your CV; be ready to explain your impact and choices.
- ✓ Prepare strong HR answers (Why PM?, Why this company?, Strengths/weaknesses)

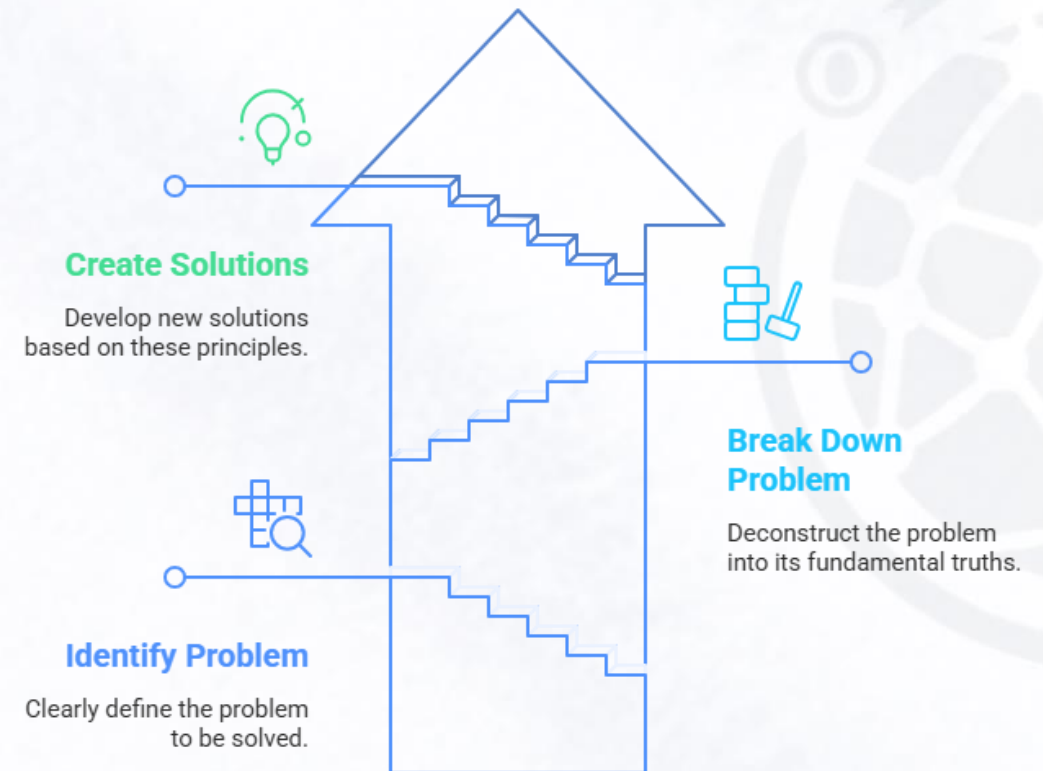
Develop Technical & Product Awareness

- ✓ Understand how common digital tech works (APIs, cloud, databases) and how it generates value
- ✓ Observe and critique everyday products, connect features to business outcomes
- ✓ Stay updated on tech trends (AI, SaaS, fintech)

Sharpen Structured Thinking

- ✓ Apply first principles thinking break problems into basics and rebuild solutions
- ✓ Think customer-first empathize with user needs

First Principles Thinking



Learn Core Case Types

- Product Design: Build from user needs to features
- Product Improvement: Identify metrics and improvements
- Root Cause Analysis: Hypothesize and dissect logically
- Other cases (e.g., market entry) are lower priority

Practice Cases

- ✓ Practice cases daily, essential for developing logic and comfort with various case types

Mock Interviews or P2Ps

- ✓ Do regular mocks with peers/seniors; simulate pressure and refine delivery
- ✓ Reflect and iterate after every mock

Learn from Other Domains preparation

- Consulting: Structured thinking, guesstimates
- Marketing: Go-to-market, pricing, customer research
- Finance: Breakeven, unit economics



Consulting

Structured thinking

Guesstimates

Case Practice



Marketing

Go-to-market strategy

Pricing strategies

Customer research



Finance

Breakeven analysis

Unit economics

Viability

CORE CONCEPTS



Launching a new product involves a systematic process that guides the journey from ideation to market entry. The following pipeline is specific to new product launches, focusing on steps to **identify customer needs, validate ideas, and prepare for scaling**. Note that scaling existing products requires a different approach.

Customer Discovery

Customer Discovery involves engaging with potential customers to deeply understand their challenges and needs. The focus here is on empathy and active listening, typically through open-ended interviews. The goal is to validate that the problem is significant and worth solving. Outputs like *Jobs to be Done* and *Agile User Stories* provide structured ways to articulate these insights.

This stage culminates in the creation of a Minimum Viable Product (MVP) - it is the simplest version of a product that delivers enough core functionality to solve a key customer problem and validate an idea with real users

Customer Validation

Customer Validation involves using the MVP to test how well the proposed solution resonates with customers. This stage is characterized by gathering user feedback, measuring key metrics (e.g., engagement, retention), and iterating based on these insights. The goal is to validate assumptions about the product, refine its features, and either succeed or pivot quickly and cost-effectively.



The aim is to succeed or fail fast and cheaply. So keep on iterating.

By the end of this phase, teams aim to achieve Product-Market Fit, where the value proposition aligns strongly with customer needs.

Customer Creation

Customer Creation focuses on testing and validating the customer's willingness to pay and the effectiveness of go-to-market strategies. This involves refining pricing models, identifying the most receptive customer segments, and iterating to fine-tune the product-market alignment. Simultaneously, the foundation for scaling is laid by exploring product growth strategies, assessing potential distribution channels, and planning how to reach broader customer bases effectively.

Company Building

Till now the product was being nurtured by a small team that tested technical and business feasibility. Company Scaling involves transitioning from a small, agile team to building a larger organization capable of supporting growth.

This phase includes hiring specialized talent, scaling infrastructure, and formalizing processes like customer support and sales. It also requires aligning the team with strategic goals, setting measurable KPIs or OKRs, and implementing tools to manage the complexities of a growing customer base. At this stage, the focus shifts to sustaining growth while maintaining the product's core value proposition.

The Product Life Cycle (PLC) describes the stages a product goes through from its inception to its eventual withdrawal from the market. Understanding the PLC helps businesses strategize for marketing, pricing, resource allocation, and innovation.

Though most resources mostly talk about only 4 steps we can also include the development stage as the initial step.

Development Stage

Characteristics: Focused on ideation, research, and prototyping (e.g., MVPs). High costs with no revenues. Activities include market research, feasibility studies, and securing funding.

Strategies: Prioritize thorough customer validation, iterative testing, and a strong go-to-market plan.

Challenges: High uncertainty, risk of poor validation, funding issues, and stakeholder misalignment.

Introduction Stage

Characteristics: Product launch with low sales, high marketing costs, and minimal/negative profits. Focus on building awareness and educating early adopters.

Strategies: Invest in marketing, refine pricing (penetration or skimming), establish distribution channels, and incorporate user feedback.

Challenges: Convincing customers, managing high expenses, and responding swiftly to market feedback.

Growth Stage

Characteristics: Rapid sales growth, increasing profitability, economies of scale, and market share expansion.

Strategies: Expand into new markets, add features, improve quality, and strengthen brand differentiation.

Challenges: Managing competition, maintaining quality, and avoiding inefficiencies from over-scaling.

Maturity Stage

Characteristics: Sales peak; market saturation leads to slower growth. Intense competition and price pressures.

Strategies: Focus on differentiation, cost optimization, exploring new uses/markets, and partnerships.

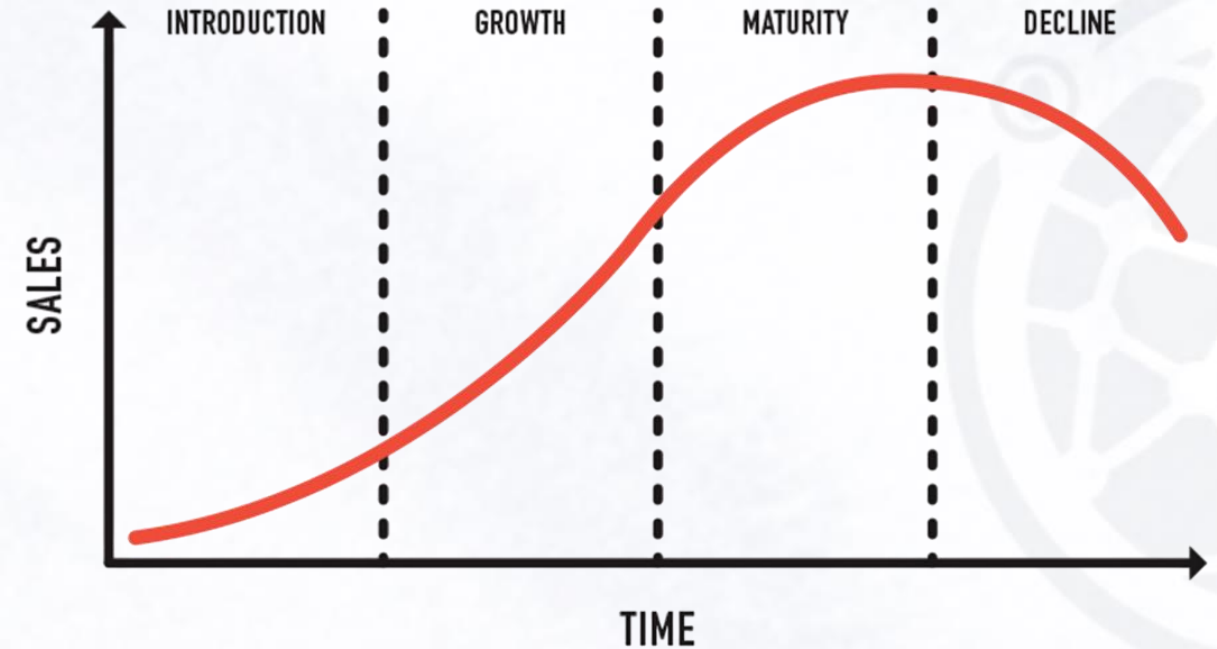
Challenges: Retaining customers amidst competition and maintaining profitability in a saturated market.

Decline Stage

Characteristics: Sales and revenues decline due to obsolescence, customer shifts, or market saturation.

Strategies: Attempt revitalization (e.g., redesign, niche targeting) or adopt harvesting/discontinuation strategies to maximize profitability.

Challenges: Deciding between rejuvenation or discontinuation and managing brand reputation.



Product Development Methodologies are structured approaches used to manage and guide the development of a product from concept to completion. There are two primary approaches to project management: Agile and Waterfall.

Agile Methodology

Overview:

Iterative, flexible, and collaborative approach emphasizing rapid delivery and stakeholder involvement.

Process:

Development occurs in sprints, allowing for incremental progress and adaptability to changing requirements.

Benefits:

- Quick adaptation to changes.
- Frequent delivery of functional components.
- Enhanced collaboration and innovation.

Challenges:

- Difficult for teams unfamiliar with iterative processes.
- Requires constant stakeholder engagement, which may not always be feasible.

Agile vs. Waterfall

Characteristic	Agile	Waterfall
 Approach	Iterative and flexible	Sequential and rigid
 Process	Sprints with frequent releases	Phases completed sequentially
 Flexibility	High	Low
 Customer Involvement	Ongoing involvement	Minimal
 Delivery	Incremental	Final product at the end
 Documentation	Lightweight and flexible	Comprehensive and fixed
 Best For	Complex, evolving requirements	Stable, well-defined projects

Waterfall Methodology

Overview:

Linear, sequential approach with distinct phases (Requirements → Design → Development → Testing → Deployment).

Process:

Each phase is completed before moving to the next; scope and requirements are fixed upfront.

Benefits:

- Easy to manage with well-defined milestones.
- Effective for projects with stable requirements.
- Comprehensive documentation supports future reference.

Challenges:

- Inflexible to changes once development begins.
- Late discovery of issues due to lack of interim deliverables.
- The final product is delivered only at the end.

Design is one of the **three core pillars of product management** alongside technology and business. A strong design sense enables product managers to create interfaces and experiences that **delight users** while meeting business objectives. Great design goes beyond aesthetics; it shapes how users interact with and feel about a product.

Design Thinking: The Foundation of Product Design

Design Thinking is a human-centered, iterative approach to solving problems and building products. It emphasizes:

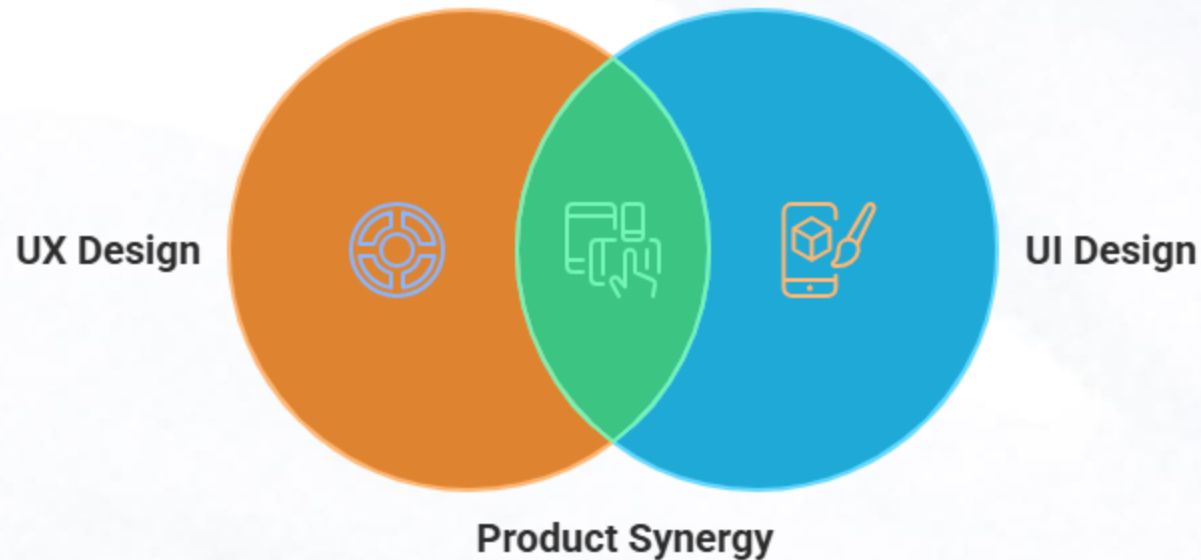
- ✓ Deeply understanding user needs
- ✓ Redefining problems
- ✓ Generating innovative, practical solutions
- ✓ Continuously integrating user feedback throughout development

This approach ensures the product stays aligned with user expectations and market demands.

Design Thinking Process Cycle



UI & UX: Creating Seamless Experiences



In product development, **User Experience (UX)** and **User Interface (UI)** design work hand-in-hand to create engaging, functional products. Though related, they focus on distinct aspects:

User Experience (UX)

UX design is about the **end-to-end journey** of the user, focusing on functionality, flow, and satisfaction. Key elements:

- ✓ Understanding user needs and pain points
- ✓ Mapping user journeys and use cases
- ✓ Prioritizing usability and accessibility
- ✓ Iterative improvements based on feedback

User Interface (UI)

UI design deals with the **look and feel**, the visual and interactive elements that users engage with directly. Key elements:

- ✓ Visual design: color schemes, typography, icons
- ✓ Interactivity: buttons, menus, animations
- ✓ Consistency: uniform styles, layouts, and patterns

User Persona

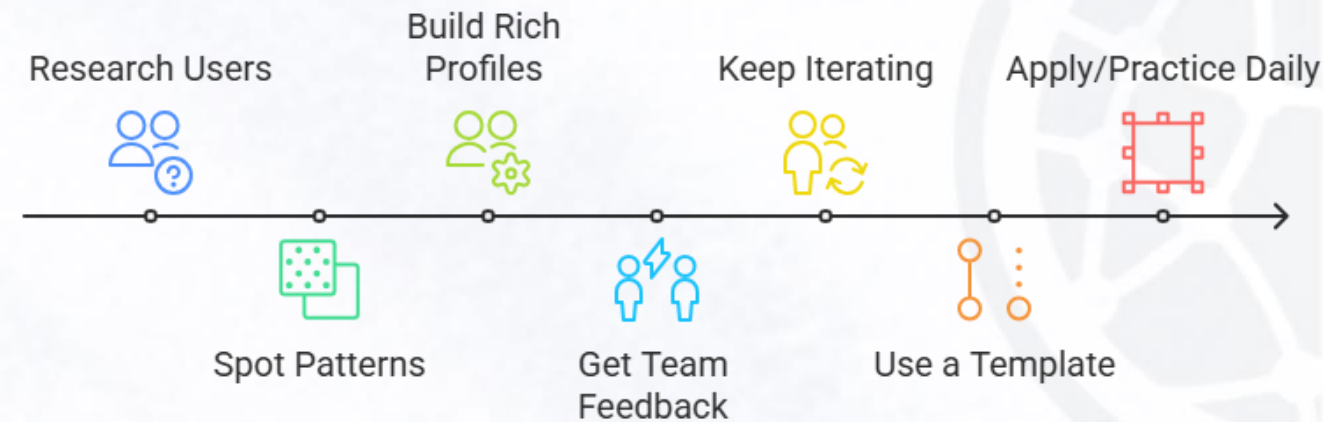
User personas are **strategic representations** of your target users — capturing their behaviors, motivations, goals, and pain points. For product managers, personas serve as essential tools to **humanize data** and guide design, development, and decision-making.

Rather than relying on abstract demographics, user personas help you **step into your user's world** and design solutions that genuinely resonate.

Why Personas Matter in Product Management

- ✓ **Bring clarity to user needs:** Personas move beyond "age 25, urban male" to describe *why* a user behaves a certain way.
- ✓ **Guide product decisions:** From feature prioritization to UI choices, personas help teams stay user-focused.
- ✓ **Align cross-functional teams:** Designers, marketers, and engineers can rally around a shared understanding of the user

Building Effective User Personas



User Habits

Understanding **user habits** is essential for product managers. It helps in building products that aren't just used once but become part of users' routines. Habit-forming products drive **higher retention, engagement, and sometimes network effects**, making them central to long-term product success.

Two widely-used frameworks to understand and design for user habits are:

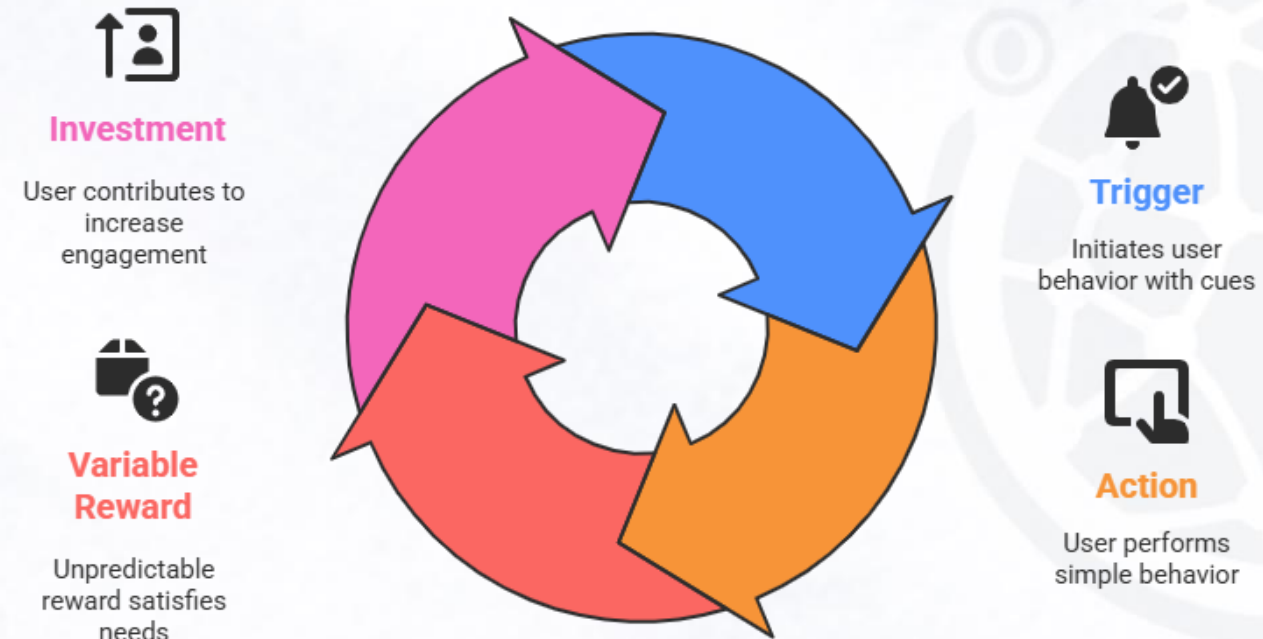
1. The Hook Model (Nir Eyal)
2. The Fogg Behaviour Model (BJ Fogg)

Both models explain **how products drive repeat behavior** — especially in consumer apps (think Instagram, Duolingo, Spotify).

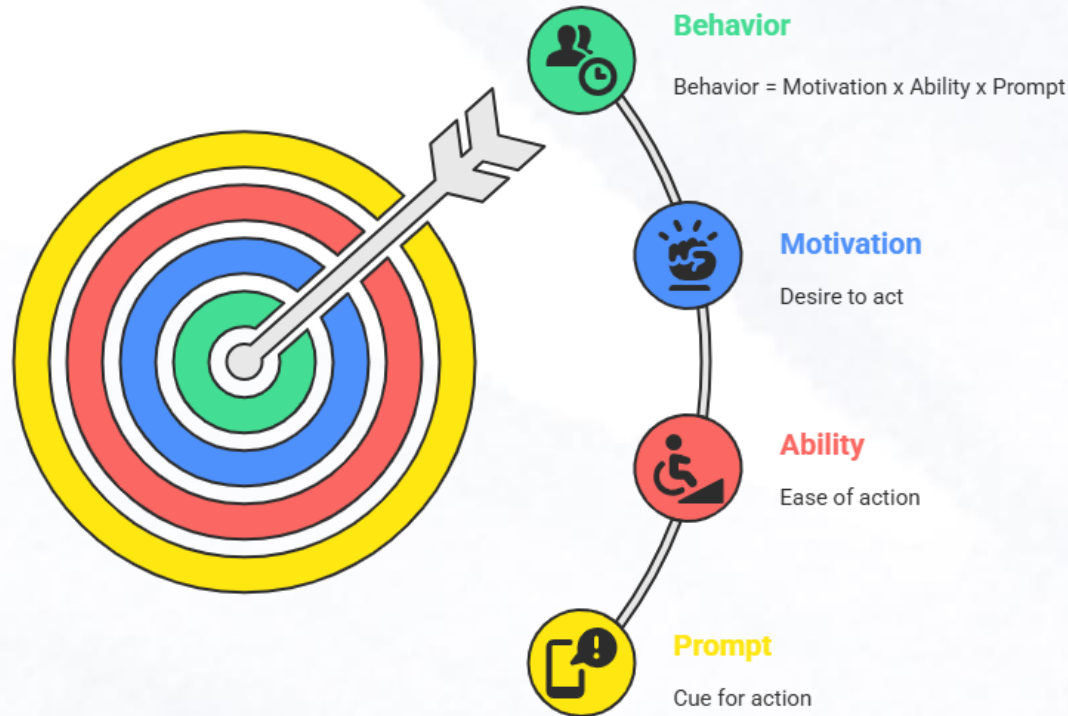
Use these models to frame your thoughts in:

- Product design cases:** How would you make X more engaging?
- Root cause analysis:** Why are users dropping off?
- Feature prioritization:** Which features drive habits?

The Hook Model Cycle



Fogg Behavior Model

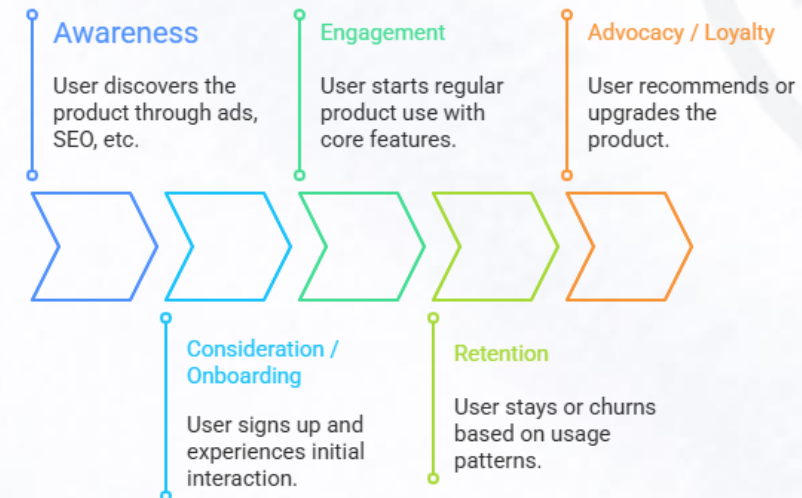


User Journey

The **user journey** maps the full experience a user has while interacting with a product, from the first touchpoint to long-term retention. It helps product managers understand not just *what* users do, but *how they feel*, *where they struggle*, and *what keeps them coming back*.

Why the User Journey Matters in Product Management?

- ✓ Identifies **pain points and drop-offs** across the funnel
- ✓ Reveals **opportunities for delight**, not just usability
- ✓ Helps align **product features with user goals** at each stage
- ✓ Encourages **cross-functional thinking** — product, design, marketing, support



Technical Concepts

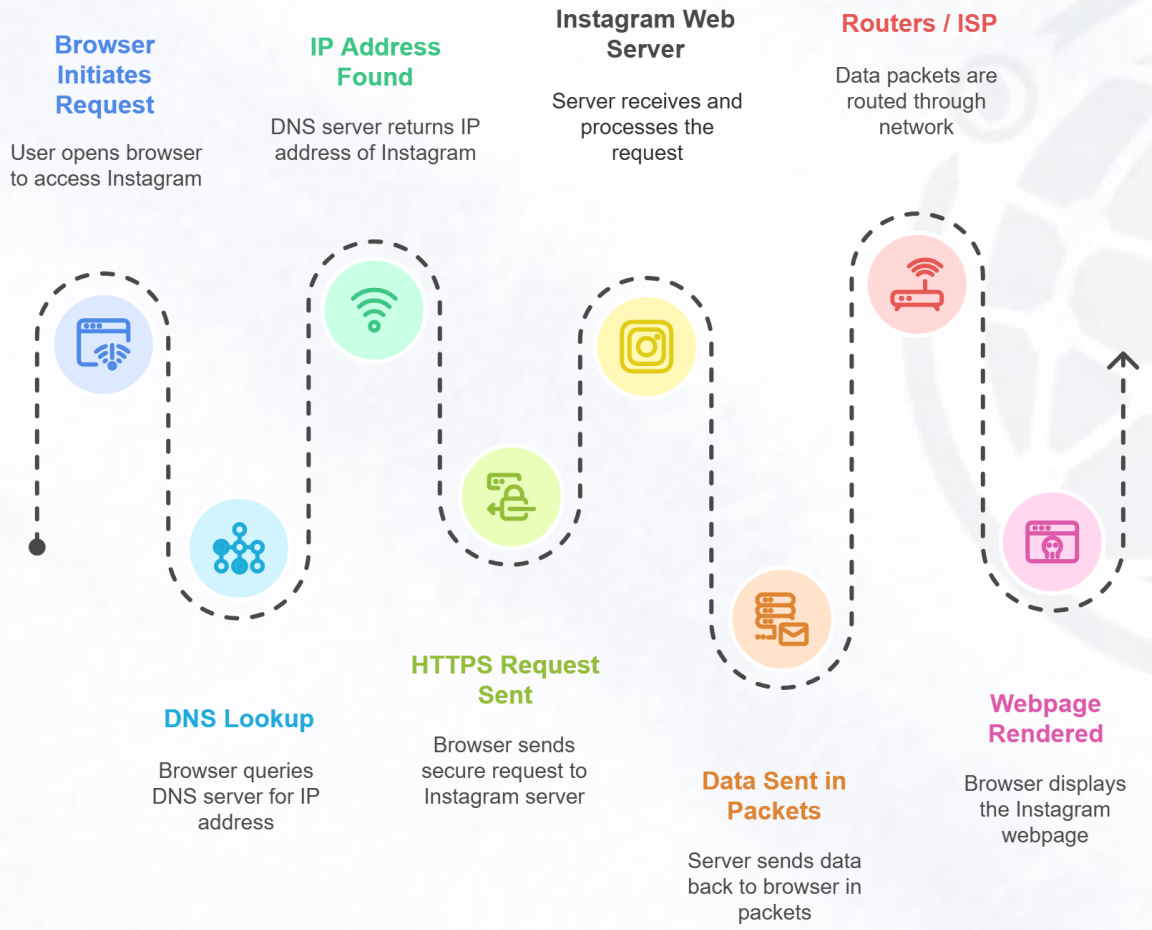


The Internet is the invisible force that powers almost everything around us today, from sending a WhatsApp message to streaming a Netflix show or making a UPI payment.

When we talk about "how the internet works," we're really talking about how billions of devices communicate across a global network, reliably, quickly, and often invisibly. This section breaks down the core building blocks of the internet, from IP addresses and DNS lookups to data packets and servers.

Concept	What It Means	Analogy
IP Address	A unique address assigned to each device connected to the internet	Like your home address.
DNS (Domain Name System)	Translates human-readable domain names into IP addresses.	Like a phonebook or contacts list.
Client & Server	The browser (client) requests a web page; the server sends the response.	Like ordering food in a restaurant.
HTTP / HTTPS	Protocols that define how data is transferred over the web. HTTPS is secure.	Like postal envelopes.
Packets	Data is broken into small pieces (packets) and sent across different paths.	Courier parcels sent via different routes.
Router	Connects networks and decides the best path for data packets.	Like a traffic controller.
ISP (Internet Service Provider)	The company that gives you access to the internet.	Electricity provider, but for data.

Accessing Instagram Webpage



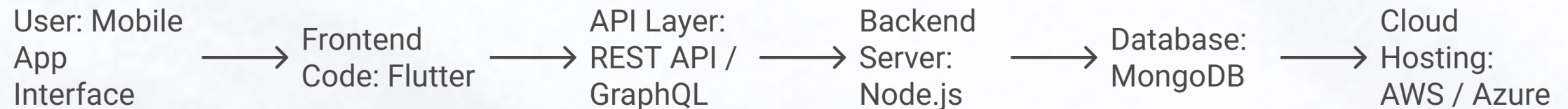
A **tech stack** is the combination of technologies used to build and run a software application. It includes the programming languages, frameworks, libraries, databases, tools, and cloud infrastructure required to develop, deploy, and maintain the app.

While tech stacks can vary based on the platform (mobile vs. web), they typically include a frontend (what the user sees), a backend (what handles the logic and data), and infrastructure components that support scalability, security, and performance.

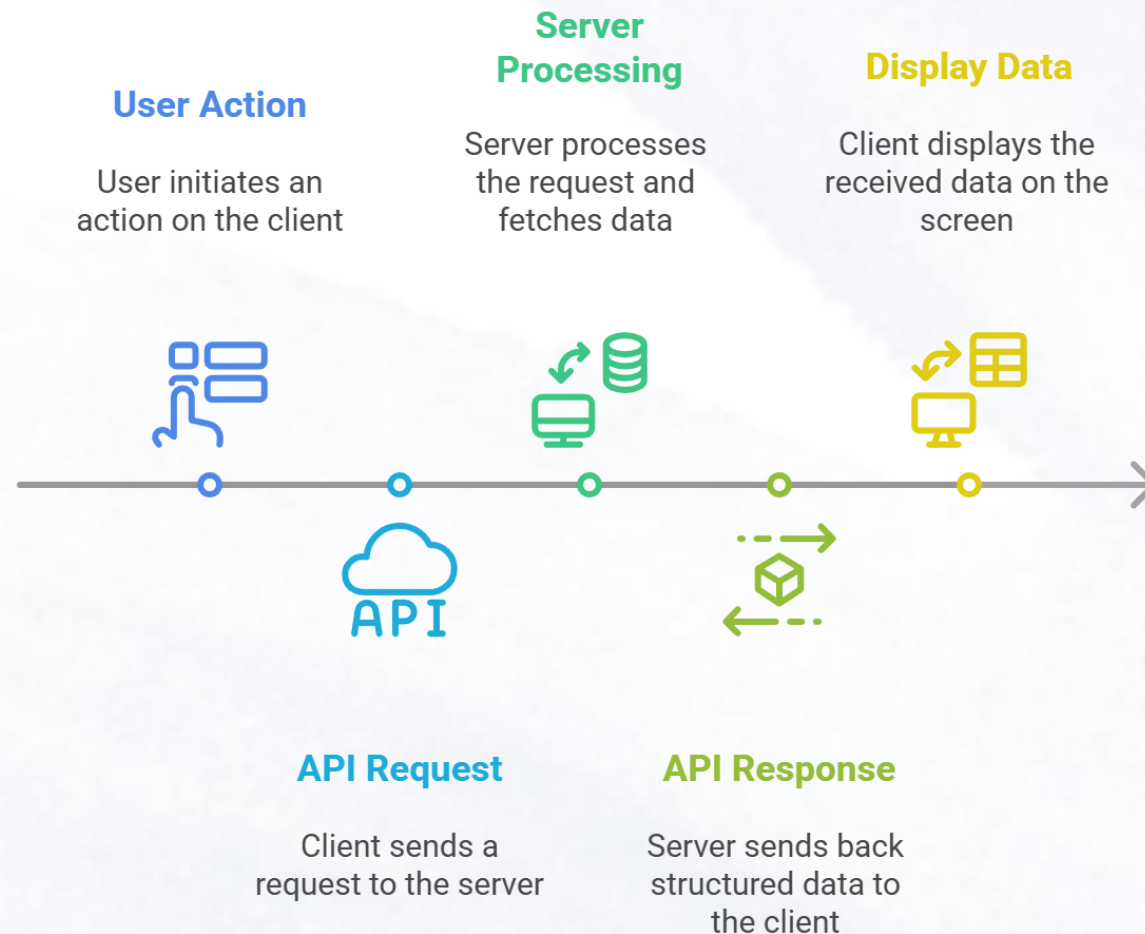
The choice of technologies at each layer depends on factors like platform (iOS, Android, Web), real-time requirements, expected traffic, and future scalability. Together, these layers form the backbone of any digital product.

Layer	What It Does	Common Tools / Technologies
Frontend (Client-Side)	Everything the user interacts with on the app – visuals, layout, buttons, screens.	iOS: Swift, Objective-C Android: Kotlin, Java
Backend (Server-Side)	Translates human-readable domain names into IP addresses.	Node.js, Django, Flask, Ruby on Rails.
Database	Stores user data, product info, order history, etc.	PostgreSQL, MongoDB, MySQL.
APIs (Application Programming Interfaces)	Bridge between frontend and backend. Fetches and sends data in a structured format.	REST APIs, GraphQL, gRPC
DevOps & Infrastructure	Deployment, monitoring, scaling, and CI/CD pipelines. Ensures the app is stable and available.	Docker, Kubernetes, AWS, Azure, Google Cloud, Jenkins.

Mobile App Tech Stack



API Request/Response Flow



Modern apps rarely work in isolation. Whether it's enabling Google login, showing weather data, or processing payments through Razorpay, applications often rely on other systems to exchange data and trigger actions. This is made possible by **APIs** and **Webhooks** — two core technologies that allow different software systems to communicate with each other.

APIs are request-based, one app asks another for something. Webhooks are event-based, one app notifies another when something happens.

API

An **API (Application Programming Interface)** is a structured way for one software application to communicate with another by sending a request and receiving a response. APIs define what data or services can be accessed, and how that access should happen. They are essential for tasks like fetching user data, placing orders, or integrating third-party services.

It is like ordering food at a restaurant. You (the client) ask for a dish from a menu (the API), and the kitchen (the server) prepares it and brings it back to you.

Webhook

A **Webhook** is a way for one application to automatically send data to another application when a specific event occurs. Unlike APIs, which require a request to get data, webhooks push data on their own, in real time, to a pre-defined URL (called the callback URL).

For example receiving a text message when your pizza is out for delivery. You don't ask — the system notifies you when the event happens.

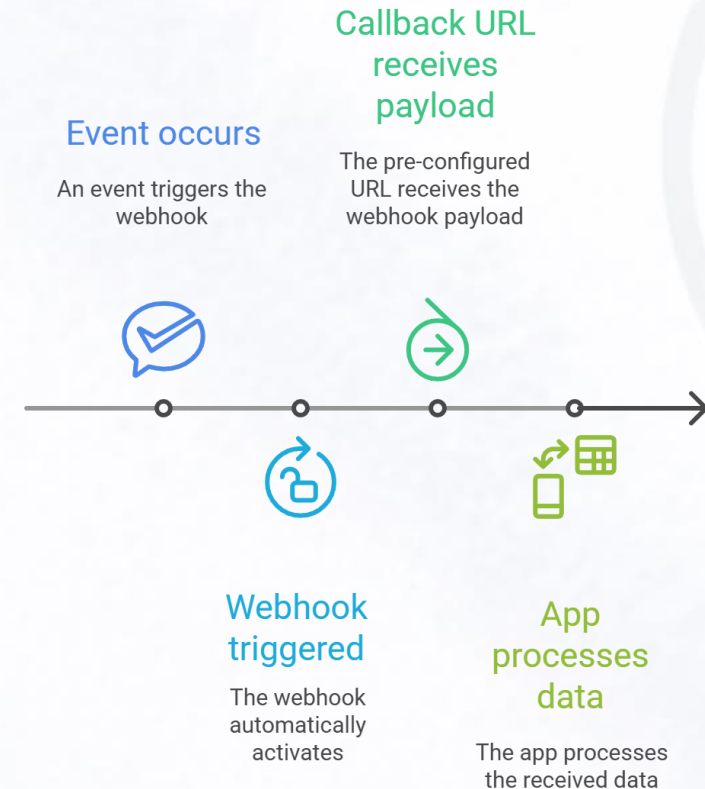
Terms

Endpoint - An endpoint is a specific URL in an API that performs a specific task, such as fetching a user profile or placing an order. It is like different counters at a food court, one for pizzas, one for desserts.

Request and Response - APIs work using requests (like GET or POST) and responses (often in JSON format). The client sends a request and receives a response from the server.

Callback URL - The pre-configured address where webhook notifications are sent. It must be set up in advance. It is like giving your phone number to the delivery agent so you can be notified when your food arrives.

Webhook Flow Illustration



Cloud Computing Process Cycle



Cloud computing is the delivery of computing services including servers, storage, databases, networking, software, and analytics over the internet (“the cloud”). Instead of owning and maintaining physical hardware or software, businesses and individuals can rent computing resources on demand from cloud providers like Amazon Web Services (AWS), Microsoft Azure, or Google Cloud Platform (GCP).

It allows apps, websites, and services to scale quickly, operate from anywhere, and handle large amounts of data without requiring massive upfront infrastructure investment.

Key Dimensions

Service Models

IaaS (Infrastructure as a Service) – Provides virtualized computing resources over the internet (servers, storage, networking). *Example:* AWS EC2 instances for hosting an app.

PaaS (Platform as a Service) – Provides a platform with built-in tools for building, deploying, and managing applications without handling server management. *Example:* Google App Engine .

SaaS (Software as a Service) – Delivers fully functional software over the internet, accessible via web or mobile app. *Example:* Gmail, Dropbox, Slack.

Deployment Models

Public Cloud – Services offered over the public internet and shared across organizations. *Example:* AWS, Azure public services.

Private Cloud – Dedicated cloud infrastructure for a single organization, offering more control and privacy. *Example:* VMware-based internal cloud.

Hybrid Cloud – Combination of public and private clouds, enabling data and app portability. *Example:* A retail chain storing sensitive customer data in a private cloud but running analytics in a public cloud.

Key Characteristics

On-Demand Self-Service – Users can provision computing resources as needed without human intervention.

Scalability – Resources can scale up or down automatically based on demand.

Pay-As-You-Go Pricing – Customers pay only for the resources they use.

Global Accessibility – Services can be accessed from anywhere via the internet.

Example: Cloud in a Food Delivery App

A food delivery app might:

Host its website on **AWS EC2** (IaaS).

Store order history and menus in **Google Cloud Firestore** (Database as a Service).

Use **Twilio (SaaS)** for sending SMS order updates.

Run promotional analytics on **Azure Machine Learning** (PaaS).

Imagine you're managing a large e-commerce platform, and millions of customer orders, product listings, and transaction details are generated daily. How do you organize all this data efficiently and make sure it's always accessible when needed? Enter the **Database Management System (DBMS)**.

A DBMS is a software system designed to manage, store, and organize data, ensuring that it's easy to retrieve, update, and secure. It acts as the bridge between end-users and the database, offering an efficient way to store, retrieve, and manipulate data.

Example (Orders App):

An app uses PostgreSQL to keep tables like Users, Products, Orders, and OrderItems. When a user places an order, the app sends SQL statements to the DBMS. The DBMS validates constraints (e.g., every OrderItem.product_id must exist in Products), writes the change to durable storage, and makes the new order visible to subsequent reads.

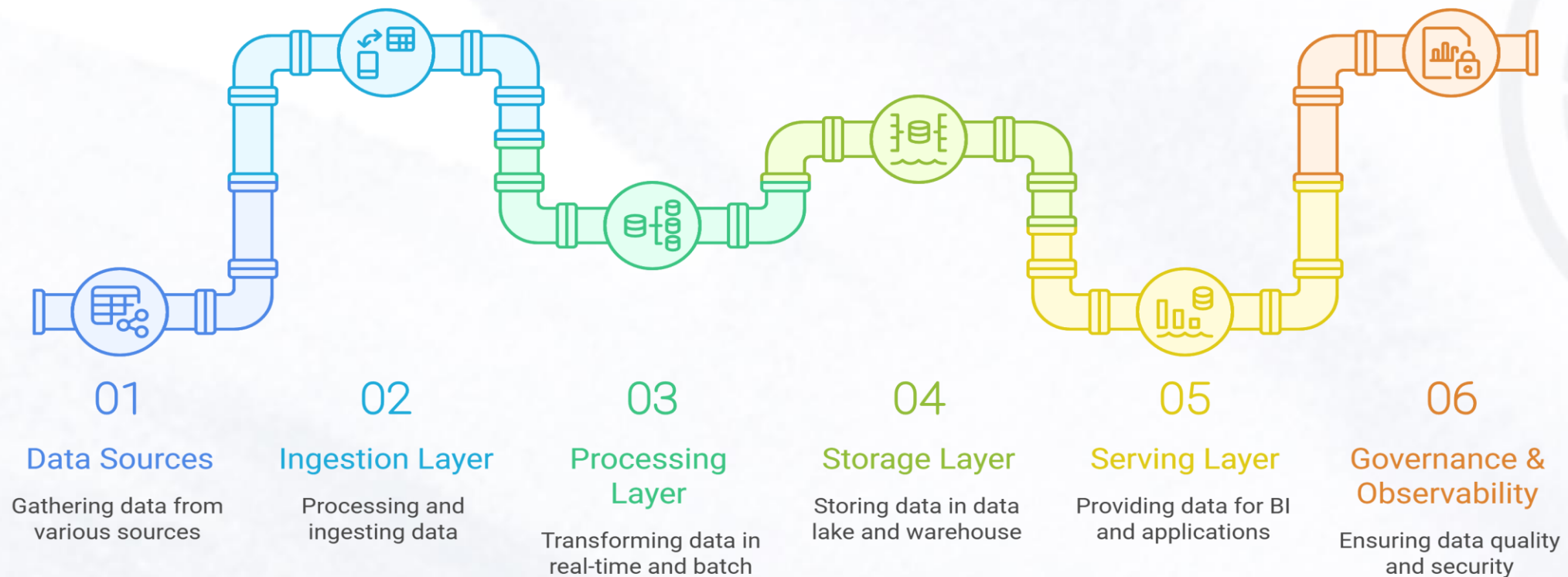
Properties of a DBMS

- **ACID transactions**
 - Atomicity: all steps of a transaction succeed or none do.
 - Consistency: constraints are preserved (no orphan OrderItem).
 - Isolation: concurrent transactions don't corrupt each other (controlled via isolation levels).
 - Durability: once committed, data survives crashes (write-ahead logs, checkpoints).
- **DBMS Engine:**
 - The DBMS engine is the core software component that provides the interface between the database and the applications using it. It manages data access, retrieval, and storage.
- **Database Scheme:**
 - A schema defines the structure of the database, including the tables, relationships, and constraints. It's the blueprint that dictates how the data is organized.
- **Query Processor:**
 - The query processor translates SQL queries into executable commands that the DBMS can process. It optimizes the queries for efficiency and ensures they are executed correctly.

Big Data refers to datasets that are too large, fast, or varied for traditional databases and single servers to handle. It focuses on how to **collect, store, process, and analyze** massive streams of information (clicks, app logs, sensor events, payments) so that useful results can be produced **quickly and reliably**.

In practice, this means combining special storage (data lakes/warehouses), high-throughput pipelines (Kafka), and distributed processing (Spark/Flink).

Big Data Pipeline



5Vs

- **Volume** – The sheer amount of data generated, often in terabytes or petabytes, that requires scalable storage solutions.
- **Velocity** – The speed at which data is created, collected, and processed, often in real time or near real time.
- **Variety** – The range of data types and formats, from structured tables to unstructured text, images, and video.
- **Veracity** – The trustworthiness and accuracy of data, which can be affected by noise, errors, or incompleteness.
- **Value** – The potential usefulness of data in generating insights, improving decisions, or creating products.

We also have 2 more Vs

- **Variability** – The change in data meaning or patterns over time, which can impact interpretation and analysis.
- **Visualization** – The ability to represent large datasets in clear, interpretable charts, dashboards, and visual tools.

Big Data Technologies

- **Hadoop:** Hadoop is an open-source framework that enables distributed storage and processing of large datasets across clusters of computers. It's based on the **MapReduce** programming model, where tasks are divided into smaller chunks and processed in parallel.
- **Apache Spark:** A faster, more advanced alternative to Hadoop's MapReduce, Spark is an open-source unified analytics engine for large-scale data processing. Unlike Hadoop, Spark processes data in-memory, making it much faster for real-time analytics.
- **NoSQL Databases:** Traditional SQL databases are designed for structured data, but NoSQL databases like MongoDB, Cassandra, and Couchbase are more suited for handling unstructured and semi-structured data. These databases provide flexibility and scalability for Big Data applications.

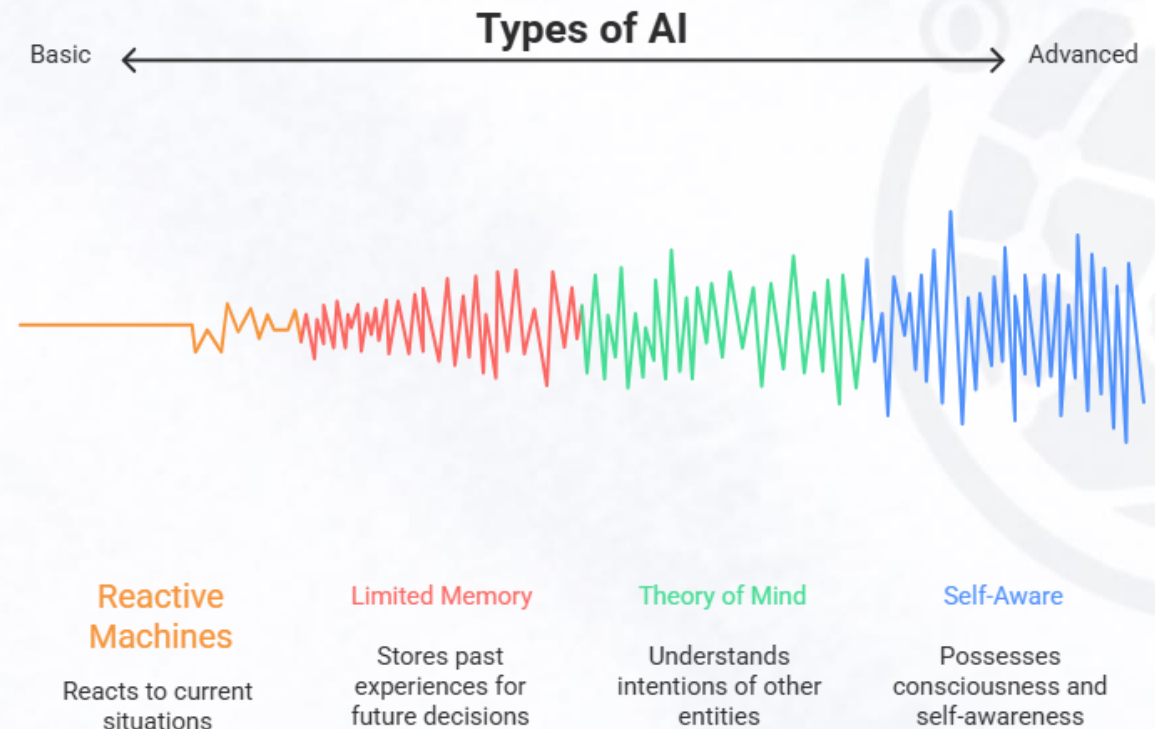
The world is undergoing a transformative shift, thanks to the rapid advancements in **Artificial Intelligence (AI)**, **Machine Learning (ML)**, and **Large Language Models (LLMs)**. From voice assistants to self-driving cars, these technologies are shaping how we interact with the world around us. But how do they work, and why are they so powerful? Let's break them down step by step, and explore their significance in today's technological landscape.

What is Artificial Intelligence (AI)?

At its core, **Artificial Intelligence** is the simulation of human intelligence in machines. The goal of AI is to enable computers to perform tasks that typically require human intelligence, such as decision-making, problem-solving, and learning from experience. AI is a broad field, encompassing various sub-disciplines and techniques designed to build intelligent systems.

Narrow AI: Also known as **Weak AI**, this type of AI is designed to perform a specific task, such as speech recognition (like Siri), image recognition, or recommendation systems.

General AI: Strong AI aims to create machines that can perform any cognitive task that a human can. While this type of AI remains theoretical and has not been fully realized, researchers are still working toward achieving it



Machine Learning

Machine Learning is a subset of AI that focuses on the development of algorithms that allow computers to **learn** from and make decisions based on data, rather than being explicitly programmed. In ML, systems use data to detect patterns, make predictions, and improve their performance over time without human intervention.

ML systems follow a simple loop:

Input Data → Learn Patterns → Make Predictions → Improve with Feedback

For example:

An email spam filter sees 1,000 labelled emails (spam vs not spam), learns patterns (e.g., keywords, sender behaviour), and starts classifying new emails with increasing accuracy over time.

Type	Description	Example
Supervised Learning	Learns from labeled data (input + known output)	Spam filter, loan approval
Unsupervised Learning	Finds hidden patterns in unlabeled data	Customer segmentation, product bundling
Reinforcement Learning	Learns via trial and error by receiving rewards/punishments	Self-driving cars, game-playing AIs (e.g., AlphaGo)

Large Language Models (LLMs)

Large Language Models (LLMs) are advanced AI systems trained to understand, generate, and manipulate human-like text. They use deep learning (specifically transformer architectures) to analyze large volumes of data and generate coherent, context-aware responses. For product managers, understanding LLMs is key to building AI-powered features, evaluating feasibility, and spotting opportunities in conversational AI, automation, and personalization.

How LLMs Work

Training Data: LLMs are trained on massive datasets (books, websites, articles) to learn language patterns, grammar, logic, and facts.

Model Architecture: Most modern LLMs use the **Transformer** architecture, which enables them to process and generate long, coherent text sequences.

Output: Given a prompt (e.g., a question), the model predicts and generates the most likely next word — repeated until it forms a full response.

How AI, ML, and LLMs Fit Together

AI is like intelligence, the broad goal of machines acting smart

ML is like learning, how machines get smarter from data

LLMs (like ChatGPT) are specialists in language, trained using ML, but focused on text-based understanding and generation

Real-World Analogy

Imagine teaching a child:

AI is the *goal*: You want the child to make good decisions.

ML is *how they learn*: They observe examples, learn from mistakes, and improve.

LLMs are like giving the child *millions of books and conversations* — now they can speak like an expert, even though they don't always know the full truth.

Some widely used LLMs:



E-commerce refers to the buying and selling of goods and services via online platforms. It enables consumers to discover and purchase products without visiting a physical store. Over time, advancements in logistics, payments, and mobile technology have significantly improved the user experience.

Key characteristics of E-Commerce:

- 1) Large product catalogues (10,000–100,000+ SKUs)
- 2) Centralized fulfilment and third-party logistics
- 3) Delivery time: typically 1–3 days
- 4) Suited for planned, higher-value purchases
- 5) Average order value (AOV): ₹800–₹2000

Key Players in India:



Quick Commerce (Q-commerce), a more recent evolution of e-commerce pushes the boundaries of convenience by offering delivery of daily-use items within 10–20 minutes. This model is designed around impulse behaviour, frequent orders, and hyperlocal fulfilment.

Key characteristics of Quick Commerce:

- 1) Curated assortment (~2,000–5,000 SKUs) based on high-velocity items
- 2) Fulfilled via dark stores or micro-warehouses
- 3) Delivery time: under 20 minutes
- 4) Suited for impulse or top-up purchases
- 5) Average order value (AOV): ₹200–₹500

Key Players in India:



From a PM’s Perspective

Parameters	E-Commerce	Q-Commerce
User Behavior	Considered Purchase	Instant needs
Metrics Used	Conversion rate, cart abandonment, return rate etc.	Delivery time, fill rate, frequency, SKU velocity etc.
Design Thinking	Reduce friction in search and cart flow	1-tap checkout, product bundling, real time tracking
Margins	Relatively better margins	Thin margins, SKU rationalization must be priority

“E-commerce solves for scale and selection; Q-commerce solves for speed and habit. As a PM, the challenge is to align tech, logistics, and user experience — optimizing for either access or immediacy, depending on context and constraints.”

User Journey E-Commerce and Quick-Commerce



What is IoT?

IoT refers to a network of physical devices that connect to the internet and to each other, making them capable of **sensing, processing, and acting** in real time. From smart thermostats in homes to machinery on factory floors, IoT is transforming how we live, work, and interact with our environment.

Key Components

- 1) **Devices & Sensors:** Physical objects like wearables, smart fridges, traffic cameras, collecting real-time data.
- 2) **Connectivity:** Communication protocols like Wi-Fi, Bluetooth, Zigbee, or LPWAN that transmit the data.
- 3) **Data Processing:** Cloud platforms or edge computing units analyze incoming data to extract actionable insights.
- 4) **Action Layer:** Based on insights, the system takes autonomous action; sending alerts, triggering processes, or adjusting settings.

IoT Data Flow and Decision Making

Sensing → Transmission → Analysis → Decision → Action

Real-World Use Cases

Smart Homes: Alexa, Google Nest, and other systems control lights, security, and appliances.

Smart Cities: IoT enables traffic signal optimization, smart parking, and efficient waste management.

Healthcare: Wearables like Apple Watch track vitals and send real-time updates to doctors.

Manufacturing (IoT): Predictive maintenance and real-time supply chain tracking improve efficiency.

The Future of IoT

As 5G and edge computing mature, IoT will become even faster, more reliable, and widespread. Expect growth in autonomous vehicles, precision agriculture, remote surgeries, and connected factories.

By 2030, the world is projected to have **50+ billion IoT devices**. India is witnessing rapid adoption in **agritech, logistics, healthcare, and energy monitoring**.

PM Interview Insight

“IoT extends product design beyond the screen. A good IoT product manages edge conditions, minimizes user friction, and creates tangible value through automation, without overwhelming the user.”

Blockchain is a **distributed digital ledger** that records transactions across a network of computers. Instead of a single central authority, each participant (or “node”) in the network holds a copy of the ledger. Every time a transaction occurs, it's grouped into a “block,” which is validated and added to the chain forming an immutable, time-stamped history of all actions.

Key Characteristics

Decentralized: No single authority controls the data; all nodes are equal participants.

Immutable: Once recorded, data on the blockchain cannot be altered or deleted.

Transparent: Every transaction is visible to all participants, improving accountability.

Secure by Design: Transactions are encrypted and verified through consensus mechanisms

PM Interview Insight

“Blockchain replaces centralized trust with math and code. As a PM, you must ask does this use case really need decentralization? If yes, how do I design around complexity and create user trust in a trustless system?”

Product Perspective

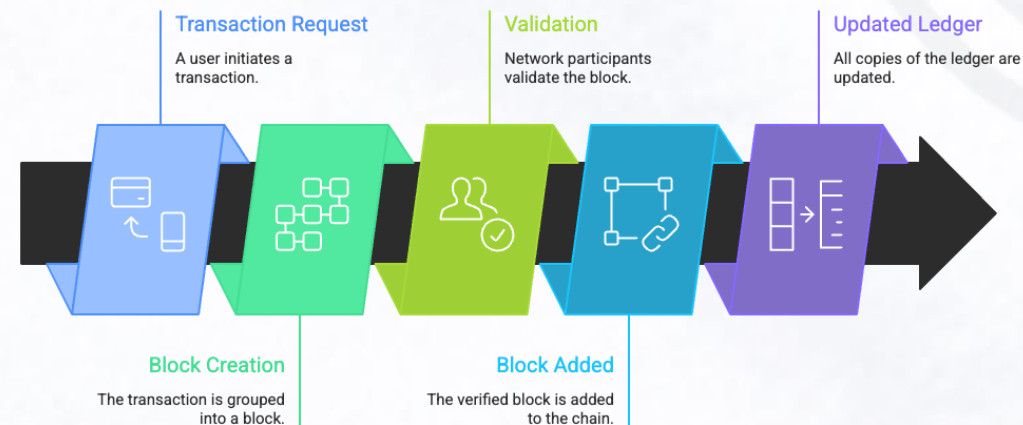
Design Considerations:

- User onboarding: abstracting complexity (e.g., private keys, wallets)
- Scalability: high latency and energy costs in some chains
- Regulation & compliance: especially in fintech, healthcare
- Ecosystem dependence: protocols evolve rapidly (e.g., Ethereum upgrades)

Key Metrics:

- Transaction throughput (TPS), gas fees (cost per transaction), latency
- User retention in decentralized apps (dApps)
- Wallet activation and usage rates

Blockchain Transaction Process



A revenue model outlines **how a company earns revenue**; who pays, what they pay for, how much, and how often. It differs from a business model (which includes cost structure, value proposition, etc.) by focusing specifically on **monetization**.

Common Revenue Models

1) Transaction-Based:

Revenue from each individual sale or transaction

Examples: Amazon (per item sold), Razorpay (per transaction fee)

2) Subscription-Based:

Recurring payments (monthly/yearly) for continued access

Examples: Netflix, Spotify, SaaS tools (e.g., Notion, Figma)

3) Freemium:

Free tier to attract users; paid tier unlocks premium features

Examples: Duolingo, Zoom, LinkedIn Premium

PM Consideration: Balance between value in free vs upsell incentive

4) Advertising-Based:

Revenue from showing ads to users (often free to use)

Examples: Google, Facebook, YouTube

Key metric: DAUs, engagement, ad impressions/clicks

5) Commission/Aggregator:

Platform takes a cut from transactions between two parties

Examples: Swiggy, Uber, Airbnb (commission from service providers)

6) Licensing & Royalties:

Revenue from intellectual property or one-time licensing

Examples: Oracle, Shutterstock, Dolby

7) Usage-Based (Pay-as-you-go):

Revenue depends on consumption (e.g., data used, API calls)

Examples: AWS, Twilio, Snowflake

8) Product Bundling / Cross-Selling:

Monetize by bundling multiple services or driving usage of other units

Examples: Apple (iCloud + devices), Microsoft 365

PM Lens – Why This Matters

- ✓ **Pricing impacts positioning**
- ✓ **Model influences retention mechanics:** Subscriptions demand consistent value
- ✓ **Unit economics must align:** $LTV > CAC$ is non-negotiable.
- ✓ **Metrics change by model:** For example, ARPU (average revenue per user), churn rate, conversion to paid.

Product interviews often test your ability to think in structured terms and not just solve problems, but **speak the language of product, tech, and business**. Here's a concise roundup of key definitions every aspiring PM should know, explained with clarity and relevance.

- ❖ **Product**:- A product is anything that delivers value to users by solving a need or problem. This could be physical (e.g., iPhone), digital (e.g., Gmail), or a service (e.g., Uber). A product should have a clear **value proposition**, a target user, and defined business goals.
- ❖ **Problem statement**: It identifies the gap or pain point the product aims to solve. A good statement is **user-centric, specific, and measurable**.
Example: "Busy professionals struggle to find healthy meals during work hours."
- ❖ **MVP (Minimum Viable Product)**: An MVP is the **simplest version of a product** that delivers core value and can be launched to validate assumptions with real users. It reduces time-to-market and helps teams learn quickly.
Goal: Learn > Perfect.

- ❖ **Product-Market Fit (PMF)**: Product-market fit means your product is being used and loved by your **target audience**, and demand is organically growing. It's a signal that you've built something people **want and are willing to pay for**.
Popular test: If 40% of users say they'd be "very disappointed" if the product disappeared, you're close to PMF.

- ❖ **North Star Metric** : A **North Star Metric** is the single most important metric that best captures the **core value** your product delivers to users. It helps align teams and prioritize work.

For Spotify: Time spent listening
For Airbnb: Nights booked

- ❖ **KPIs vs Metrics**:

Metric: A quantifiable measure (e.g., number of users, revenue)

KPI (Key Performance Indicator): A strategic metric that tracks success against a business or product goal (e.g., churn rate, NPS)

All KPIs are metrics, but not all metrics are KPIs.

- ❖ **Churn**: Churn is the **rate at which users stop using your product or cancel subscriptions**. It's a critical retention metric, especially for SaaS or subscription products.
- ❖ **CAC & LTV**: **CAC (Customer Acquisition Cost)**: Total cost to acquire one customer (marketing + sales spend). **LTV (Customer Lifetime Value)**: Total revenue a customer brings over their engagement span. *A healthy product usually ensures $LTV > 3x CAC$.*



Frameworks



The Business Model Canvas is a strategic management tool that helps visualize and articulate a business model. It's a single-page document divided into nine building blocks that cover the four main areas of a business: customers, offer, infrastructure, and financial viability.

The canvas provides a holistic view of a company's business model, making it easier to understand and communicate. It's an excellent tool for product managers because it forces you to think beyond the product itself and consider the entire business ecosystem in which it exists. It helps product managers understand how their product fits into the larger business context.

The Nine Building Blocks

The canvas is read from right to left, starting with the customer and moving toward the financial aspects.

1. Customer Segments

This block defines the different groups of people or organizations an enterprise aims to reach and serve. These are your target customers.

For example, a streaming service might have customer segments for young adults, families, and seniors.



2. Value Propositions

This describes the bundle of products and services that create value for a specific customer segment. What problem are you solving? What need are you satisfying? This is the core offering of your business. For instance, a coffee shop's value proposition isn't just coffee; it's a convenient location, a cozy atmosphere, and quality beverages.

3. Channels

This block describes how a company communicates with and reaches its customer segments to deliver a value proposition. Channels can be direct (e.g., a company's website) or indirect (e.g., a retail store).

4. Customer Relationships

This describes the types of relationships a company establishes with specific customer segments. This could be anything from automated self-service to personal assistance. For a software product, this might include community forums, email support, or dedicated account managers.

5. Revenue Streams

This represents the cash a company generates from each customer segment. How do you make money? This could include one-time sales, subscription fees, licensing, or advertising revenue.

6. Key Resources

These are the most important assets required to make a business model work. This can be physical (e.g., a factory), intellectual (e.g., patents), human (e.g., skilled engineers), or financial.

7. Key Activities

These are the most important things a company must do to make its business model work. For a software company, this might include software development, platform maintenance, and customer acquisition.

8. Key Partnerships

This block describes the network of suppliers and partners that make the business model work. These are the relationships you build to optimize your business model, reduce risk, or acquire resources.

9. Cost Structure

This describes all the costs incurred to operate a business model. This could include fixed costs (e.g., rent) and variable costs (e.g., raw materials).

The Value Proposition Canvas is a tool that helps ensure a product or service is positioned around a customer's values and needs. It is an extension of the Business Model Canvas, specifically focusing on the "Customer Segments" and "Value Propositions" blocks. It consists of two main parts: the Customer Profile and the Value Map.

Customer Profile (The Circle)

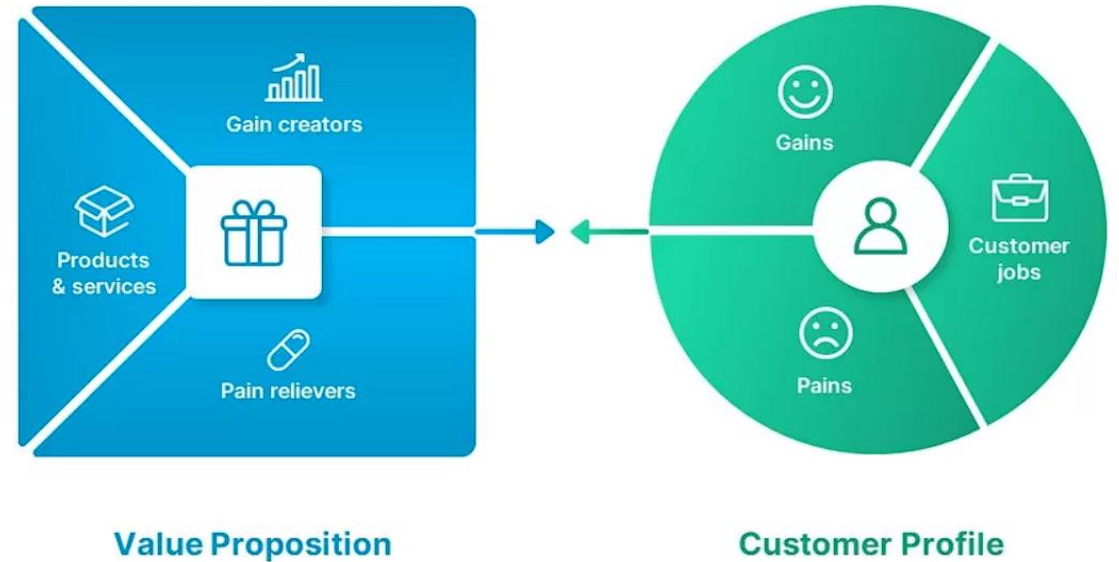
This part of the canvas focuses on understanding your target customer in detail.

Customer Jobs

These are the things your customers are trying to get done in their work or personal life. This could be a task they are trying to perform, a problem they are trying to solve, or a need they are trying to satisfy. For example, a "job" for someone using a ride-sharing app might be "getting from point A to point B efficiently."

Pains

These are the negative experiences, emotions, and risks that the customer experiences while trying to get a job done. Pains could be anything from a high cost to a negative social consequence. For the ride-sharing user, a "pain" might be "waiting a long time for a ride" or "the ride is too expensive."



Gains

These are the positive outcomes and benefits the customer wants to achieve. They can be functional utility, social gains, positive emotions, or cost savings. A "gain" for the ride-sharing user could be "getting a discount on the ride" or "arriving at their destination feeling relaxed."

Value Map (The Square)

This part describes the value you are creating for your customer.

Products & Services

This is the list of items your value proposition is built around. This can be anything from a physical product to an intangible service.

Pain Relievers

This describes how your products and services alleviate specific customer pains. For the ride-sharing app, a "pain reliever" could be a "predictable fare estimate" or "real-time tracking of the driver."

Gain Creators

This describes how your products and services produce customer gains. A "gain creator" could be "loyalty points for frequent rides" or "a premium option with better cars."

Alignment

The ultimate goal of the Value Proposition Canvas is to achieve a "fit" between the two sides. This is when the products and services you offer (the Value Map) directly address the jobs, pains, and gains of your customer (the Customer Profile). By visualizing this relationship, you can create products that truly resonate with your target audience.



Product-market fit is the degree to which a product satisfies a strong market demand. It is often described as the point where your product meets the needs of a large enough market, resulting in a sustainable and scalable business. The framework is not a single, universally defined model, but rather a set of principles and metrics used to assess this "fit". It's a critical concept for product managers, as achieving product-market fit is the primary goal of any early-stage product.

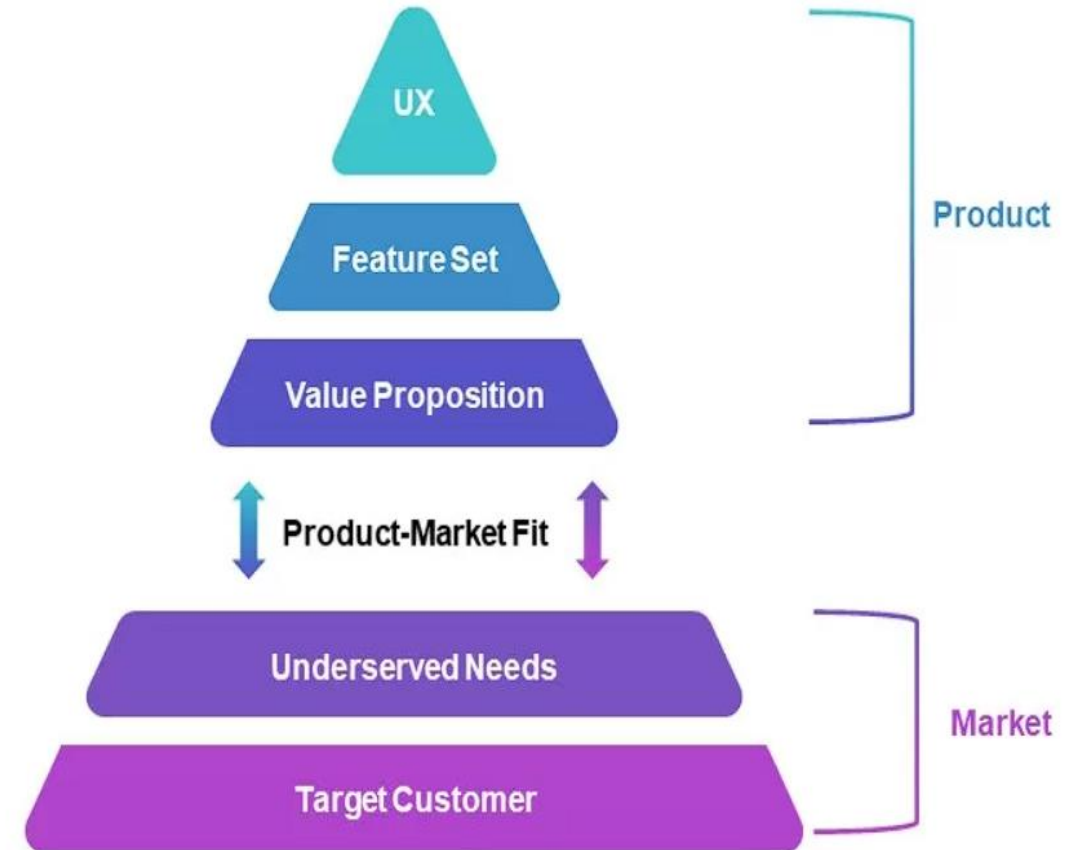
The Three Core Components

The framework can be broken down into three key components, often represented as a Venn diagram.

The Market

This is your target audience and the problem they have. A large and well-defined market is a prerequisite for product-market fit.

- **Market Need:** Is there a clear, urgent problem that needs to be solved?
- **Market Size:** Is the market large enough to sustain a business? (e.g., total addressable market).



The Product: This is your solution to the market's problem.

- **Value Proposition:** What unique value do you offer that solves the market's problem?
- **Features & Functionality:** Does your product have the necessary features to deliver on that value proposition?
- **User Experience (UX):** Is the product easy and enjoyable to use?

The Fit: This is the alignment between the market and the product. You've achieved "fit" when the market's need is so strong that your product sells itself.

Metrics and Signals of Product-Market Fit

How do you know when you've achieved it? While there is no single metric, a combination of qualitative and quantitative signals can help you assess the fit.

Quantitative Metrics (The "Hard" Data):

- **Retention Rate:** High user retention is a key indicator. Are users coming back and using your product regularly without being prompted?
- **Usage Frequency & Engagement:** Are users engaging with your product frequently and deeply?
- **Organic Growth:** Are new users coming to your product through word-of-mouth rather than paid marketing? This is often the most significant signal.

- **Revenue Growth:** For paid products, strong, consistent revenue growth with low churn.

Qualitative Signals (The "Feelings" Data):

- **User Testimonials:** Are customers providing unsolicited, enthusiastic praise?
- **Low Churn:** Customers are not leaving your product.
- **Unsolicited Positive Feedback:** You're getting positive reviews and feedback without even asking for them.
- **Viral Loops:** The product is so good that users are sharing it and inviting others to use it, creating a natural growth cycle.

Marc Andreessen, who popularized the term, famously said, "You can always feel product-market fit when it's happening. The customers are buying the product just as fast as you can make it, or the usage is growing just as fast as you can add more servers."

Problem-solution fit is the stage where you've clearly identified a **problem** that is worth solving and have a well-defined **solution** that effectively addresses it. It's the first step in the journey toward achieving product-market fit, acting as a crucial bridge between understanding a customer and building a product for them. It focuses on validating that your proposed solution truly resonates with the problem your target customers face.

The Core Components

The framework is a simple but powerful exercise that can be visualized as two interlocking parts:

The Problem

This is about deeply understanding your customer and the specific issue they face.

- **The Problem Itself:** What is the core pain point or unfulfilled need? Is this problem significant enough that people would pay for a solution?
- **Customer Pains:** What are the frustrations, obstacles, and risks associated with the problem?
- **Existing Alternatives:** How are customers currently solving this problem? What are the shortcomings of

those alternatives? This helps you identify what a good solution must do better.

The Solution

This is your proposed answer to the problem.

- **Solution Features:** What specific features and functionalities will your product have to solve the problem?
- **Pain Relievers:** How will your solution directly alleviate the customer's pains?
- **Unique Value Proposition:** What makes your solution better than the existing alternatives?

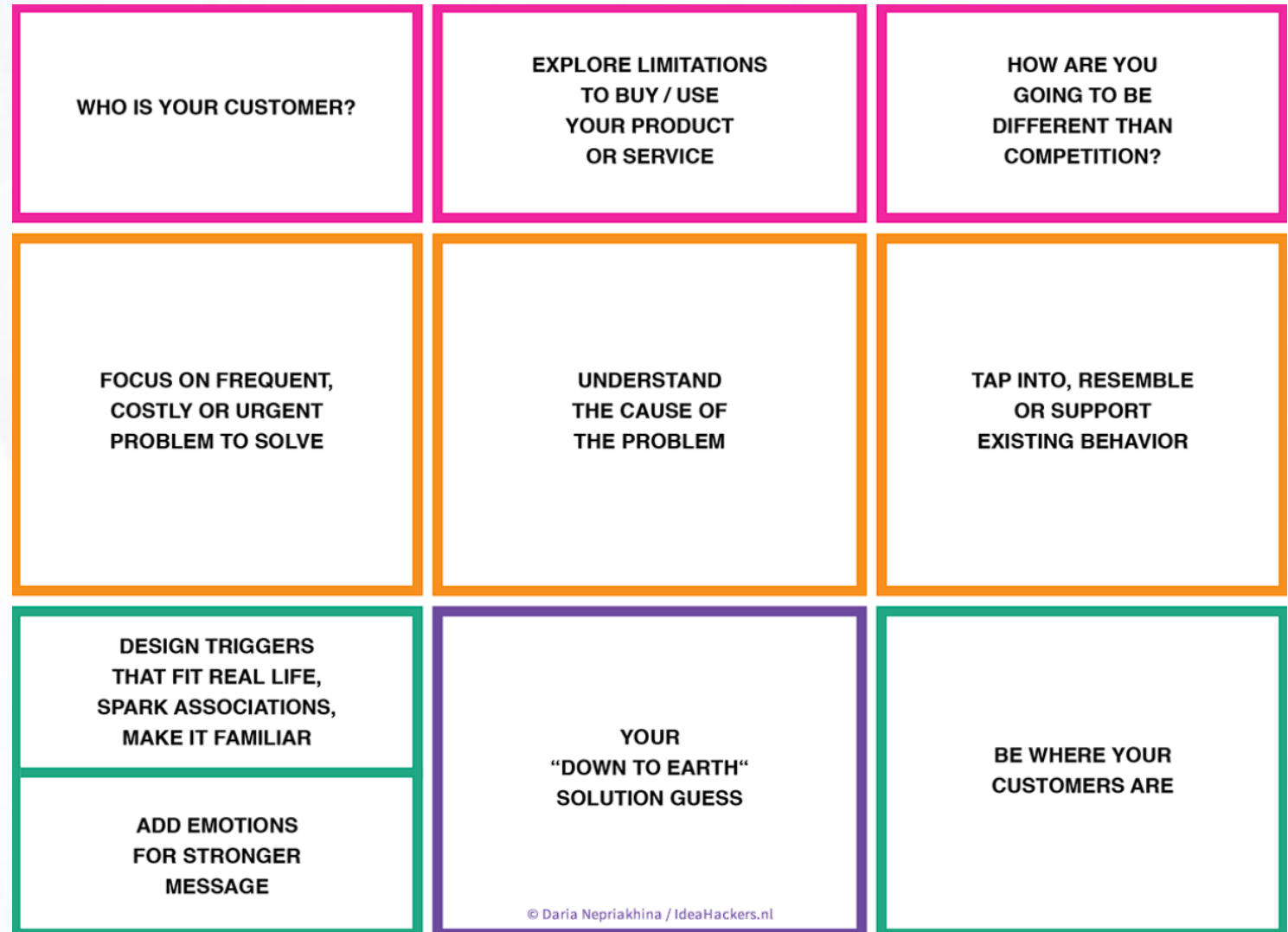
How to Achieve and Validate Problem-Solution Fit

Achieving problem-solution fit is less about building a final product and more about conducting research and validation. It's a hypothesis-driven process that involves talking to customers, not just for a product idea, but for the problem itself.

- **Interviews and Surveys:** Conduct one-on-one interviews and surveys with your target audience to understand their daily lives, problems and what they're willing to pay for.

- **Creating a Customer Journey Map:** Visualize the steps a customer takes to achieve a goal. This helps you pinpoint specific pain points at each stage.
- **Problem Statement:** Formulate a clear, concise problem statement that is validated by your research.
- **Hypothesis Testing:** Create a hypothesis (e.g., "Customers are frustrated with the slow process of X, and we believe a tool that does Y will solve this frustration") and test it using a **Minimum Viable Product (MVP)**.
- **Validation through Prototypes or Mockups:** Show customers early prototypes or mockups of your solution and get their feedback. Do they believe this solution would effectively solve their problem? Their enthusiastic "yes" is a strong signal of fit.

A strong problem-solution fit ultimately lays the foundation for building a product that people will actually want to use.



The CIRCLES Framework is a structured approach to answering product design and improvement questions, particularly in product management interviews. Developed by Lewis C. Lin, it provides a step-by-step method for breaking down complex problems and presenting a logical, well-thought-out solution.

The Six-Step Framework

Each letter in the acronym **C.I.R.C.L.E.S.** represents a key step in the process.

C - Comprehend the Situation: Begin by asking clarifying questions to understand the context. Don't jump to a solution immediately. Who is the user? What is their current situation? What are the constraints (e.g., time, budget, technology)? This step ensures you're solving the right problem.

I - Identify the Customer: Define who you are designing the product for. You can segment the user base and choose a specific segment to focus on. For instance, for a new music app, you might focus on "Gen Z students" or "professional musicians."

R - Report the Customer's Needs: Based on your customer

segment, identify their pain points and needs. What are they trying to achieve? What are their frustrations with current solutions? Prioritize the top 2-3 needs to focus your solution on.

C - Cut Through Prioritization: This is where you prioritize the customer needs you just identified. You can use a matrix (e.g., urgency vs. importance) or a scoring model to determine which needs are most critical to solve first.

L - List the Solutions: Brainstorm a wide range of potential solutions to the prioritized needs. Don't be limited by feasibility at this stage; think broadly. For example, for a dating app, a solution might be a new "ice-breaker" feature, a new way to match, or a group chat function.

E - Evaluate the Solutions: Analyze your list of solutions against key criteria. This could include technical feasibility, business impact, and user desirability.

S - Summarize the Recommendation: Conclude your answer by summarizing your recommendation and the rationale behind it. Briefly restate the problem, your recommended solution and why you believe it will succeed, showing a clear, logical progression from problem to solution.

Design a New Product - CIRCLES Framework

- Don't monologue for too long
- Be concise no need to explain all the technology deeply
- At the end give a success matrix and also implementation strategy
- Here you can take a break between sections to collect thoughts.
- Problem statement should remain your guiding light for any decision

Comprehend Situation

Narrow down the question as much as possible
Not all questions need be asked, use your best judgement

- Understand all the terms given in the question
- Know about the company, target and any other background.
- The objective, why us, why now
- Any limitations

Sumarise the question

Identify Customer

- Use Personas - cover both customer & consumer
- Define in terms of - Behavior, Demographics, Needs and Goals
- Can be role based
- Include underrepresented sectors, name your personas (not always male and pale)
- Can ask which persona to pick
- Can use a matrix to determine

Report Needs

- Need to find pain points
- Use customer journey
- Can use JTBD or Agile user stories - recommended only when pain points are too limited
- Also can do Gap analysis (what customers want - what is available) - done when a competitor already exists in the market
- Can ask which one to pick

Cut through Prioritization

- A simple matrix can be the size of the cohort (scope) and severity
- Or can use any own matrix
- Give no. or ordinal values
- Keep a focus on the problem statement

List Solutions

- Map solutions to pain points/user needs
- Use tech knowledge and pick from products around you
- Can even provide moonshot ideas

Evaluate Tradeoffs

- Impact, Confidence and Ease can be adopted
- A 2*2 matrix b/w value and effort
- RICE, Kano Model
- MoSCoW for multilevel model
- Can use the weighted average
- Keep a focus on the problem statement

- Before applying any matrix once confirm with interviewer

Summarize

- Give an MVP and an advanced version if asked
- A statement about how our product will help in achieving the (business) objective
- If the implementation timeline can use MoSCoW

Success Matrix

- Usually before Summary
- Use logic or AARRR or HEART framework

Additional

- Make it conversational - many times interviewer is looking for very narrow pain points or solutions - so just ask
- Don't just throw solutions out - know about the technology and at least 1 use case
- Use case solving learnings & keep in mind the business side of things

The HEART Framework is a user-centric method for measuring the quality of user experience on a product. Developed by Google, it helps product teams define and track metrics that directly relate to user goals and product success. Unlike traditional business metrics (like revenue or user count), HEART focuses on the actual user experience.

The Five Categories of Metrics

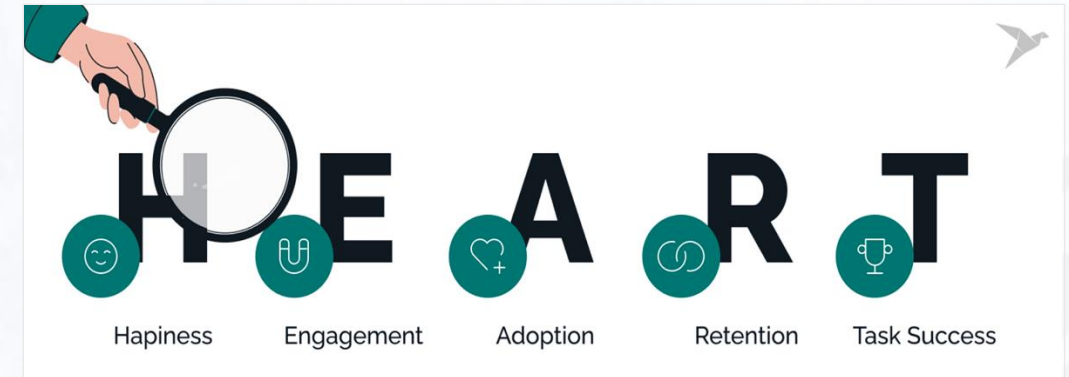
Each letter in the acronym **H.E.A.R.T.** represents a key category for measuring user experience.

H - Happiness: This category measures the users' attitudes and satisfaction. It's about how users feel about your product.

- **Examples:** User satisfaction (e.g., surveys), Net Promoter Score (NPS), and customer service ratings.

E - Engagement: This measures the user's level of involvement with your product. It's about how often and how deeply they interact with it.

- **Examples:** Number of sessions per user, session length, number of visits per day, week, or month and key feature usage.



A - Adoption: This measures how many new users are acquiring and starting to use a product or a specific feature.

- **Examples:** Number of new sign-ups, new users of a feature, or users who have completed the onboarding process.

R - Retention: This measures how many existing users are returning to the product over time. It's the inverse of churn.

- **Examples:** Daily/weekly/monthly active users, churn rate and resurrection rate (users who have returned after being inactive).

T - Task Success: This measures the efficiency and effectiveness of the user in completing a specific task.

- **Examples:** Time to complete a task, error rate, search success rate and conversion rate

Using the Framework

To apply the HEART framework, you don't just pick a metric for each category. Instead, you follow a process that links user goals to product metrics:

- **Start with a Goal:** Begin by defining a high-level goal for a specific feature or the entire product (e.g., "Help users find relevant content quickly").
- **Identify Signals:** Brainstorm the user actions that would indicate success for that goal. For the content example, a signal might be "Users are searching for and clicking on articles."
- **Create Metrics:** Finally, turn those signals into measurable metrics using the HEART categories. The signal above could lead to metrics like **Task Success** (e.g., search-to-click ratio) and **Engagement** (e.g., average articles read per session).



The AARRR Framework, also known as the Pirate Metrics, is a popular growth hacking model that helps product managers and marketers track the entire customer lifecycle. Created by Dave McClure, the framework provides a simple and effective way to measure the performance of a product and identify areas for growth. Each letter in the acronym represents a crucial stage in a user's journey.

The Five Stages of the Customer Lifecycle

Each letter in the acronym **A.A.R.R.R.** represents a key metric category.

A - Acquisition: This stage focuses on how users find your product. It's about where they come from and how they first hear about you.

- **Questions:** Where do users come from? What channels are most effective for acquiring new users?
- **Examples:** Website visitors, app installs, new sign-ups, and sources like search engines, social media, or referrals.
- **Key Metrics:** Cost per acquisition (CPA), visitor-to-sign-up conversion rate.

A - Activation: This stage measures whether users have a great first experience. It's not enough to acquire a user; you need to get them to perform a key action that signifies they've found value.

- **Questions:** What constitutes an "activated" user? What is the "aha!" moment for your product?
- **Examples:** A social media user sending their first message, a streaming user watching their first video, or an e-commerce user completing their first purchase.
- **Key Metrics:** Percentage of new users who complete a key activation event within the first day or week.

R - Retention: This stage measures how many users are returning to your product over time. A high retention rate is a strong indicator of product-market fit.

- **Questions:** Are users coming back? How often are they returning?
- **Examples:** Daily/weekly/monthly active users (DAU/WAU/MAU), user churn rate.
- **Key Metrics:** Cohort retention analysis, stickiness ratio (DAU/MAU).

R - Referral: This stage measures how users are spreading the word about your product. It's about turning satisfied

customers into your best marketing channel.

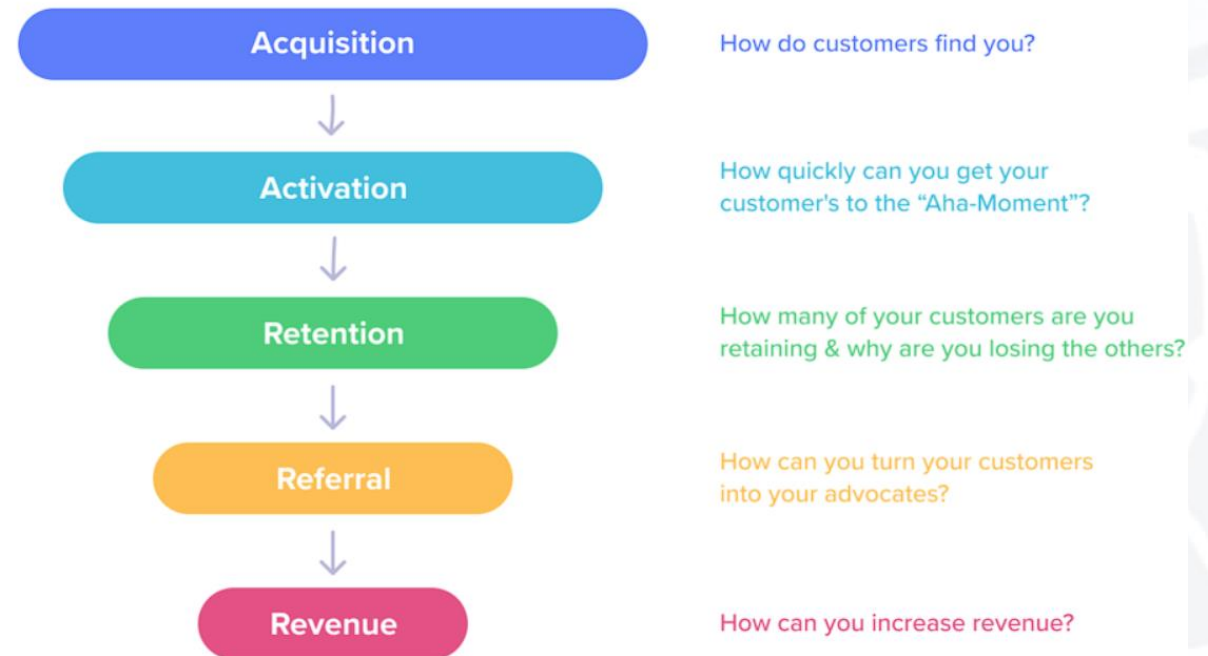
- **Questions:** Are users inviting others to use the product? Is there a built-in viral loop?
- **Examples:** Users sending invites, sharing content, or leaving positive reviews.
- **Key Metrics:** Viral coefficient, Net Promoter Score (NPS), and number of invites sent per user.

R - Revenue: This stage measures how your business makes money. It's the ultimate indicator of business viability.

- **Questions:** How are you monetizing your user base? What's the value of a user?
- **Examples:** One-time purchases, subscriptions, ad revenue.
- **Key Metrics:** Average revenue per user (ARPU), customer lifetime value (LTV), and churn rate (of paying customers).

Why It's a Framework, Not Just Metrics

The AARRR framework provides a narrative for a company's growth strategy. By focusing on each stage, a product manager can identify bottlenecks in the user journey. For example, a high acquisition rate but low activation rate suggests the onboarding process needs improvement, while a high activation rate but low retention rate points to a lack of long-term value. This structure allows teams to systematically address the most pressing issues for growth.



Solution prioritization is a systematic process used by product managers and teams to rank and select which features, ideas, or initiatives to work on. Its primary goal is to ensure that the team focuses its efforts on the solutions that will deliver the most value to the customer and the business, given limited time and resources. There are various frameworks to prioritise features.

RICE

The RICE framework prioritizes features by scoring them on four factors: **Reach** (how many people it will impact), **Impact** (how much it will move a key metric), **Confidence** (how certain you are of your estimates), and **Effort** (how much work is required). The formula is $(\text{Reach} \times \text{Impact} \times \text{Confidence}) / \text{Effort}$, which provides a single score to rank features objectively. This model helps minimize bias and is great for making informed decisions.

ICE

A simpler version of RICE, the ICE framework is ideal for making quick prioritization decisions, particularly in a fast-paced or startup environment. Features are scored based on **Impact** (how much it will affect a key metric), **Confidence** (how

certain you are it will work), and **Ease** (how easy it is to implement). The scores are multiplied together ($\text{Impact} \times \text{Confidence} \times \text{Ease}$) to get a total score, making it a "good enough" estimation tool.

Value vs. Effort Matrix

Also known as the Impact vs. Effort matrix, this is a visual framework that plots features on a two-by-two grid. The horizontal axis represents **Effort** (or cost), and the vertical axis represents **Value** (or impact). This results in four quadrants:

- **Quick Wins** (High Value, Low Effort): Prioritize these first.
- **Big Bets** (High Value, High Effort): These are strategic projects.
- **Fill-ins** (Low Value, Low Effort): Do these if you have extra time.
- **Time Sinks** (Low Value, High Effort): Avoid these. The matrix provides a straightforward way to visualize and discuss priorities with stakeholders.

Kano

The Kano model prioritizes features based on how they affect customer satisfaction. It classifies features into four categories:

- **Must-be:** These are basic, expected features. If you don't have them, users will be highly dissatisfied, but having them doesn't increase satisfaction.
- **Performance:** The more of this feature you have, the more satisfied customers are. For example, faster load times.
- **Attractive (Delighters):** These are unexpected features that, if present, delight customers but don't cause dissatisfaction if absent.
- **Indifferent:** These features don't affect user satisfaction either way. This model helps product managers focus on the features that provide the most delight for the investment.

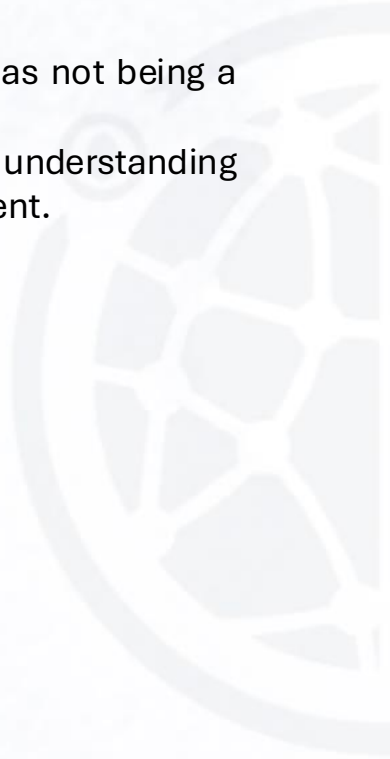
MoSCoW

The MoSCoW method is a simple prioritization technique that categorizes requirements into four levels of importance, helping to set expectations and manage scope with stakeholders.

- **Must-have:** These are non-negotiable, critical features for the product to function.

- **Should-have:** Important, but not essential. The product can launch without them, but they add significant value.
- **Could-have:** Nice-to-have features that can be included if time and resources permit.
- **Won't-have (this time):** Features agreed upon as not being a priority for the current release.

This framework is great for reaching a common understanding and managing scope, especially in agile development.



Cases



Case 1: Design an app that helps individuals cook their own meals

INTERVIEWER: Many individuals want to cook their own meals but often lack the time, knowledge, or inspiration to do so effectively. Design a food app that helps individuals cook their own meals

CANDIDATE: Just to clarify, we are designing an app for individuals who want to cook their meals on their own, correct?

INTERVIEWER: Yes, that's correct.

CANDIDATE: Alright, before we begin designing the app, I have a few clarifying questions. (ask them one by one to keep it organized, but mentioning them here itself)

- Could you please elaborate on what we mean by "individuals" in this context?
- On which platforms are we planning to launch this app?
- Can you provide some insights into our company's current activities? Have we previously launched a similar app?
- What are our primary targets and objectives for this project?

- What is our target launch date for this app?
- Are there any constraints we should be aware of, such as time, budget, or resource limitations?

INTERVIEWER: (Answered when asked)

- When we say "individuals," we mean anyone interested in cooking, without any specific segmentation.
- We plan to launch the app on both iOS and Android platforms.
- Our company specializes in software development, and while we have launched other apps, this will be our first venture into this particular category.
- Our primary goal is to acquire as many customers as possible and achieve profitability as quickly as possible.
- There is no specific deadline or time constraint for the launch.
- There are no constraints to be concerned about at the moment.

CANDIDATE: Okay, please give me a minute to think about this.

(After one minute.)

CANDIDATE: Okay, I would like to start first by discussing who the possible users of this product will be, then possible use

case scenarios, a few solutions and wrap up with my recommendation.

User Personas

CANDIDATE: Let's discuss the potential user base:

- Bachelor's and students living alone who prefer cooking their meals instead of ordering food online or eating out.
- Gym enthusiasts who aim to prepare meals aligning with recommended dietary intake and their personal taste preferences.
- Housewives with culinary skills, looking to experiment with new recipes for their families.
- Food enthusiasts and chefs with a passion for exploring diverse cuisines from around the world.
- Working professionals seek quick and easy-to-prepare meals to enjoy during their breaks.

Pain Points

CANDIDATE: In all these scenarios, it's clear that our product should cater to various user needs, encompassing a range of features and tasks such as:

- Lack of clear and organized recipe displays, making it difficult to follow.
- Insufficient nutritional information per serving, leaving users uncertain about the health aspect.

- Absence of demonstration videos, potentially causing confusion during meal preparation.
- Inability to estimate and filter recipes by cooking time, leading to time-consuming searches.
- Limited filters for dietary preferences, causing frustration for vegetarian or non-vegetarian users.
- Inability to customize taste preferences, resulting in dissatisfaction with meal options.
- Missing inbuilt timers or cook-along timers, leading to overcooking or undercooking.
- Unclear indications of cooking complexity, making it challenging for users to find suitable recipes.
- Lack of images showcasing the finished dish and presentation tutorials, affecting the overall cooking experience.

Prioritization

CANDIDATE: (We could use RICE matrix and other methods to prioritize and gauge product success, but in this case I am providing solutions to all the pain points)

Solutions

CANDIDATE: Here is a list of solutions:

- Providing a comprehensive recipe display, including ingredients and quantities.

- Offering nutritional information per serving of the dish.
- Featuring demonstration videos illustrating how to prepare the meal.
- Displaying an estimated cooking time and enabling users to filter recipes based on cooking time.
- Offering filters for vegetarian & non-vegetarian options, country of origin, and cuisine type.
- Allowing users to select taste preferences like sweet, spicy, and others.
- Incorporating an inbuilt timer or providing cook-along timers for each recipe.
- Indicating the level of cooking complexity or skill required, with filtering options for beginners, intermediate, and advanced users.
- Provide cooked dish images and how to present tutorials.
- Save dishes for future use, and option to review.
- Suggest similar dishes based on past likes and preferences

CANDIDATE: For an MVP, I would prioritize the ability to view recipe and quantity, veg and nonveg filter, level of difficulty and taste preference filter, and dish images.

INTERVIEWER: Can you suggest some KPI to measure the success of this app?

CANDIDATE: To gauge the success of our food app, we'll closely monitor several key performance indicators (KPIs). These include active user numbers, session durations, recipe engagement metrics like views and saves , and conversion rates if we offer premium features. User satisfaction will be assessed through Net Promoter Scores and user feedback. Additionally, we'll track community engagement & retention rates, ensuring a strong & supportive user base. Technical performance, market expansion, sustainability impact, and operational efficiency will also be measured to ensure our app's long-term success and its positive impact on users' cooking habits.

INTERVIEW:
That's great, thank you.

Sample Case Sheet

- **Objective:** to acquire as many customers as possible and achieve profitability as quickly as possible
- **Who are we:** specialize in software development and launched other apps, this will be first venture into this particular category
- **Individuals:** anyone interested in cooking, without any specific segmentation
- **Platform:** Both iOS and Android
- **Deadline:** No
- **Other Constraints:** No
- **Geography:** India

User Personas

- Bachelor's and students living alone who prefer cooking their meals instead of ordering food online or eating out.
- Gym enthusiasts who aim to prepare meals aligning with recommended dietary intake and their personal taste preferences
- Housewives with culinary skills, looking to experiment with new recipes for their families.
- Food enthusiasts and chefs with a passion for exploring diverse cuisines from around the world.

- Working professionals seek quick and easy-to-prepare meals to enjoy during their breaks.

Pain Points

1. Lack of clear and organized recipe displays
2. Insufficient nutritional information per serving, leaving users uncertain about the health aspect.
3. Absence of demonstration videos, potentially causing confusion during meal preparation.
4. Inability to estimate and filter recipes by cooking time, leading to time-consuming searches.
5. Limited filters for dietary preferences, causing frustration for vegetarian or non-vegetarian users.
6. Inability to customize taste preferences, resulting in dissatisfaction with meal options.
7. Missing inbuilt timers or cook-along timers, leading to overcooking or undercooking.
8. Unclear indications of cooking complexity, making it challenging for users to find suitable recipes.
9. Lack of images showcasing the finished dish and presentation tutorials,

Pain Point	1	2	3	4	5	6	7	8	9
Severity	4	3	3	4	4	3	4	3	2
Frequency	5	4	3	5	4	3	4	3	3

Solutions

1. Providing a comprehensive recipe display, including ingredients and quantities.
2. Offering nutritional information per serving of the dish.
3. Featuring demonstration videos illustrating how to prepare the meal.
4. Displaying an estimated cooking time and enabling users to filter recipes on it
5. Offering filters for vegetarian and non-vegetarian options and cuisine type.
6. Allowing users to select taste preferences like sweet, spicy, and others.
7. Incorporating an inbuilt timer or providing cook-along timers for each recipe.
8. Indicating the level of cooking complexity or skill required, with filtering options
9. Provide cooked dish images and how to present tutorials.
10. Save dishes for future use, and review.
11. Suggest similar dishes based on preferences

Solution	1	2	3	4	5	6	7	8	9	10	11
Impact	5	4	3	5	4	3	4	4	2	3	4
Effort	2	4	5	3	3	4	4	3	4	2	2

Case 2: Design a social media platform for children

CANDIDATE: I'll need to ask some clarifying questions first. First of all, what is the age-range of the children who are going to be the users?

INTERVIEWER: 9-13 years.

CANDIDATE: What kind of social media platform is it going to be educational or recreational?

INTERVIEWER: It's going to be recreational.

CANDIDATE: What is the objective of designing this app?

INTERVIEWER: The company wants to acquire new users for their business and also retain them.

CANDIDATE: Does the company have any other business?

INTERVIEWER: Yes, they actually develop video games as their primary business.

CANDIDATE: So can I assume that they want to acquire users for

their primary business as well?

INTERVIEWER: Yes.

CANDIDATE: Alright. Is there any specific location I should focus on?

INTERVIEWER: Consider globally.

CANDIDATE: Are there any monetary or time constraints?

INTERVIEWER: No constraints.

CANDIDATE: What platform should I focus on? iOS, Android? Since it is a gaming company should I also focus on any gaming consoles?

INTERVIEWER: Yes, consider all.

CANDIDATE: Alright, I am going to focus on what kind of users or stakeholders the app can expect.

INTERVIEWER: Go ahead.

CANDIDATE: The first and primary users would be the young

children, teens, and tweens who want to interact with their friends after school and play video games with them, either inperson or remotely. The second would be the parents who are concerned about their children's safety and well-being. They would want to monitor not just their online presence but also their daily life.

Third would be the gaming influencers who would want to share their gaming experience online and would want to target a specific age group and build their audience. Fourth would be the young content creators in the age range of 9-13 who would want to explore creative arenas and would want a comfortable and safe space to showcase that. These are the user personas that I have considered, should I explore more or should I go ahead with these?

INTERVIEWER: Prioritize the users from this list, and specify the reason why.

CANDIDATE: Alright, I would like to prioritize the young children, teens and tweens since they are going to be the primary users and are going to engage with the app the most. They are the primary stakeholders. Secondly I would prioritize the parents since they are going to monitor their children and have the power in the matter of their children using the app in the first place, so we'd need to consider them as well.

INTERVIEWER: What pain points do these user groups face?

CANDIDATE: First of all, the children don't have a lot of experience when it comes to browsing on the internet. They are also prone to being exposed to age-inappropriate content. Since they are unfamiliar with the internet, they might also face the problem of sharing their gaming moments with their friends or other people. They would also be unaware of the latest trends that are happening and are also followed by their peer group- so they'd want to fit in. They would also be unwilling to try new things since they might fear that their peer group might make fun of them. When it comes to parents, they would be concerned about how they are not as connected to their children since they are growing up and have different interests. Also, they don't want their children to indulge in play for long periods of time. Should I explore more pain points?

INTERVIEWER: That's enough. What pain points will you focus on?

CANDIDATE: I would like to prioritize them on the basis of impact and frequency. I would focus mainly on children having limited online experience and the possibility of them being exposed to age inappropriate content. Also, on children's

users and are going to engage with the app the most. They are the primary stakeholders. Secondly I would prioritize the parents since they are going to monitor their children and have the power in the matter of their children using the app in the first place so we'd need to consider them as well.

INTERVIEWER: What pain points do these user groups face?

CANDIDATE: First of all, the children don't have a lot of experience when it comes to browsing on the internet. They are also prone to being exposed to age-inappropriate content. Since they are unfamiliar with the internet, they might also face the problem of sharing their gaming moments with their friends or other people. They would also be unaware of the latest trends that are happening and are also followed by their peer group- so they'd want to fit in. They would also be unwilling to try new things since they might fear that their peer group might make fun of them. When it comes to parents, they would be concerned about how they are not as connected to their children since they are growing up and have different interests. Also, they don't want their children to indulge in play for long periods of time. Should I explore more pain points?

INTERVIEWER: That's enough. What pain points will you focus on?

CANDIDATE: I would like to prioritize them on the basis of impact and frequency. I would focus mainly on children having limited online experience and the possibility of them being exposed to age inappropriate content. Also, on children's concern on how to find the latest trends followed by their peer group. For the parents, their concern on how much time their children spend on play would constitute a bigger impact. I would also focus on how parents are concerned about the growing disconnect between them and their children.

INTERVIEWER: And what features would you implement to solve these pain points?

CANDIDATE: To solve the problem of children's limited online experience, guides and tutorials could be introduced to onboard them with a simple and user-friendly interface. There could be heavy content moderation to ensure that there is only age appropriate content. To allow parents to know how much their children are indulging in play, there could be tools to monitor the time and also to control the usage and put restrictions or timer. For children, they could have the option to directly share their gaming experience in the form of screenshots or videos to the app quickly and conveniently. To address parents' concerns over growing disconnect, there could be a split screen option to engage with the child and also to play together on games in the

app. There could also be an in-app gaming store with games sorted according to their popularity.

INTERVIEWER: What features would you finally implement?

CANDIDATE: To decide which features to implement, I would prioritize them on the basis of impact and ease of implementation. Guides and tutorials would be easy to implement and they would help the young users immensely in understanding how the app works. Heavy content moderation would be difficult to implement since it would need AI as well as human moderators but it will have a huge impact and will make the app a safer place for the users. Tools for monitoring and usage control would be easy to implement and will have a high impact since it will ensure parents confidence. Direct game to app sharing will be a bit difficult to implement since both platforms will need to be compatible and other factors will also come into play, such as device and internet connection. For the split screen solution, it will need to be compatible with the platform and might also have external factors to depend on. The in-app gaming store will be easy to implement since it can be linked to the game which can be hosted outside but it ranks low on the impact.

INTERVIEWER: How would you determine the success of these

features?

CANDIDATE: First of all, the DAU and MAU will let us know how many people are active on the app. The number of completed games on the app will also let us know if the people are spending time on both playing and sharing. To determine how safe the platform is we can look at the number of reports or accounts blocked. Also, NPS to see if people are liking it enough to recommend it to other users. And finally, reach and impressions to determine the amount of activity and engagement.

INTERVIEWER: That's great, thank you.

Case 3: Design a chair

INTERVIEWER: Your case is to design a chair.

CANDIDATE: Before I begin, I'd like to clarify a few things. Is there a specific context in which this chair will be used: like home, office, or outdoors? And are we only designing it for sitting, or can it have additional use cases?

INTERVIEWER: Let's assume it's for urban settings, home or compact apartments. The idea is that the chair should not just be for sitting; it can include features like posture support, storage, maybe even be foldable or serve other purposes.

CANDIDATE: Got it. So the key goal here is to design a space-efficient, multi-functional chair that enhances comfort and daily usability. I'd say the success criteria would be how comfortable and useful it is in tight spaces, customer satisfaction, and adoption in small urban homes. Is that correct?

INTERVIEWER: That works. How would you proceed?

CANDIDATE: I'd start by thinking about the kinds of users who would most benefit from this. Some personas that come to mind are:

- Remote workers spending hours at their desks
- Urban couples with limited furniture space
- Elderly people who need posture support
- Students in hostels or small PG rooms
- Freelancers or creators working from home

INTERVIEWER: Who among them would you prioritize?

CANDIDATE: Based on both frequency of chair use and size of segment, I'd go with:

- Remote workers as they use chairs the most
- Urban couples as they're short on space and prefer multi-purpose furniture
- Elderly users as they need ergonomic support

INTERVIEWER: Makes sense. What problems are you solving for them?

CANDIDATE: Here are the common pain points:

- Back and neck pain from poor posture
- Limited space and furniture clutter
- No flexibility — can't convert the chair into something else
- Lack of storage options
- Chairs not aesthetically aligned with modern home decor
- Top 3 pain points based on severity and prevalence would be:

- Poor posture support
- Lack of space-saving or multi-use functionality
- No in-built storage or adaptability

INTERVIEWER: What kind of solutions would you propose?

CANDIDATE: I’d propose the following:

- Ergonomically designed chair with adjustable lumbar and neck support
- Posture-correcting memory foam with embedded pressure sensors and slouch alerts
- Foldable chair that doubles up as a lounge or mini-table
- Convertible design that transforms into a step ladder or ironing surface
- Hidden storage compartment under the seat

INTERVIEWER: How would you prioritize what to build first?

CANDIDATE: I’d use the RICE framework:

Top 3 priorities are:

- Ergonomic lumbar support
- Under-seat storage
- Foldable lounge-table combo

Solution	Reach	Impact	Confidence	Effort	RICE Score
Ergonomic chair	9	9	8	4	162
Memory foam + slouch alert	6	7	6	5	50.4
Foldable lounge/table	8	8	7	5	89.6
Convertible ladder/ironing board	6	6	6	6	36
Storage under seat	7	7	8	3	130.6

INTERVIEWER: What would you measure to determine success?

CANDIDATE: North Star Metric: Average daily usage duration per chair

Supporting metrics:

- % reduction in self-reported back/neck pain
- CSAT scores from early adopters
- Repeat purchases or referrals

Case 4: Design an ATM for the Blind

INTERVIEWER: The case is to design an ATM for blind users.

CANDIDATE: Got it. I'd like to ask a few clarifying questions before I begin. Is this ATM designed exclusively for the blind or should it also be usable by the general public? Are there any constraints like infrastructure limitations or implementation timelines? And what would the success criteria look like?

INTERVIEWER: Good questions. The ATM should be inclusive but optimized for blind users. Think of an urban/semi-urban setting in a developing country. Cost-effectiveness, scalability, and enhancements without requiring full redesign accessibility are important. The goal is to allow visually impaired users to use ATMs independently and securely. Success criteria: increased accessibility, transaction security, and user satisfaction.

CANDIDATE: Understood. I'll begin by identifying the potential user personas who would benefit from accessibility improvements:

- Totally blind individuals
- Partially visually impaired individuals
- Elderly people with deteriorating eyesight
- Rural users with low tech literacy

- Bank staff assisting disabled users

Before prioritizing, I want to explain the rationale. Prioritizing user personas ensures we focus our design efforts on those facing the highest friction and with the most to gain. It increases both social impact and usability.

INTERVIEWER: Makes sense. Who are your top 3 personas?

CANDIDATE: Based on severity of challenges and how frequently they use ATMs, I'd go with:

- Totally blind users: they face the highest barrier to independence
- Elderly users: a large and growing demographic that often struggles with visibility
- Partially blind users

INTERVIEWER: Alright. Let's talk pain points.

CANDIDATE: Here are the major pain points:

- Inability to see or navigate the screen
- Dependence on others, risking PIN exposure and loss of privacy
- No audio instructions or inconsistent voice feedback
- Lack of tactile or Braille-based navigation
- Inconsistent ATM layouts across locations

- No confirmation of entered inputs (e.g., PIN, amount)
- Fear or lack of confidence in using ATMs independently
- No pre-transaction orientation or guidance

INTERVIEWER: Prioritize the top pain points.

CANDIDATE: Based on how severe and how commonly these issues arise, I'd prioritize:

Pain Points	Severity(1-5)	Frequency(1-5)	Total Score
Inability to see or navigate the screen	4	5	9
Dependence on others, leading to privacy/security risks	4	4	8
Lack of tactile/Braille-based navigation	3.5	4	7.5

- Inability to see or navigate the screen
- Dependence on others, leading to privacy/security risks
- Lack of tactile/Braille-based navigation

INTERVIEWER: Great. What solutions would you recommend?

CANDIDATE: I'd recommend the following:

- Voice-guided interface with multilingual audio support and head phone jack
- Tactile keypads with Braille labelling and embossed directional markers
- Screen-free PIN entry using tactile confirm buttons with audio confirmation
- Haptic or voice feedback after every user action (e.g., "₹500 entered")
- Standardized ATM layout across branches for muscle memory
- Pre-authentication via mobile app or SMS

INTERVIEWER: Prioritize the solutions.

CANDIDATE: Top 3 solutions would be:

- Braille keypad & tactile navigation
- Voice-guided interface
- Tactile confirmation with audio PIN feedback

Here's how I have evaluated the solutions:

Solution	Reach	Impact	Confidence	Effort	RICE Score
Braille keypad & tactile navigation	9	9	9	3	243
Voice-guided interface	9	9	8	4	162
Tactile confirm + audio PIN feedback	8	8	8	5	102.4
Haptic/audio feedback	7	7	7	4	85.75
Standardized layout	6	6	6	5	43.2
Mobile pre-authentication	5	6	6	6	30

INTERVIEWER: And how would you measure success?

CANDIDATE: North Star Metric: % of visually impaired users completing ATM transactions without assistance.

- Supporting metrics:
- Reduction in average transaction time
- CSAT/NPS from blind and elderly users
- Drop in PIN compromise or privacy-related complaints

INTERVIEWER: Nicely done.

Case 5: Design a Turf Booking App

INTERVIEWER: Your case is to design a turf booking app.

CANDIDATE: Alright! I'd like to clarify a few things first. Is this app meant for individuals or group users? Are we looking at city-based users only? And should the app focus only on booking, or should it also enable payments, team coordination, and ratings?

INTERVIEWER: It's meant for group users like friend circles or college teams in cities. Yes, the app should cover real-time availability, payments, group coordination, and even ratings. The goal is to make turf booking seamless. Success criteria: increase in bookings, repeat usage, and ease of coordination.

CANDIDATE: Understood. I'll start by outlining the potential user personas:

- Casual friend groups playing on weekends
- College/university sports teams
- Freelancers or community organizers hosting paid games
- Office teams scheduling weekly matches
- Turf owners managing availability

Before prioritizing, let me share the rationale. Prioritizing personas helps us ensure we are designing for the highest-need, most frequent users. These are the ones most likely to engage deeply and validate our product quickly.

INTERVIEWER: Makes sense. Who are your top 3?

CANDIDATE: I'd go with:

- Casual friend groups- largest volume, but coordination is tricky
- College sports teams - recurring and loyal users
- Turf owners - supply side is critical to ensure demand fulfilment

INTERVIEWER: Good. What are the major pain points?

CANDIDATE: Here's what I see across these personas:

- No real-time visibility of available slots
- Difficulty collecting or splitting payments from friends
- Last-minute cancellations or member no-shows
- Poor coordination among team members
- Lack of transparency on turf quality, amenities, and rules
- Turf owners struggle with double-bookings or idle slots

INTERVIEWER: Prioritize the top 3 pain points.

CANDIDATE: Based on both severity and how frequently users encounter them, I'd pick:

- No real-time visibility of available slots
- Payment and cost-splitting coordination issues
- Difficulty in team coordination and confirming attendance

INTERVIEWER: Go ahead with the solutions.

CANDIDATE: To solve these, here’s what I’d suggest:

- Live calendar view with color-coded slot availability
- Integrated split-payment system with UPI/wallet integration
- In-app RSVP and group coordination feature with notifications
- Turf profiles with images, rules, and user ratings
- Real-time cancellation waitlist and resell system
- Turf owner dashboard with auto-scheduling and analytics

INTERVIEWER: Prioritize your solutions.

CANDIDATE: Sure. Here’s how I’d map them:

Solution	Value (1–10)	Effort (1–10)	Priority
Live slot availability	9	3	High
Split payments	8	5	Medium
Group RSVP and notifications	7	4	Medium
Turf profiles (images/ratings)	6	3	Medium
Cancellation waitlist system	6	6	Low
Turf owner auto-scheduling	5	7	Low

Top 3 to build first would be:

- Live slot availability
- Split payments
- Group RSVP and reminders

INTERVIEWER: Good work. What metrics would you track?

CANDIDATE: North Star Metric: Number of completed turf bookings per month

Supporting metrics:

- Team return rate (repeat bookings)
- % of successful split payments
- Time to fill cancelled slots

Case 6: Design a product for Software Developers (Zapier-like Tool)

INTERVIEWER: Your case is to design a productivity tool for software developers, something similar to Zapier.

CANDIDATE: Interesting. Let me ask a few clarifying questions. Are we targeting individual developers or teams? Are we looking to automate only integration workflows or broader tasks like deployment, testing, etc.? And what's the success metric?

INTERVIEWER: Target is individual developers and small teams: freelancers, early-stage startups, junior devs. It should help automate repetitive developer tasks: integrations, testing, notifications. Success criteria are task efficiency, reduced, manual effort and developer satisfaction.

CANDIDATE: Got it. I'll begin by identifying user personas:

- Freelance developers juggling multiple client projects
- Junior developers building MVPs and side projects
- QA engineers/testers running repeated test suites
- Backend developers managing microservices
- DevOps engineers handling CI/CD pipelines

INTERVIEWER: Who are your top 3 user persona?

CANDIDATE:

- Freelancers: they deal with context switching and multiple tools
- Junior developers: need guidance and productivity boosts
- QA engineers: high repetition, benefit from workflow automation

INTERVIEWER: Go ahead.

CANDIDATE: Here are the common pain points:

Time wasted switching between platforms (GitHub, Slack, Jira, etc.)

- Manual repetition of common workflows like testing or deployment
- Lack of low-code/no-code automation tools for non-senior devs
- Poor notification and alert systems for task status
- Setting up integrations and pipelines takes too long
- Errors due to missing automated triggers or delays

INTERVIEWER: Pick the top pain points.

CANDIDATE: Based on frequency and friction, the top pain points are:

- Manual repetition of workflows
- Difficulty setting up integrations
- Lack of smart notifications/alerts

INTERVIEWER: What would you build to solve these?

CANDIDATE: Here are my proposed solutions:

- Visual, low-code workflow builder for automating common dev tasks
- One-click plug-and-play integrations with GitHub, Slack, Jira, Jenkins, etc.
- Trigger-based automation for builds, tests, or notifications
- Unified notification center with customizable rules and routing
- Template library of popular workflows (e.g., deploy & notify)

INTERVIEWER: Prioritize the solutions.

CANDIDATE: Sure.

Solution	Reach	Impact	Confidence	Effort	RICE Score
Visual workflow builder	9	9	8	5	129.6
Plug-and-play integrations	9	8	9	4	162
Trigger-based automation	8	9	8	4	144
Notification center	7	7	8	4	98
Template library	6	6	8	3	96

Thus the top solutions to build would be:

- Plug-and-play integrations
- Trigger-based automation
- Visual workflow builder

INTERVIEWER: And what would you track?

CANDIDATE: North Star Metric: Number of automated workflows executed per user per week

Supporting metrics: Integration success rate, Time saved per developer, retention rate per user.

Case 7: Design an Airport

INTERVIEWER: The case is to design an airport.

CANDIDATE: That's definitely a challenging one! Before diving in, a few clarifying questions. Are we designing a new airport from scratch, or improving an existing one? Are we focusing on passengers, airlines, or staff? Any geographic context or traffic volume assumptions?

INTERVIEWER: Good clarifications. Let's assume we're designing a new airport in a Tier-2 city with rising passenger traffic. Focus is on passenger experience, inclusion and operational efficiency. Success criteria would include reduced wait times, improved passenger satisfaction, and efficient throughput.

CANDIDATE: Got it. I'll begin by identifying the potential user personas:

- Domestic leisure travelers
- Frequent business travelers
- Elderly passengers or passengers with reduced mobility
- Families with kids
- Airline and airport ground staff

Airports cater to thousands of people daily, so prioritizing the most common and vulnerable user types

helps design an inclusive, high-utility experience that balances comfort with scale.

INTERVIEWER: Makes sense. Go with your top 3 user persona?

CANDIDATE:

- Domestic leisure travelers: the majority in Tier-2 cities
- Elderly and mobility-constrained passengers: need accessibility support
- Ground staff: enabling their efficiency improves overall passenger flow

INTERVIEWER: Go ahead.

CANDIDATE: Here are the pain points these personas typically face:

- Long security check and immigration queues
- Poor signage and confusing navigation inside the terminal
- Inadequate assistance for elderly and differently-abled
- Delays in baggage claim
- Lack of real-time information on boarding gates and flight changes
- Limited food and seating availability in peak hours
- Ground staff overwhelmed by passenger queries and lost items

INTERVIEWER: Prioritize the top 3 pain points.

CANDIDATE: Based on how disruptive they are and how frequently they occur:

- Long queues at security and immigration
- Poor signage and terminal navigation
- Inadequate support for elderly/mobility-constrained passengers

INTERVIEWER: What are your proposed solutions?

CANDIDATE: Here are some ideas to address those issues:

- Smart queue system with digital tokens and live wait-time updates
- Indoor wayfinding app with turn-by-turn navigation and multilingual voice guidance
- Dedicated fast-track lanes and assistance kiosks for elderly and special-needs passengers
- Real-time flight status boards at key decision points (cafes, restrooms, etc.)
- Automated baggage carts and self check-in counters to reduce reliance on staff
- Unified helpdesk chatbot accessible via airport Wi-Fi

INTERVIEWER: How would you prioritise your solutions?

CANDIDATE: I'd prioritize them using value vs. effort

Solution	Value (1–10)	Effort (1–10)	Priority
Smart queue + wait-time updates	9	4	High
Indoor wayfinding app	8	6	Medium
Fast-track lanes & assistance kiosks	7	5	Medium
Real-time flight info boards	7	3	High
Automated baggage carts	6	8	Low
Unified chatbot helpdesk	6	4	Medium

Top 3 solutions based on value vs. effort:

- Smart queue with wait-time updates
- Real-time flight info boards
- Assistance kiosks and fast-track lanes

INTERVIEWER: And what would your success metrics be?

CANDIDATE: North Star Metric: Average passenger transit time from check-in to gate

Supporting metrics:

- Queue length and average wait time
- CSAT and elderly passenger satisfaction score
- App adoption rate for indoor navigation

INTERVIEWER: Sounds good!

Case 8: Design a product for homestay owners

INTERVIEWER: The case is to design a product for homestay owners.

CANDIDATE: I'd like to ask a few clarifying questions. Are we designing a software product or a physical solution? Is the focus on homestay owners in urban tourist areas or rural destinations? And are we solving for guest experience, host management, or profitability?

INTERVIEWER: Let's go with a digital solution: primarily software. Target is small-to-mid-size homestay owners in tourist regions. The focus is on improving operations, guest experience, and helping them increase revenue. Success criteria: host satisfaction, operational ease, and guest retention.

CANDIDATE: Understood. Let me list out some potential user personas:

- Individual homestay owners managing 1–2 properties
- Rural families listing a spare room seasonally
- Boutique property managers running 5–10 properties
- Hosts who rely on OTAs (like Airbnb/Oyo) for bookings
- Digitally illiterate hosts managing manually or offline

Before prioritizing personas, here's my rationale. It's important to focus on those who are most underserved and yet have the most to gain from a structured, scalable solution: typically those managing 1–2 properties who lack staff or automation.

INTERVIEWER: Makes sense. Who are your top 3?

CANDIDATE:

- Individual owners with 1–2 listings — they struggle the most with time and tools
- Hosts reliant on OTAs — they need visibility and revenue control
- Rural hosts — often lack tech awareness but benefit most from digitization

INTERVIEWER: Alright. Go ahead.

CANDIDATE: Based on conversations and user behaviour, here are some common pain points:

- Difficulty managing bookings from multiple platforms (Airbnb, Booking.com, etc.)
- Poor communication and coordination with guests
- Manual tracking of payments, expenses, and taxes
- Lack of visibility into guest preferences or repeat visits
- Struggles with dynamic pricing & low off-season occupancy

- No simple dashboard to monitor reviews, ratings and feedback.

INTERVIEWER: Prioritize the top 3 pain points.

CANDIDATE: Based on how often they occur and their business impact:

- Managing bookings across multiple platforms
- Handling payments and finances manually
- Inconsistent or delayed communication with guests

INTERVIEWER: What would you build?

CANDIDATE: Here are my suggested solutions:

- Unified calendar and channel manager for syncing all booking platforms
- Integrated ledger for automated revenue, payout, and tax tracking
- Automated guest messaging system with templates and scheduling
- Dashboard with review analytics and guest feedback tracking
- Basic CRM for returning guest data and preferences
- Off-season occupancy booster with local partnership recommendations

INTERVIEWER: How would you prioritise your solutions?

CANDIDATE: Here's how I'd rate them:

Solution	Reach	Impact	Confidence	Effort	RICE Score
Channel manager for multi-platform syncing	9	9	9	4	182.25
Integrated ledger for revenue and tax	8	8	8	5	102.4
Automated guest messaging	9	7	8	3	168
Review dashboard	7	6	8	4	84
Guest CRM	6	5	7	4	52.5
Off-season occupancy tool	5	6	6	5	36

Hence, the order of priority would be:

1. Channel manager for platform syncing
2. Automated guest messaging
3. Integrated ledger for revenue/tax management
4. Review dashboard
5. Guest CRM
6. Off-season occupancy tool

INTERVIEWER: Nicely done. What metrics would you track?

CANDIDATE: North Star Metric: Monthly booking volume per property

Supporting metrics:

- % reduction in double bookings
- Guest response time
- Repeat booking rate
- Host NPS

INTERVIEWER: Excellent.

Case 9: Design a Mode of Transport for Long Distance (Human Cannon Concept)

INTERVIEWER: Your case is to design a new mode of transport for long-distance travel, something innovative and out of the box. For example, imagine a cannon that shoots people to their destination.

CANDIDATE: That's a fun and futuristic problem! Let me begin with a few clarifying questions. Are we aiming for mass adoption or niche use? Should this be an alternative to flights, or something supplementary? What constraints do we need to consider: safety, cost, environmental impact?

INTERVIEWER: Good questions. Let's assume it's a high-speed, next-gen alternative to air travel that can potentially become mainstream. Safety, time efficiency, and sustainability are important. Success criteria: adoption rate, cost-efficiency, travel time reduction, and perceived safety.

CANDIDATE: Got it. I'll begin by identifying user personas who might use such a mode of travel:

- Business travelers looking to save time
- Budget-conscious travelers looking for cheap but fast travel

- Environmentally conscious travelers seeking sustainable transport
- Adventure seekers open to unconventional experiences
- Emergency service providers (e.g., organ transport, disaster response)

Before I prioritize personas, I'd like to explain the rationale. Prioritizing helps us focus on designing for high-frequency users and early adopters, which is critical for validating viability and scaling new tech like this.

INTERVIEWER: That makes sense. Who are your top 3?

CANDIDATE:

- Business travelers: value time more than comfort, likely early adopters
- Budget travelers: if pricing is right, they'd switch quickly
- Adventure seekers: more open to testing unusual transport methods

INTERVIEWER: Let's move to pain points.

CANDIDATE: Sure. Here are some potential pain points across those personas:

- Safety concerns: fear of malfunction or injury
- High G-forces and physical discomfort

- Lack of control or ability to change route/make stops
- Anxiety due to lack of familiarity with the technology
- Absence of supporting infrastructure (landing pods, terminals)
- Legal or regulatory challenges in allowing such travel
- Accessibility concerns for elderly or physically vulnerable users

INTERVIEWER: Which ones would you prioritize?

CANDIDATE: Based on severity and prevalence among our top personas, I’d prioritize:

- Safety concerns: absolutely non-negotiable for adoption
- Physical discomfort during transit
- Anxiety or trust gap with the new mode of travel

INTERVIEWER: What solutions would you suggest?

CANDIDATE: I’d recommend the following:

- Shock-absorbent, enclosed personal pods with biometric locking and crash-proof material
- G-force dampening systems to reduce physical strain inside pods
- Onboarding simulation zones to familiarize passengers with the launch experience

- Emergency landing parachute system in each pod
- Pre-screening system for users based on age and health metrics
- Real-time pod tracking for users and their families
- AI-assisted routing to adjust trajectory dynamically if needed

INTERVIEWER: Go ahead. Now prioritize your solutions.

CANDIDATE: Here’s how I’d evaluate:

Solution	Reach	Impact	Confidence	Effort	RICE Score
Crash-proof personal pods	9	10	8	6	120
G-force dampening system	8	9	7	5	100.8

Simulation onboarding experience	7	7	9	3	147
Emergency parachute per pod	7	8	8	6	74.7
Health-based eligibility screening	6	7	7	4	73.5
Real-time pod tracking	6	6	9	3	108
AI-assisted dynamic routing	5	8	6	6	48

The order of priority would be:

- Simulation onboarding experience
- Real-time pod tracking
- Crash-proof personal pods

- Real-time pod tracking
- G-force dampening system
- Emergency parachute per pod
- Health-based eligibility screening
- AI assisted dynamic routing

INTERVIEWER: And how would you measure success?

CANDIDATE: North Star Metric: Successful long-distance trips completed per month

Supporting metrics:

- Passenger NPS after each journey
- Number of incidents per million miles travelled
- Repeat usage rate
- Time savings vs. flight/train travel

INTERVIEWER: Very creative and structured approach. Well done!

Case 10: Design an App for FMS MBA 1st Year Students

INTERVIEWER: Your case is to design an app specifically for FMS MBA 1st year students.

CANDIDATE: Great! A few clarifying questions before I begin. Should this be a general-purpose campus utility app or something focused on academics, placements, or networking? Are we designing for Android/iOS or web as well?

INTERVIEWER: Let's assume it's a mobile-first app that supports academics, preparation, scheduling, networking, and peer-to-peer learning. It should enhance productivity, streamline information, and reduce chaos. Success criteria: adoption rate, engagement, and student satisfaction.

CANDIDATE: Got it. Here are the potential user personas within the 1st-year batch:

- Academic-focused students
- Placement-prep focused students
- Committee/society members juggling multiple events
- Students coming from non-business backgrounds needing catch-up resources
- Exchange aspirants who need centralized info and deadlines

Before prioritizing, the rationale is that identifying personas allows us to tailor the MVP toward the most frequent, high-impact use cases. The value delivered to these users will influence word of mouth and overall app adoption.

INTERVIEWER: So who would you prioritize?

CANDIDATE:

- Placement-focused students - they engage heavily with prep material and calendars
- Committee/society members - they're always coordinating and need tools
- Non-business students - need for academic support

INTERVIEWER: Alright. What pain points do these users face?

CANDIDATE: These are the common pain points I'd expect:

- Scattered communication across WhatsApp, email, Google Classroom, etc.
- No centralized calendar for academic and committee deadlines
- No peer-sharing platform for notes, resources, or prep materials
- Repetition of queries regarding SOPs, shortlists, deadlines, etc.

- Difficulty accessing past year interview experiences or case prep resources

INTERVIEWER: How would you prioritise the top pain points?

CANDIDATE: Based on prevalence and disruption:

- Scattered communication and too many channels
- Lack of centralized calendar
- No peer-to-peer sharing of prep resources

INTERVIEWER: What would your solutions be?

CANDIDATE: I'd suggest the following:

- Centralized dashboard that aggregates deadlines, announcements and class schedules
- Smart calendar integration with push notifications for classes, committee events, and SIP deadlines
- Resource hub with peer-uploaded casebooks, notes, and prep materials
- Discussion board for SOP/case queries to reduce repetitive questions
- Searchable archive of past interview experiences
- Anonymous feedback system to raise academic or scheduling concerns

INTERVIEWER: Prioritize your solutions.

CANDIDATE: Sure. Here's my assessment based on value added and effort required to build:

Solution	Value (1–10)	Effort (1–10)	Priority
Centralized dashboard	9	5	High
Smart calendar with notifications	8	4	High
Peer-to-peer resource hub	8	6	Medium
SOP/case Q&A board	6	4	Medium
Past interview archive	7	5	Medium
Anonymous feedback feature	5	3	Low

INTERVIEWER: Looks good. What metrics would you use?

CANDIDATE: North Star Metric: Weekly Active Users (WAU)
Supporting metrics: Task interaction rate , Average Session Time

Case 1: How would you improve Paytm?

CANDIDATE: To understand the context better, could you clarify what specific improvement aspect we are targeting? Are we looking to boost user acquisition, engagement, revenue, user retention, experience, or something else?

INTERVIEWER: We are looking to improve the app in general, so you can choose any aspect you believe would be most impactful.

CANDIDATE: I suggest we focus on improving user experience, as it directly impacts user satisfaction, and brand loyalty and gives a competitive advantage as well. Since Paytm already has a substantial user base, improving the experience would likely lead to enhanced user retention.

INTERVIEWER: That sounds reasonable.

CANDIDATE: Before we proceed - I would like to ask some questions to gain better clarity on the problem statement.

- Since we have decided to focus on improving user experience, what is the measure of user experience/satisfaction?

- Are we focusing on Paytm for Consumers or Paytm for Business?
- Are we focusing on a specific offering like Payments or Paytm Money or Paytm Shopping, or all of them?
- Are we focusing on any particular geographical area?
- Is the improvement for both Android and iOS apps?
- Are there any budgetary, time or technological constraints?

INTERVIEWER:

- Customer ratings are the qualitative measure of user experience that we are considering right now.
- Focus on whichever you prefer.
- Focus on whichever you prefer.
- Let's focus on India for now.
- Yes, both Android and iOS.
- Zero constraints

CANDIDATE: The user base for Paytm could be divided as follows:

- Consumers: Use Paytm for everyday transactions, bill payments, and online purchases.
- Merchants: Use Paytm to accept payments from customers and manage their finances.

- Investors: Use Paytm to invest in mutual funds, stocks, and other financial products.
- Service Providers: Entities using Paytm to receive payments for services like travel services or insurance.

The consumer segment is likely the largest user base of Paytm, who primarily use the platform for making payments.

Hence, I would like to focus on **Payments** offering in the **Paytm for Consumers** app. Improvements made for consumers will impact a broader audience, thus maximizing the benefits of any enhancements.

Does this sound good?

INTERVIEWER: Yes, that makes sense. How do you plan to proceed?

CANDIDATE: Here's how I would approach this challenge:

- Understanding Consumer Segments
- Identifying Pain Points
- Proposing and Prioritizing Solutions

INTERVIEWER: That sounds like a plan. Let's start.

CANDIDATE: Here are some of the consumer user personas:

- Young college students who live away from home, in shared apartments, and use Paytm for daily payments (rent, food,

- commute, leisure, etc.)
- Elderly people, who use Paytm to pay their utility bills, are not very tech-savvy, and can use any feature with trust only when taught.
- A farmer in a rural area, uses Paytm to purchase farming supplies online from distant markets. Occasionally receives payments for his produce via Paytm.
- A Freelance content creator, who travels frequently, uses Paytm for doing his own transactions and to handle domestic and international payments from clients.

INTERVIEWER: Sounds good, please go ahead.

CANDIDATE: These users could have the following pain points:

- Frequent disruptions during transactions in rural areas or areas with poor internet infrastructure, leading to transaction failures or multiple deductions.
- Difficulty in monitoring online expenditures due to unclear categorization/labeling of expenses leading to budget mismanagement.
- Because of several offerings in one app, less tech-savvy users find it difficult to navigate the app, especially less frequently used services.
- Difficulty in staying updated on Banking and Regulatory Information/Actions that are relevant to users.

- Predefined templates and one-sided conversation in chat-support don't give much control to users who want to seek help.
- Too many features and ads together lead to slower response time even in primary features.

INTERVIEWER: These pain points seem relevant. Which ones would you prioritize solving and why?

CANDIDATE: The problems can be prioritized based on the impact that they would have on consumers :

- Internet connectivity failure while making payments - Impact: High
- Difficulty in managing personal finances and tracking spending - Impact: Medium
- Difficulty in navigation - Impact: Low
- Difficulty in staying updated on compliance and regulations - Impact: Medium
- A two-way communication missing in Chat Support - Impact: Low
- Slow response time due to too many features - Impact: Low

Since pain points 1, 2, and 4 have high and medium impacts, we can prioritize these for improvement.

Here are my proposed solutions:

- USSD Integration for Payments: A feature that allows users to make payments using USSD codes, which do not require internet connectivity. This ensures that users can complete transactions even in areas with poor internet coverage.
- Spend Analytics: A comprehensive tool that would offer detailed spend analysis, categorize expenses, and provide insights into spending patterns.
- Compliance Page: A dedicated section within the app that provides users with comprehensive information on compliance and regulations related to banking and other services. This would include related content in the form of videos, blogs, news articles, or infographics.

INTERVIEWER: Which one would you implement first and why?

CANDIDATE: The solutions can be judged based on the potential impact and the reach they would have:

- USSD Integration for Payments - Impact: High, Reach: High
- Spend Analytics - Impact: Medium, Reach: High
- Compliance and Regulation Hub - Impact: Medium, Reach: Medium

Based on this, the USSD Integration for Payments offers high impact and high reach and hence should be prioritized.

INTERVIEWER: How would you measure the success of this feature?

CANDIDATE: To measure the success of the USSD Integration we can use the following metrics:

1. Adoption Rate:

- Number of Transactions via USSD
- Percentage of Users Adopting USSD

2. Transaction Success Rate:

- Reduction in Transaction Failures due to internet connectivity issues
- Success Rate of USSD Transactions

3. User Satisfaction:

- CSAT (Customer Satisfaction) score
- Net Promoter Score (NPS)

INTERVIEWER: Those metrics seem comprehensive. Thank you. We can end the case.

Sample Case Sheet

- **Objective:** Improve user experience
- **Measure of UX:** Customer Ratings
- **Paytm for consumers/business:** Consumers
- **Payments/Paytm Money/paytm Shopping:** Payments
- **Geography:** Pan-India
- **Platform:** Both Android/iOS
- **Constraints:** None

User Personas

- Young college students who live away from home, in shared apartments, and use Paytm for daily payments (rent, food, commute, leisure, etc.)
- Elderly people, use Paytm to pay their utility bills, are not very tech-savvy, and can use any feature with trust only when taught.
- A farmer in a rural area, uses Paytm to purchase farming supplies online from distant markets. Occasionally receives payments for his produce via Paytm.
- A Freelance content creator, who travels frequently, uses Paytm for doing his own transactions and to handle domestic and international payments from clients.

How would you improve PayTM?

Pain points

1. Frequent disruptions during transactions in rural areas or areas with poor internet infrastructure, leading to transaction failures or multiple deductions.
2. Difficulty in monitoring online expenditures due to unclear categorization/labeling of expenses leading to budget mismanagement.
3. Because of several offerings in one app, less tech-savvy users find it difficult to navigate the app, especially less frequently used services.
4. Difficulty in staying updated on Banking and Regulatory Information/Actions that are relevant to users.
5. Predefined templates and one-sided conversation in chat-support don't give much control to users who want to seek help.
6. Too many features and ads together lead to slower response time even in primary features.

Pain Point	1	2	3	4	5	6
Severity	H	M	L	M	L	L

Solutions

1. *USSD Integration:* allows users to make payments using USSD codes, which do not require internet connectivity.
2. *Spend Analytics:* Would offer detailed spend analysis, categorize expenses, and provide insights into spending patterns.
3. *Compliance Page:* A dedicated section that provides users with information on compliance and regulations related to banking and other services.

Solution	1	2	3
Impact	High	Medium	Medium
Reach	High	High	Medium

Metrics

Adoption Rate:

- Number of Transactions via USSD
- 2. Transaction Success Rate:
 - Success Rate of USSD Transactions
- 3. User Satisfaction:
 - CSAT score; NPS

Case 2: Improve YouTube

CANDIDATE: To start, there are two broad categories of users on YouTube: content creators and content consumers. Which category would you like me to focus on?

INTERVIEWER: Let's focus on content creators.

CANDIDATE: Understood. Creators can be categorized by their content type: such as vloggers, educators, entertainers, etc. Several common overarching pain points affect their content creation and audience engagement.

INTERVIEWER: Could you detail these common pain points?

CANDIDATE: Absolutely. Key pain points include:

1. Creators often spend a substantial amount of time and effort creating YouTube shorts that relate to a longer video they've already produced.
2. Creators frequently miss out on opportunities to collaborate with peers creating similar content due to a lack of visibility or networking tools.
3. Creators struggle to understand what their audience wants, which, if known, could guide their content strategy.

4. Many creators face challenges in finding relevant help for research, graphic design, editing, etc., which limits their production capacity and quality.

INTERVIEWER: These challenges seem critical. How would you prioritize addressing them?

CANDIDATE: Prioritization should focus on impact:

1. Time-Consuming Production: Low Impact.
 2. Lack of networking tools for collaboration: Medium impact, influences growth and community building.
 3. Insufficient Insights on Audience Interests: Medium impact, crucial for budding creators, directly relates to content relevance and viewer engagement.
 4. Difficulty in Finding Production Assistance: High impact, crucial for budding creators, affects content quality.
- Therefore, we should prioritize pain points 2, 3, and 4, as they are high and medium impacts.

INTERVIEWER: Based on these priorities, what solutions do you propose?

CANDIDATE: To address these effectively, I suggest:

- Creator Collab Platform: A feature to connect creators with similar content interests for potential collaborations,

enhancing community engagement and content diversity.

- **Trending Search Tool:** This tool would provide real-time data on trending searches, particularly of their subscribers, allowing creators to better understand viewer interests and tailor their content accordingly.
- **Creator Support Network:** Establishing a network where creators can connect with professionals like researchers, graphic designers, and editors who can assist in the video production process.

INTERVIEWER: Which solution would you implement first?

CANDIDATE: The solutions can be judged based on the potential impact and the reach they would have:

1. Creator Collab Platform: Impact – High, Reach – Medium.
2. Trending Search Tool: Impact - Medium, Reach – High.
3. Creator Support Network: Impact – High, Reach - High.

Hence, I would start with the Creator Support Network feature, as this offers both high reach and high impact.

INTERVIEWER: How would you measure success?

CANDIDATE: Success metrics would include:

1. Adoption Rate: Tracking how many creators use the Creator Support Network.

Calculated as:

$$\left(\frac{\text{Number of Creators Using the Support Network}}{\text{Total Number of Creators on the Platform}} \right) * 100$$

2. **Creator Feedback:** Conducting surveys to assess satisfaction and gather qualitative feedback.

3. **Enhancement in Content Quality:** Assessment through qualitative reviews and ratings by viewers.

INTERVIEWER: That sounds comprehensive. We can conclude the case here. Thank you for your insights.

Case 3: Improve the Operational Efficiency of an Airport

INTERVIEWER: Your case is to improve airport flow. Think from an operational lens.

Candidate: Got it. A few clarifying questions: Are we looking to improve passenger flow, baggage handling, or staff operations? Are we focusing on a large international airport or a regional domestic one?

INTERVIEWER: Think large domestic airport with growing traffic. Focus on optimizing passenger flow from entry to gate. Success criteria include faster transit, fewer bottlenecks and better user experience.

CANDIDATE: Understood. I'll begin by identifying the different types of users who interact with airport operations.

- First-time fliers or infrequent travelers
- Frequent business travelers
- Elderly passengers or those with mobility needs
- Airport staff (security, cleaners, baggage handlers)
- Families with kids

Before prioritizing, I will explain the rationale: selecting the right personas helps us optimize operations around those most likely

to cause or experience congestion, which impacts overall flow.

INTERVIEWER: Fair. Who are your top three user persona?

CANDIDATE:

- First-time travelers: often slowest through process, create delays
- Elderly passengers: need additional support, may slow lanes
- Frequent travelers: expect efficiency, demand reliability

INTERVIEWER: Walk me through the user journey.

CANDIDATE: Sure. A passenger's typical journey looks like this:

- Entry & Security Checkpoint Queue
- Check-in Counters/Baggage Drop
- Security Check
- Terminal Navigation to Boarding Gate
- Waiting Area Congestion
- Boarding

At every step, flow can be interrupted due to process inefficiencies, lack of information or layout issues.

INTERVIEWER: What are the pain points?

CANDIDATE: Here's what typically slows things down:

•

- Long queues at security and check-in
- Poor signage or confusing terminal layout
- Manual ID checks causing bottlenecks
- Repetitive verification across multiple touchpoints
- Inadequate staff at crowd-prone areas
- Waiting area congestion near gates
- Delays in boarding coordination

INTERVIEWER: Pick the top pain points.

CANDIDATE: Based on severity and prevalence of the issue, the top pain points would be:

- Security queue delays
- Confusing terminal layout and signage
- Manual verification redundancies

INTERVIEWER: What improvements do you suggest?

CANDIDATE: I'd recommend:

- Pre-security slot-booking system for crowd control
- Dynamic digital signage with gate directions, queue wait times
- Facial recognition kiosks for automated ID and boarding pass checks
- Real-time crowd heatmaps to alert staff on bottlenecks
- QR-based boarding to skip repetitive ID scans
- Staff reallocation dashboard powered by footfall analytics

INTERVIEWER: How would you prioritise the solutions?

Solution	Value (1–10)	Effort (1–10)	Priority
Pre-security slot booking	9	4	High
Dynamic signage & wait-time display	8	4	High
Facial recognition kiosks	9	8	Medium
Crowd heatmaps & alert system	7	6	Medium
QR-based ID verification	6	5	Medium
Staff reallocation dashboard	5	7	Low

- Top 3 priorities would be:
- Slot-booking at security
 - Digital signage with wait times
 - Facial recognition for ID/boarding pass

INTERVIEWER: What are some relevant metrics?

CANDIDATE:

Primary metric: Average passenger transit time from entry to gate

Supporting metrics:

- Queue abandonment rate
- Number of on-time boardings
- Passenger NPS

INTERVIEWER: Good! Strong operational thinking.

Case 4: Improve WhatsApp

INTERVIEWER: Your case is to improve WhatsApp.

CANDIDATE: Sure! Before jumping in, I'd like to ask a few clarifying questions. Are we focusing on the core WhatsApp app or WhatsApp Business? And should I prioritize engagement, retention, monetization or something else?

INTERVIEWER: Good questions. Focus on the core WhatsApp app, especially in the Indian context. Try to improve user experience while staying aligned with Meta's mission to enable private, simple, and reliable communication. Success will be measured by retention, usage, and user satisfaction.

CANDIDATE: Got it. I'll begin by identifying user personas. This will help in targeting the improvements meaningfully.

- Family group users: share a lot of media, rely on the app daily
- College students: use it intensively for coordination, quick chats
- Professionals: often rely on WhatsApp for work updates
- Small business owners: use the app as an informal storefront
- Older adults: prefer simple, reliable communication

INTERVIEWER: Great. Who would you prioritize?

CANDIDATE: Prioritizing the right personas helps us focus our efforts on users who interact with the product the most or experience the most friction. Based on that, I'd go with:

- Family group users - they're the largest and most media-heavy segment
- College students - they engage frequently and drive stickiness
- Small business users - they often operate on the boundary of the core and business app

INTERVIEWER: Alright. Let's focus on college students. Walk me through their user journey.

CANDIDATE: Sure. Let's take a college student:

- Opens WhatsApp
- Checks personal chats and multiple group chats
- Sends messages, memes, files, links
- Gets overwhelmed by unread messages or noisy groups
- Occasionally uses voice notes or statuses
- At times, struggles to find old messages or clear storage

INTERVIEWER: What pain points do you see here?

CANDIDATE: Here are the main ones:

- Cluttered storage - memes, videos, and duplicate files pileup
- Poor search/navigation - can't easily find past messages or media
- Group chaos - hard to keep up with relevant messages in large groups
- Overwhelming notifications - especially from low-importance groups
- No way to sort or prioritize chats

INTERVIEWER: Pick the top three.

CANDIDATE: Based on severity and prevalence, I'd choose:

- Media clutter and storage overload
- Ineffective message search/navigation
- Group chat noise and lack of organization

INTERVIEWER: What solutions would you propose?

CANDIDATE:

- Smart media cleaner where AI detects duplicates, memes, large videos, and suggests deletion
- Advanced search filters to search by file type, sender, keyword or time period
- Chat folders/tags to sort chats by work, family, study groups, etc.

- Highlight messages in groups to flag important info
- Broadcast-only group mode for announcements in large groups
- Optional read receipts in groups that helps users feel less pressured to reply instantly

INTERVIEWER: How would you prioritize these?

CANDIDATE: I'll use the parameters reach, impact, confidence and effort.

Solution	Reach	Impact	Confidence	Effort	RICE Score
Smart media cleaner	9	9	9	4	182.25
Advanced search filters	8	8	8	4	128
Chat folders/tags	9	7	8	5	100.8
Group message highlights	6	7	7	4	73.5
Broadcast-only mode	7	6	7	5	58.8
Optional group read receipts	6	5	7	5	42

INTERVIEWER: Looks good. And how would you measure success?

CANDIDATE:

Primary metric: Daily Active Users engaging in search and media management features

Supporting metrics:

- Storage cleaned per user per month
- Frequency of search feature usage
- Group engagement rates (replies to pinned messages)
- Drop in group mute rate



Case 5: Improve Spotify

INTERVIEWER: Let's say the case is to improve the app you use the most on your phone.

CANDIDATE: Sure. That would be Spotify for me. I use it every single day: for music, podcasts, and sometimes even background noise while working. Do you want me to focus on just the music experience or podcasts too?

INTERVIEWER: Let's keep it broad: both music and podcast experiences. Assume this is for the Indian market, and success would be measured in terms of increased engagement, user retention, and overall satisfaction.

CANDIDATE: Got it. I'll start by identifying the key user personas, which will help us ground the improvements better.

- Casual listeners who just want to hit play and enjoy music
- Podcast binge-listeners who prefer long-form content
- Runners/gym-goers who need hands-free, distraction-free use
- Music curators who enjoy creating and sharing playlists
- Smart speaker users who primarily use voice navigation

INTERVIEWER: Sounds good. Who do you think are the top three to prioritize?

CANDIDATE: Let me quickly explain why we need to prioritize. Different personas engage with the app in very different ways: by focusing on the most high-frequency, high-friction users, we can deliver the biggest impact. Based on that, the top 3 would be:

- Casual listeners – They make up the majority of daily users
- Podcast users – They spend the most time in the app
- Runners/gym users – They're a niche but highly vocal group, and improving their experience adds convenience

INTERVIEWER: Great. Walk me through the journey of a typical Spotify user.

CANDIDATE: Sure. Here's a simplified journey:

- Open the app
- Either scroll through recommendations or search directly
- Select a playlist, song, or podcast
- Use features like like/dislike, download, add to playlist
- Often switch to background or lock screen playback

INTERVIEWER: And what are some common pain points they might experience?

CANDIDATE: Across the top personas, here are the major ones:

- Poor podcast discovery: It's hard to find good content unless it's trending

- Shuffle feels repetitive: Even large playlists repeat quickly
- Bad in-motion usability: Runners and drivers struggle to change songs safely
- Lack of contextual playlists: The app doesn't always feel personalized by mood or time
- Curation feels overly algorithmic: Not enough human-touch playlists

INTERVIEWER: Okay, which ones would you prioritize?

CANDIDATE: I'd go with these top three based on severity and prevalence:

- Podcast discovery
- Shuffle redundancy
- In-motion usability

INTERVIEWER: Great. What kind of solutions are you thinking of?

CANDIDATE: Here's what I'd propose:

- Podcast tagging & categorization – Add tags like "Tech," "TrueCrime," or "Short 15-min episodes" to make browsing easier
- Dynamic shuffle – Use recent skip behavior, time of day, and mood to vary songs better
- Workout-friendly mode – A toggle that activates large

- buttons, voice commands, and easy gestures
- Smart context playlists – "Morning Vibes," "Focus Rainy Day," or "Workout Beats" based on time and weather
- Offline auto-sync – Downloads your top 25 most played tracks weekly without user input

INTERVIEWER: Prioritize these and pick the top three.

CANDIDATE: I'll use the **Value vs. Effort** model for this one:

Solution	Value	Effort	Priority
Podcast tagging & discovery	9	4	High
Dynamic shuffle	8	5	High
Workout-friendly UI	7	4	Medium
Smart context playlists	6	6	Medium
Offline auto-sync	5	5	Low

Thus, the top three to implement would be:

- Podcast tagging & categorization
- Dynamic shuffle
- Workout-friendly interface

INTERVIEWER: And what metrics would you track?

CANDIDATE:

Primary metric: Average listening time per user per day
Supporting metrics: Search-to-play conversion rate for podcasts, Skip rate in shuffled playlists, User Retention

Case 6: Improve a Dating App

INTERVIEWER: Your next case is to improve a dating app.

CANDIDATE: Alright. I'll go with **Bumble** because I've used it and I also appreciate that it positions itself as a safer, women-first platform. Just to clarify, should I focus more on user retention, safety, monetization, or experience?

INTERVIEWER: Think from the lens of improving user experience and alignment with Bumble's mission: safe, empowering connections. Monetization is important, but don't let it drive your decisions. Assume Indian urban users for context.

CANDIDATE: Got it. I'll start by listing the relevant user personas. Dating apps have very different users with different needs.

- Women looking for relationships
- New city movers looking to date or socialize
- Casual users browsing or dating for fun
- LGBTQ+ users with specific community needs
- First-time users unfamiliar with dating app etiquette

INTERVIEWER: Who are your top three and why?

CANDIDATE: Before I get into that: prioritizing personas helps us focus on users with the highest friction and expectations. These

are the people most likely to churn or evangelize the product. Based on that, I'll prioritize:

- Women users – because Bumble is built around making them feel safe and in control
- New city movers – they have high intent and urgency to connect
- First-time users – first impressions make or break retention

INTERVIEWER: Makes sense. Tell me how a typical user journey flows.

CANDIDATE: Sure. For most users:

- Download the app and create a profile
- Set preferences (gender, age, distance)
- Start swiping
- Get a match, then initiate conversation
- Decide whether to move to an in-person meeting
- Optionally, delete app or go dormant if matches fail

INTERVIEWER: What pain points do you see in that journey?

CANDIDATE: Quite a few. These are the big ones:

- Swipe fatigue – After a point, swiping feels like a chore
- Ghosting – You match, say hi, then silence
- Creepy or unverified profiles – Especially a concern for women

- Lack of clarity in user intent – Long-term vs casual, etc.
- Bad onboarding for first-time users

INTERVIEWER: Pick your top three.

CANDIDATE: Based on severity and prevalence:

- Swipe fatigue
- Ghosting and shallow conversations
- Unverified profiles and safety concerns

INTERVIEWER: Great. Now give me some solutions.

CANDIDATE:

- Intent-based filtering — Users select what they’re looking for:serious, casual, friends, etc.
- Prompt-based icebreakers — Conversation starters like “Let’s debate pineapple on pizza” help avoid ghosting
- Profile verification through selfie/video challenge
- Daily quality match limit — Less swiping, more meaningful interaction
- Safety score indicator — Similar to Uber ratings, based onbehavior (optional and private)

INTERVIEWER: Can you prioritize these and pick the top three?

CANDIDATE: Absolutely.

Solution	Reach	Impact	Confidence	Effort	RICE Score
Prompt-based icebreakers	8	8	9	4	144
Intent-based filtering	9	9	8	5	129.6
Selfie/video verification	7	9	8	5	100.8
Daily match limit	6	6	9	2	81
Safety behavior indicator	6	6	7	5	50.4

Top three would be:

- Prompt-based icebreakers
- Intent-based filtering
- Profile verification

INTERVIEWER: What about metrics?

CANDIDATE:

Primary metric: Match-to-chat conversion rate
 Other metrics: % of users who send a message post-match, Profile verification opt-in rate, Repeat app opens per day/week, Ghosting rate (3-message drop-off)

Case 7: Improve Teaching in Tier 2 Cities in India

INTERVIEWER: How would you improve the quality of teaching in Tier 2 cities in India?

CANDIDATE: A very meaningful case. Quick clarification — are we looking at public schools, low-fee private schools, or both?

INTERVIEWER: Focus on government and semi-private schools, where teachers have limited support and digital tools. Assume constraints like patchy internet, minimal budget, and mixed tech literacy. The goal is to improve learning outcomes at scale.

CANDIDATE: Got it. I'll start by identifying the key stakeholders or personas.

- Government school teachers: often undertrained and overworked
- Students (Grades 6–12): the direct beneficiaries
- Parents with low digital literacy
- School administrators: decision-makers for adopting tools
- NGOs/ed-tech partners: support with training and technology

INTERVIEWER: Who would you prioritize?

CANDIDATE: Let me first explain why persona prioritization is important. In a system like this, improving teacher experience has a cascading effect. If the teacher is empowered, students benefit more directly than if we just build tools for students.

So, top 3 personas would be:

- Teachers: they drive daily learning
- Students: ultimate beneficiaries
- School administrators: enablers of adoption and implementation

INTERVIEWER: Makes sense. What pain points are teachers facing?

CANDIDATE:

- Lack of accessible, engaging teaching content
- Low exposure to digital pedagogy
- No real-time visibility into student progress
- Language barriers when using national ed-tech content
- Low motivation and isolation among teachers

INTERVIEWER: Which would be the top pain points?

CANDIDATE:

- Lack of good, ready-to-use content
- Low digital/pedagogical training
- No real-time feedback on student performance

INTERVIEWER: What are your proposed solutions?

CANDIDATE:

- Offline-compatible content libraries in local languages aligned to the curriculum
- Modular digital pedagogy courses: mobile-first and gamified
- Mobile-friendly student performance tracker with dashboards
- Peer learning circles via WhatsApp/Telegram groups for idea sharing
- Audio-visual lesson bank for daily use in classrooms

INTERVIEWER: Prioritize your solutions.

CANDIDATE: Absolutely.

Solution	Reach	Impact	Confidence	Effort	RICE Score
Offline content library	9	9	9	4	182.25
Digital pedagogy training modules	8	9	8	5	115.2
Performance tracking dashboards	7	8	7	6	65.3

Peer learning forums	6	6	8	3	96
Audio-visual lesson bank	6	6	7	5	58.8

INTERVIEWER: What success metrics would you track?

CANDIDATE:

Primary metric: Increase in student learning outcomes (e.g., scores in state board or baseline assessments)

Other metrics:

- % of teachers completing digital training
- Daily usage rate of lesson content
- Student attendance and engagement
- Peer forum activity levels

Case 1: Decrease in order fulfillment on the Amazon delivery app

INTERVIEWER: Let's start with the case. We've seen a 25% decrease in order fulfillment on the Amazon delivery app in the past month. Can you walk me through your thought process on identifying the root cause?

CANDIDATE: Thank you. To start, I'd like to clarify a few points. First, could you please elaborate on what you mean by "order fulfillment"? Are we talking about the point when a customer pays for an item or when the product is actually delivered?

INTERVIEWER: Great question. Let's define order fulfillment as the point when the customer pays for the product.

CANDIDATE: Okay, understood. Secondly, is this decrease in order fulfillment observed across all product categories or is it specific to certain segments?

INTERVIEWER: It's primarily affecting the apparel segment.

CANDIDATE: Got it. And lastly, is this issue geographically specific or is it happening nationwide?

INTERVIEWER: It's happening nationwide.

CANDIDATE: Thank you for the clarification. Given the sudden drop in order fulfillment specifically within the apparel category, I believe it's crucial to understand if there have been any external factors impacting the overall market or if the issue lies within our product or processes.

To begin, I'd like to explore potential economic changes, competitor activities, or industry trends that might be influencing this decline. For instance, have there been any significant disruptions in the supply chain or logistics industry that could be impacting order fulfillment? Or have competitors introduced new fulfillment services that might be drawing customers away from Amazon?

INTERVIEWER: There haven't been any significant disruptions in the supply chain or major competitor moves.

CANDIDATE: Okay, let's consider other potential external factors. Have there been any changes in consumer behaviour related to apparel shopping? For example, has there been a shift in purchasing habits towards offline stores or other online platforms?

INTERVIEWER: There has been no significant change in

consumer behavior.

CANDIDATE: Understood. Let's explore other potential external factors. Have there been any changes in government regulations or policies that could impact the apparel industry or order fulfillment process? For example, new taxes, shipping regulations, or labor laws?

INTERVIEWER: There have been no significant changes in government regulations.

CANDIDATE: Okay, given that we haven't found any significant external factors, let's delve deeper into internal factors. To understand this better, I'd like to understand the typical customer journey for an apparel order. It usually involves browsing the app, selecting items, proceeding to checkout, payment, order confirmation, and finally, delivery. Is this understanding correct?

INTERVIEWER: Yes, that's correct.

CANDIDATE: Could you please confirm if there have been any recent changes to the app, such as updates, new features, or design modifications?

INTERVIEWER: Yes, there was a major update about a month ago that introduced an AR try-on feature for apparel products.

CANDIDATE: Interesting. Could you elaborate on the specific changes made in this update?

INTERVIEWER: The update focused on enhancing the apparel shopping experience by allowing customers to virtually try on clothes using augmented reality.

CANDIDATE: Thank you. Based on the information provided, it seems plausible that the recent update might be the culprit. Introducing new features, especially those that are computationally intensive like AR, can sometimes impact app performance.

If the app is loading slowly or crashing frequently, it could frustrate users and drive them away. Additionally, the AR feature might be diverting resources away from core functionalities, affecting overall app performance.

INTERVIEWER: That's a good point. The product page load time has indeed increased significantly after the update, and we've received numerous complaints about app crashes. Can you suggest any approach to tackle this situation?

CANDIDATE: Given the situation, I suggest a two-pronged approach. In the short term, we should prioritize fixing the performance issues by optimizing the AR feature or temporarily disabling it if necessary. We can also implement a loading indicator to improve user experience while the app loads.

In the long term, we need to conduct a thorough performance analysis to identify the root cause of the performance degradation and implement sustainable solutions. Additionally, we should invest in improving our app monitoring and alerting system to prevent such issues from recurring.

INTERVIEWER: Great suggestions. Thank you for your analysis.

Sample Case Sheet

Clarifying Questions

- **Order Fulfillment Definition:** When the customer pays for the product
- **Affected Segments:** Primarily the apparel segment
- **Geographical Scope:** Nationwide impact
- **Sudden or Gradual Drop:** Sudden

External Factors

1. **Economic Changes:** No significant economic disruptions impacting the market.
2. **Supply Chain or logistics changes:** No significant disruptions in the supply chain
3. **Competitor Activities:** No new entrant in the market and no major moves from existing competitors.
4. **Consumer Behavior:** No change in behavior and no notable shift towards offline stores or other online platforms.
5. **Government Regulations:** No new taxes, shipping regulations, or labor laws affecting the apparel industry or order fulfillment process.

Internal Factors

- **Customer Journey:**
Browsing the app → Selecting items → Proceeding to checkout → Payment → Order confirmation → Delivery
- **Recent Changes to the App:**
 - Major update introduced about a month ago.
 - **New AR try-on feature for apparel products.**
 - **Increased product page load time and app crashes post-update**

Recommendations:

Short-term Actions:

1. Optimize the AR feature to improve performance.
2. Temporarily disable the AR feature if optimization is not immediately feasible.
3. Implement a loading indicator to enhance user experience during app load times.

Long-term Actions:

1. Conduct a thorough performance analysis to identify and address the root cause of performance degradation.
2. Implement sustainable solutions to improve app stability and performance.
3. Invest in better app monitoring and alerting systems to prevent recurrence of similar issues.

Case 2: Decrease in weekly active users (WAUs) on Spotify

INTERVIEWER: We've observed a 30% decrease in weekly active users (WAUs) on Spotify in the past three weeks. Can you walk me through your thought process on identifying the root cause?

CANDIDATE: Thank you. To start, I'd like to clarify a few points. First, how is WAU defined? Is it based on streaming at least one song, or does it include other user interactions like creating playlists or searching for music?

INTERVIEWER: WAU is defined as a user who has listened to at least one song in a week.

CANDIDATE: Okay, understood. Secondly, is this decrease in WAUs observed across all user segments or is it specific to certain demographics?

INTERVIEWER: It's affecting all user segments.

CANDIDATE: Got it. Thirdly, is this decline in WAUs sudden or gradual?

INTERVIEWER: It's been gradual over the past three weeks.

CANDIDATE: Okay, understood. Lastly, could you confirm if the BI tool we're using to measure WAUs is functioning correctly?

INTERVIEWER: Yes, the BI tool is working fine.

CANDIDATE: Thank you for the clarification. Given the gradual decline in WAUs across all user segments, it's crucial to understand if there are any external factors impacting overall music streaming behavior.

CANDIDATE: Thank you for the clarification. Given the gradual decline in WAUs across all user segments, it's crucial to understand if there are any external factors impacting overall music streaming behavior.

To begin, I'd like to explore potential economic changes, competitor activities, or industry trends that might be influencing this decline. For instance, have there been any significant economic downturns that could be impacting consumer spending on entertainment? Or have competitors introduced new features or pricing models that might be drawing users away from Spotify?

INTERVIEWER: There haven't been any significant economic downturns or major competitor moves.

CANDIDATE: Okay, let's consider other potential external factors. Have there been any changes in consumer behavior related to music consumption? For example, has there been a shift in listening habits towards short-form video platforms or podcasts?

INTERVIEWER: There has been a slight increase in short-form video consumption, but it doesn't seem significant enough to account for the WAU decline.

CANDIDATE: Understood. Let's explore other potential external factors. Has there been any significant change in the overall music industry? Any major events or trends that could impact music consumption?

INTERVIEWER: There have been no major events or trends in the music industry.

CANDIDATE: Given that we haven't found any significant external factors in the music industry or consumer behavior, let's explore other potential areas. Have there been any changes in government regulations or policies that could impact the music streaming industry or Spotify specifically? For example, new taxes, licensing fees, or data privacy regulations?

INTERVIEWER: There have been no significant changes in

government regulations.

CANDIDATE: Okay, let's consider the impact of public perception. Has Spotify faced any negative publicity or PR crisis recently? This could potentially impact user trust and loyalty.

INTERVIEWER: There have been no major negative PR incidents.

CANDIDATE: Understood. Given the absence of significant external factors, let's delve deeper into internal factors.

CANDIDATE: Before we delve deeper into internal factors, I'd like to understand the Spotify user journey better. A Spotify user journey typically begins with account creation or login, followed by exploring the platform's extensive music library through personalized recommendations, curated playlists, or genre-based browsing. Once users discover desired tracks, albums, or artists, they can add them to playlists, create stations, or listen immediately. The core experience involves streaming audio, controlling playback, and potentially sharing music with friends. Users can also engage with social features like following artists or friends, discovering new music through shared playlists, and participating in collaborative playlists. Additional features such as podcasts, audiobooks, and offline listening enhance the user experience. Is my understanding about user journey correct?

INTERVIEWER: Yes your understanding is correct.

CANDIDATE: Got it. And how does Spotify measure user engagement beyond WAUs? Are there any other metrics that could provide additional insights?

INTERVIEWER: We also track metrics like average listening time, songs played and playlist creation.

CANDIDATE: Any complaints/issues reported from users in any of the specific steps in the user journey?

INTERVIEWER: Nothing reported as such.

CANDIDATE: Are there any recent changes to the app, such as updates, new features or design modifications?

INTERVIEWER: There was a significant update three weeks ago that removed the offline listening capability.

CANDIDATE: That's interesting. The removal of offline listening capability could potentially impact WAU numbers. Could you elaborate on the reasoning behind this decision?

INTERVIEWER: The decision was made to focus on improving the streaming experience and to encourage users to rely on our network for uninterrupted playback.

CANDIDATE: I understand. However, offline listening is a valuable feature for many users, especially those in areas within consistent internet connectivity. Removing this feature could lead to decreased user satisfaction and potentially drive users to other platforms.

INTERVIEWER: We anticipated some initial backlash but believed the long-term benefits would outweigh the short-term drawbacks.

CANDIDATE: I see. To further understand the impact, it would be helpful to analyze user behavior before and after the feature removal. Are there any data points on user engagement metrics, such as average listening time or songs played for the past three weeks?

INTERVIEWER: We've seen a slight decrease in average listening time and songs played.

CANDIDATE: This data supports the hypothesis that the removal of offline listening has negatively impacted user engagement. To mitigate the issue, I suggest reintroducing the offline listening feature or exploring alternative solutions like offline caching or improved network optimization. Additionally, we should conduct user surveys to gather feedback on the feature removal and identify potential improvements.

Case 3: Zomato is facing a 10% daily drop in its transacting users.

CANDIDATE: I understand that Zomato is seeing a 10% daily drop in transacting users. Let me start by asking a few clarifying questions to frame the problem better. First, when we say “transacting user,” how exactly are we defining that?

INTERVIEWER: A transacting user is someone who actually completes a purchase or order.

CANDIDATE: Got it. And since when has this drop been visible?

INTERVIEWER: We’ve been seeing it every day for about a week now.

CANDIDATE: Interesting. Is it happening at certain times of the day, like peak hours, or is it more uniform across the board?

INTERVIEWER: It’s a generic drop, not tied to any specific time.

CANDIDATE: Okay. And does it impact all kinds of users, or just a particular segment?

INTERVIEWER: It's across all user groups.

CANDIDATE: Understood. Before I jump into hypotheses, I want to quickly rule out some obvious possibilities. First, could this be due to any changes in how we’re tracking or calculating the metric?

INTERVIEWER: No, nothing has changed in the logging or formulas.

CANDIDATE: Alright. On the competitive side—have any rivals launched big offers or new features recently that could explain the shift?

INTERVIEWER: Only small updates, nothing major.

CANDIDATE: And from an external standpoint: any issues with operating systems, push notifications, or marketing campaigns that might have disrupted user engagement?

INTERVIEWER: No, none that we’ve seen.

CANDIDATE: Makes sense. Just to be sure, does this season typically see a drop in orders, maybe due to trends or seasonality?

INTERVIEWER: Not really. This season usually has steady

volumes. The drop is unusual.

CANDIDATE: Okay, so it's not competition, not seasonality, and not data capture. That points me toward internal factors. Has the drop been consistent across platforms, say Android and iOS?

INTERVIEWER: Yes, both are affected.

CANDIDATE: Any spikes in bug reports or app crashes?

INTERVIEWER: No spikes in crashes, but support tickets have gone up.

CANDIDATE: That's worth noting. Let me quickly walk through the typical user journey to see where things might be breaking down - User opens the app - Browses restaurants - Selects one and checks the menu - Adds items to the cart - Proceeds to checkout - Confirms delivery details - Applies promo codes if available - Reaches the payment screen - Completes payment - order confirmed

INTERVIEWER: Yes, that's the flow.

CANDIDATE: Perfect. Now, were there any recent changes to this journey like new releases, feature tweaks, or UX updates that

could have affected it?

INTERVIEWER: Actually yes. A couple of weeks ago we updated the payment screen with new elements, meant to improve the experience.

CANDIDATE: That helps a lot. My hunch is that the payment screen changes might have introduced friction. For example, maybe the flow got longer, or the page is loading more slowly now, or the design tweaks disrupted the familiar process for users.

To be sure, I'd want to dig into a few things:

- User feedback specifically about the payment flow
- Load times on the updated payment screen
- Drop-off rates step by step, especially at payment
- Heatmaps or click maps comparing old vs. new behavior

INTERVIEWER: Makes sense. We can conclude.

Case 4: Flipkart is experiencing a 10% decline in the number of Add to Cart actions.

CANDIDATE: Just to clarify, an “Add to Cart” action is an event that a user performs when they add any product to the cart, regardless of the product category, before proceeding to checkout.

INTERVIEWER: Yes, that is correct!

CANDIDATE: Since when has the drop become visible?

INTERVIEWER: The drop has been visible on a daily basis since last 10 days.

CANDIDATE: Is there a specific day or time when this usually occurs, or is it a generic drop?

INTERVIEWER: It is a generic drop.

CANDIDATE: Is the drop consistent across all user segments, or are there specific segments experiencing larger drops?

INTERVIEWER: It is consistent across all user segments.

CANDIDATE: Based on the responses, I have identified four possible categories where the root cause may lie. First, we will quickly check for any system or logging-related issues. Have any changes been made to the logging systems or any formulas for metric calculations that could have affected the visibility of the drop?

INTERVIEWER: No, nothing of that sort has been affected.

CANDIDATE: Now I will quickly analyze some external factors that could also contribute to the drop. In the past few weeks, have there been any significant changes in product offerings by our competitors or have they launched any new major product?

INTERVIEWER: There have been minor product developments, but no significant changes have been observed. However, there has been a surprise sale running by one of our biggest competitors.

CANDIDATE: Thank you for bringing that up. It could indeed be a contributing factor to the decline, but I'd like to examine all potential factors before drawing a conclusion. Now, I'd like to inquire if we've observed any changes in the external environment that could directly or indirectly impact our app? This could include issues such as operating system malfunctions causing

the app to malfunction at certain times, or issues preventing push notifications or other marketing activities from reaching users, potentially contributing to the observed drop.

INTERVIEWER: No such ambiguities have been detected or recorded.

CANDIDATE: Is the “Add to Cart” event/action seasonal or dependent on factors such as festivals or sale periods? Have we had any sales or similar occasions in the last few weeks? I just want to ensure that this drop is not due to a seasonal effect or any change in market trends.

INTERVIEWER: There is usually a seasonal effect, but in this case, that is not the reason. The drop is lower than what we have during the same off-season period during any time of the year.

CANDIDATE: Is this drop consistent across all the devices and browsers?

INTERVIEWER: Yes, it has been same across Android and iOS.

CANDIDATE: Has there been any sudden rise in the number of bug reports, app crashes, or support tickets that can directly assist us in identifying the cause of declining metrics?

INTERVIEWER: No, there has been no such rise observed as everything seems normal.

CANDIDATE: Now, I would like to delve deeper into the user journey that leads to a user completing a transaction. A typical journey would look like this: the user opens the Flipkart app or website and either searches for a product or navigates through categories and recommendations until the product listing page is displayed with multiple options. From there, the user clicks on a product to open the detailed product page, where they view product images, read descriptions, check specifications, and review ratings. They may also explore partner offers, bundled deals, or EMI options before deciding on the quantity, variant such as size or color, and delivery location. Once these preferences are set, the user looks for the primary CTAs like “Add to Cart” or “Buy Now.” In some cases, the user may compare options by scrolling further or checking related products, eventually returning to the CTAs. Finally, once the user clicks “Add to Cart,” the item is added, and they are taken to the cart view to either continue shopping or proceed to checkout.

INTERVIEWER: This is correct as you have clearly mentioned all the steps in the user flow.

CANDIDATE: Thank you for confirming. Now, I would like to

analyze whether we have made any new product or feature changes in the last few weeks that are part of this user flow. Often, the launch of new features and products can impact certain product metrics. This could result from a bug, a UI/UX issue, or changes in user experience due to additions or deletions in the user flow.

INTERVIEWER: Now that you have accurately described the complete user flow, we have made a few changes in the last few weeks in the current user flow.

CANDIDATE: Thank you for the confirmation. Could you please specify what changes were made, and which steps in the user flow could have been affected by these changes?

INTERVIEWER: There have been multiple changes made within the last 28 days. Each week, the team made alterations to elements on the product page, add-to-cart page, and payment page. These changes mainly focused on positional and UI/UX.

CANDIDATE: Can you detail out the changes that were made specially on the Product Page?

INTERVIEWER: To highlight partner and banking offers, buttons such as “Add to Cart,” “Buy Now,” and “Change Qty” were moved below. Additionally, secondary CTA buttons were hidden for

certain categories like Electronics.

CANDIDATE: Thank you for the confirmation. Based on my analysis, I have a few observations: Due to internal changes in the user flow, where we moved the buttons down, there could have been an impact on the user flow, possibly leading to an increase in users dropping off before performing the “Add to Cart” event. Additionally, as Electronics is one of the primary categories, hiding secondary CTAs could also have affected the “Add to Cart” event. External factors may have also played a role, as the competitor’s sale might have led many users to visit Flipkart solely to compare prices. If they found a considerable difference or a better deal on the competitor’s app/website, they may have dropped off without completing a purchase.

However, these observations can only be confirmed with specific data points:

- Heatmaps and user scroll scores for the product page can provide insights into how user behavior has changed after the recent adjustments.
- Additionally, it’s important to note the following metrics: the number of secondary CTA clicks on the Electronics category before the changes, and the number of users dropping off from the product page.

INTERVIEWER: That makes sense. Thank you.

Case 5: A grocery e-commerce app has seen a 20% drop in sales.

INTERVIEWER: A grocery e-commerce app has seen a 20% drop in sales. Why do you think this might be happening?

CANDIDATE: Thanks for the question. I'd like to ask a few clarifying questions first. Are we looking at the Indian market?

INTERVIEWER: Yes you can assume this is for the Indian market.

CANDIDATE: And when did this problem start showing up?

INTERVIEWER: We started noticing this last week.

CANDIDATE: Got it. Can I take a minute to structure my thoughts before diving in?

INTERVIEWER: Sure, please go ahead.

CANDIDATE: To approach this, I'd like to break the problem into two buckets: external factors and internal factors. Let me start with external factors. Was there a new competitor recently?

INTERVIEWER: No, there haven't been any major new entrants.

CANDIDATE: Understood. Were there any big promotional events or sales from competitors last week, something like Amazon's Great Indian Sale or Flipkart's Big Billion Days?

INTERVIEWER: No, none that we're aware of.

CANDIDATE: Alright. Was there any disruption in the broader grocery supply chain: say, strikes by transporters or delivery staff, that could have impacted sales overall?

INTERVIEWER: Not that we've seen.

CANDIDATE: Interesting. And is this drop uniform across geographies, demographics, and product categories, or is it concentrated in a particular segment?

INTERVIEWER: No particular pattern, it looks fairly spread out.

CANDIDATE: Okay. So, for now, can we assume external factors haven't caused this?

INTERVIEWER: Yes, let's move on to internal factors.

CANDIDATE: Perfect. Let me focus on the internal side then. First, could there have been an issue with the data collection tool that tracks sales?

INTERVIEWER: No, the data tools are working fine.

CANDIDATE: Understood. Is this drop being seen in both the web app and the mobile app, or just one of them?

INTERVIEWER: It's specific to the mobile app.

CANDIDATE: That's useful. Were there any spikes in uninstallations last week, particularly on Android?

INTERVIEWER: No unusual increase in uninstallations.

CANDIDATE: Alright. Did you release any app updates recently?

INTERVIEWER: Yes, we had an update roll out last week.

CANDIDATE: Was it for Android, iOS, or both?

INTERVIEWER: Only Android.

CANDIDATE: Interesting. That seems like a good place to

investigate further. Did the update make any changes to the add-to-cart or checkout flow?

INTERVIEWER: Yes, we made some tweaks in the user flow.

CANDIDATE: Let me check step by step then. First, is the add-to-cart button working as expected?

INTERVIEWER: Yes, products are being added to carts without an issue.

CANDIDATE: And when users go to the cart, are they able to see the items correctly?

INTERVIEWER: Yes, the cart displays items properly.

CANDIDATE: That's good. How about the payment stage. Are there any bug reports around checkout?

INTERVIEWER: Yes, we did receive bug reports about payments failing.

CANDIDATE: Are these issues happening across all payment methods, or just specific ones like credit/debit cards, UPI, or wallets?

INTERVIEWER: Those methods are working fine.

CANDIDATE: Then is it linked to the Cash on Delivery option?

INTERVIEWER: Yes, there have been bugs specifically with the Cash on Delivery button.

CANDIDATE: That explains it. I think the 20% drop in sales is tied to the issues with Cash on Delivery in the Android app's payment flow. Since CoD is a major mode of payment in India, a bug there could directly lead to a significant sales drop.

INTERVIEWER: That makes sense.

CANDIDATE: To support this, let me briefly outline the typical user journey in a grocery app. A user usually begins by opening the mobile app and browsing through categories like fruits, vegetables, dairy, or packaged goods. They may also use the search bar to look for specific items or rely on recommendations and frequently bought items. Once they find a product, they check details such as price, brand, available discounts, and delivery times. After selecting the desired quantity or variant, they add the product to the cart. Most users continue shopping until their cart is filled with essentials, at which point they head to checkout. At checkout, they confirm delivery address and slot, then move to the payment screen where they choose their

preferred method, often Cash on Delivery in India. If that step doesn't work, it becomes a critical blocker, and the entire purchase journey fails despite the effort the user put into building their cart. This makes it very likely that the CoD bug explains the sales decline.

INTERVIEWER: That's a solid breakdown. Can you wrap this up with your key takeaways?

CANDIDATE: Absolutely. Based on the discussion, my diagnosis is that the 20% drop in sales is primarily due to a Cash on Delivery bug in the Android app, introduced with the recent update. Since CoD is a highly preferred payment option in India, even a short-term glitch here can cause a significant dip in completed transactions. As next steps, I would recommend: first, rolling out a quick patch fix for the bug and closely monitoring sales recovery; second, issuing proactive communication to customers, especially those who attempted failed CoD transactions, reassuring them and offering a small incentive to retry their purchase; and third, strengthening the QA process for future releases, particularly around payment flows, since these directly impact revenue. This way, we not only fix the immediate issue but also build trust with customers going forward.

Case 1: Estimating TAM and Market Size of Wearable Fitness Trackers in India

INTERVIEWER: Estimate the Total Addressable Market (TAM) and market size of wearable fitness trackers in India.

CANDIDATE: Sure, to get a more accurate picture, let's ask a few clarifying questions first.

INTERVIEWER: Yes, please do.

CANDIDATE:

- **Time Frame:** Are we looking at the current market size or projecting for a specific future year?

INTERVIEWER: We're interested in the current market size only.

CANDIDATE:

- **Product Definition:** Are we considering all types of wearable fitness trackers or only a specific segment?

INTERVIEWER: For now, let's focus on fitness tracker watches.

CANDIDATE:

- **Geographical Focus:** Are we focusing only on urban areas, or should we include rural areas as well?

INTERVIEWER: We want to consider the entire country, both urban and rural.

Great! Now, let's move on to the guesstimate.

CANDIDATE: We can estimate the TAM by applying a series of filters to the Indian population.

Filter 1: Population of India

CANDIDATE: The total population of India in 2024 is estimated to be around 1.4 billion people.

INTERVIEWER: Right.

Filter 2: Urban vs Rural Population

CANDIDATE: Approximately 35% or 490 million people live in urban areas, while the remaining 65% or 910 million live in rural areas.

INTERVIEWER: Okay.

Filter 3: Affordability Filter

CANDIDATE: Wearable fitness trackers are likely affordable primarily for middle and upper-income groups. Let's assume:

- **Urban:** 50% of the urban population (around 245 million) belong to middle and upper-income groups.
- **Rural:** Only 10% of the rural population (around 91 million) can be considered middle or upper-income.

INTERVIEWER: That's a fair assumption.

Filter 4: Age Group Filter

CANDIDATE: The target age group for fitness trackers is typically between 18 and 50 years old, which is roughly 40% of the population.

Applying the age filter to each income group:

- **Urban:** 40% of 245 million is 98 million people.
- **Rural:** 40% of 91 million is 36.4 million people.

INTERVIEWER: Agreed.

Filter 5: Gender Filter

CANDIDATE: Let's assume an equal split of males and females interested in fitness trackers within both urban and rural areas.

INTERVIEWER: Sounds reasonable.

So, here's the breakdown:

- Urban Males: 50% of 98 million, which is 49 million.
- Urban Females: Another 49 million.
- Rural Males: 50% of 36.4 million, which is 18.2 million.
- Rural Females: Another 18.2 million.

Total Addressable Market (TAM)

CANDIDATE: Now we can combine these figures to get the TAM.

Total TAM = Urban Males + Urban Females + Rural Males + Rural Females

TAM = 49 million + 49 million + 18.2 million + 18.2 million = 134.4 million people

Market Size

CANDIDATE: To estimate the market size in terms of revenue, we need to consider the average selling price of a fitness tracker watch.

INTERVIEWER: Let's assume an average selling price of ₹2,000.

CANDIDATE: Great!

Market Size = TAM * Average Selling Price

Market Size = 134.4 million * ₹2,000 = ₹268.8 billion

So, the estimated TAM for wearable fitness tracker watches in India is 134.4 million, and the market size is approximately ₹268.8 billion.

Case 2: How Many Tennis Balls Can Fit in a Boeing 747?

INTERVIEWER: This is a fun question! Let's estimate how many tennis balls you could fit inside a Boeing 747.

CANDIDATE: Sounds interesting! To get a more accurate picture, a couple of clarifying questions first.

INTERVIEWER: Sure, fire away!

CANDIDATE:

- **Space Considered:** Are we considering the entire internal volume of the plane, including the cargo hold, or just the passenger cabin?

INTERVIEWER: Let's consider every space a tennis ball could fit in, including the cargo hold.

CANDIDATE:

- **Plane Interior:** Should we account for the space occupied by seats and other fixtures, or assume an empty plane?

INTERVIEWER: Great point. Let's assume the plane is completely empty for this calculation.

Alright, with that in mind, let's get started!

CANDIDATE: We can estimate the number of tennis balls by calculating the usable space inside the plane and then dividing it by the volume of a single tennis ball.

Volume of the Boeing 747 Interior:

CANDIDATE: Imagine the plane's interior as a long cylinder with rounded ends like a capsule.

INTERVIEWER: I can picture that.

CANDIDATE: To calculate the volume, we'll need some assumptions about the size of the plane.

Assumptions:

- **Diameter:** Let's say the fuselage is wide enough for two people to stand comfortably side-by-side, which is roughly 3 meters in diameter.
- **Cabin Length:** Assuming a typical seating configuration with 30 rows, and considering the gaps between rows, each row might be about 1 meter long.
- **Cylindrical Section:** With 30 rows, the total length of the cylindrical part would be 30 meters.

Volume Calculations:

- **Cylinder Volume:** We'll use the formula for the volume of a cylinder: $V_{\text{cylinder}} = \pi \times r^2 \times h$
 - **Radius (r):** Diameter divided by 2, so $r = 1.5$ meters.
 - **Height (h):** Total length of the cylindrical section, $h = 30$ meters.
 - $V_{\text{cylinder}} \approx \pi \times (1.5)^2 \times 30 \approx 212$ cubic meters
- **Hemispherical Ends:** The rounded ends can be approximated by two hemispheres.
 - **Sphere Volume:** $V_{\text{sphere}} = \frac{4\pi r^3}{3}$
 - **Radius (r):** Same as the cylinder radius, $r = 1.5$ meters.
 - **Volume of one sphere:** $V_{\text{sphere}} \approx \frac{4\pi(1.5)^3}{3} \approx 14.14$ cubic meters

Total Interior Volume:

CANDIDATE: Now we can add the volume of the cylinder and the two hemispheres to find the total volume inside the plane.

$$V_{\text{total}} = V_{\text{cylinder}} + V_{\text{hemispheres}} \approx 212 + (14.14 \times 2) \approx 226.14 \text{ cubic meters}$$

Volume of a Tennis Ball:

CANDIDATE: The standard diameter of a tennis ball is about 0.07 meters.

INTERVIEWER: Okay.

CANDIDATE: Using the radius (half the diameter), we can calculate the volume of a single tennis ball.

Volume of a tennis ball (V_{ball}): $V_{\text{ball}} \approx \frac{4\pi(0.035)^3}{3} \approx 0.00018$ cubic meters

Packing Efficiency:

CANDIDATE: Here's a catch! We can't perfectly pack spheres like tennis balls into the plane. There will be gaps and wasted space.

INTERVIEWER: Right, that makes sense.

CANDIDATE: Scientists have determined that the most efficient way to pack spheres is around 74%. This is called packing efficiency.

How Many Tennis Balls Fit?

CANDIDATE: To account for the wasted space, we need to multiply the total usable space by the packing efficiency.

Effective volume for tennis balls: $226.14 \text{ cubic meters} \times 0.74 \approx 167.34$ cubic meters

Now we can finally divide the usable space by the volume of a single ball to find the number of tennis balls that would fit.

Number of tennis balls $\approx 167.34 \text{ cubic meters} / 0.00018 \text{ cubic meters/ball} \approx 929,68$

Case 3: How many elevators are required in a 40-storey commercial building?

CANDIDATE: Cool. Let's walk through this. I'm imagining a typical commercial building — maybe an office tower in Mumbai or Gurgaon.

Let's start with the people. Suppose each floor has 10 office suites, and each suite has about 10 employees. That gives us:
 $40 \text{ floors} \times 10 \text{ suites} \times 10 \text{ people} = 4,000 \text{ people}.$

Now, not all 4,000 people will use the elevator at the same time, but a significant number will during peak hours, like 9–10 AM and 5–6 PM. Let's assume half of them travel up in the morning, that's 2,000 people.

INTERVIEWER: Right. And elevators have capacity limits too.

CANDIDATE: Yes! Let's say one elevator holds 15 people and completes a round trip — up, unload, back — in around 3 minutes. So in an hour, one elevator can make about 20 trips, carrying:

$20 \text{ trips} \times 15 \text{ people} = 300 \text{ people}.$

To handle 2,000 people in an hour, we need: $2,000 \div 300 \approx 6.7 \rightarrow$
Round up to 7 elevators.

But that's still tight. And we haven't accounted for lunch breaks, visitors, maintenance, or service elevators. Also, in real buildings, zoning is used — different elevators serve different sets of floors to speed things up. So I'd bump that number to 10–12 elevators, arranged in 3–4 elevator banks for redundancy and efficiency.



Case 4: How many iPhones are sold in India in a year?

CANDIDATE: This one's really interesting because it blends demographics, economics, and brand perception.

Let's start with India's population — roughly 1.4 billion people. Not all of them are smartphone users, so let's focus on the urban middle class and above. Let's say roughly 10% of the population can afford a premium smartphone, so that's:
 $1.4 \text{ billion} \times 10\% = 140 \text{ million people.}$

Among smartphone users, India is very Android-dominant. iPhones have a much smaller slice of the pie. From market reports, Apple holds around 3–4% market share in India. Annual smartphone sales in India hover around 150 million units/year. So:
 $150 \text{ million} \times 4\% = 6 \text{ million iPhones.}$
So I'd estimate around 5 to 6 million iPhones sold in India each year.

INTERVIEWER: Does this include refurbished or older models?

CANDIDATE: Good question! I think this number might even go slightly higher if we include older models like the iPhone SE or refurbished imports, which are popular among students and young professionals. So a range of 6–7 million iPhones annually seems fair.



Case 5: Estimate the number of ATMs in India

CANDIDATE: Alright, since we don't have data on the number of banks, let's go with what we do know — India's population.

India's current population is around 1.4 billion. Not everyone uses formal banking services, but thanks to financial inclusion programs like Jan Dhan Yojana, mobile banking, and digital adoption, a large percentage of the population is now linked to the formal banking system. Let's assume about 60% of the population is actively using banking services, either through debit cards, ATMs, or mobile wallets. That gives us:
 $1.4 \text{ billion} \times 60\% = 840 \text{ million banked individuals.}$

Now, how many people does one ATM typically serve? From what I've observed in Indian cities and small towns, the range can vary. In rural areas, an ATM might serve 2,000–5,000 people, while in metros, it could be as low as 1,000 people per ATM due to higher transaction volumes and expectations of convenience. To keep it conservative and balanced, let's assume:
 $1 \text{ ATM serves about } 4,000 \text{ people on average}$
 $840 \text{ million} \div 4,000 = 210,000 \text{ ATMs.}$

INTERVIEWER: That feels about right. What if some areas don't have access?

CANDIDATE: That's true — there's definitely uneven distribution. Metro areas might have 1 ATM every 500–1,000 people, especially near business districts, while some rural clusters share a single ATM across several villages. So while the national average comes to about 210,000, in reality, there are dense clusters in cities and sparse coverage in rural India

Case 6: Estimate the number of flights taking off from Delhi Airport per day.

CANDIDATE: Alright, let's look at this from a ground-level, operational view — literally.

Delhi's Indira Gandhi International Airport has 4 runways, which is quite a lot for a single airport. But runways aren't all doing take-offs 24/7. Some handle landings, some are under maintenance or reserved for emergencies. So let's assume that on average, 2 of the 4 runways are actively used for departures during most of the day.

Now, from what I've read about runway throughput at busy airports, a single runway can realistically handle around 15 departures per hour when you factor in spacing between aircraft, runway clearance, and coordination with arrivals. So with 2 runways handling departures, we get:

$15 \text{ departures/hr/runway} \times 2 \text{ runways} = 30 \text{ departures per hour.}$

Now let's estimate how many hours per day this level of activity continues. While the airport is open 24 hours, peak scheduling typically spans about 18 hours per day — early morning to late night — with overnight hours having lighter traffic.

$30 \text{ departures/hour} \times 18 \text{ hours} = 540 \text{ departures per day.}$

INTERVIEWER: Seems very grounded. Are we ignoring anything?

CANDIDATE: That's a good question. We're averaging things out, for example, runway usage may shift throughout the day, and actual throughput may fluctuate during peak vs. non-peak hours. Also, this doesn't explicitly include cargo flights or private jets, which use separate slots, maybe a few dozen extra per day. So adding a little buffer for those, I'd say:
 $540 + 50 \approx 590$ total take-offs, that is around 550 to 600 departures per day.

Case 1: Category Launch Strategy for Amazon Pharmacy

INTERVIEWER: Thank you for joining us today. Could you start by giving us an overview of the strategy you would design for launching Amazon's Pharmacy category in the Delhi & NCR region?

CANDIDATE: Absolutely. To successfully launch the Pharmacy category in the Delhi & NCR region, I would outline a comprehensive strategy that encompasses several key phases: Business Feasibility Analysis, Distribution and Logistics Strategy, User Interface Design, Supplier Onboarding, Sales, Marketing, and Pricing Strategy, and After-Sales Services.

INTERVIEWER: Let's dive into each phase in detail. How would you approach the Business Feasibility Analysis for this category?

CANDIDATE: For the initial launch, I would recommend adopting a marketplace model due to the sensitive nature of medicines.

This approach minimizes costs associated with warehousing, maintenance, and hiring new employees. It also allows us to partner with multiple sources to ensure broad coverage and availability.

INTERVIEWER: Moving on to the Distribution and Logistics strategy, what plans do you have in place to ensure timely and efficient delivery?

CANDIDATE:

- I would implement a hyperlocal model to guarantee delivery within 2-3 hours post-order.
- Utilizing Amazon's extensive logistics network in Delhi for bulk orders.
- Leveraging local suppliers' logistics for small orders due to their familiarity with the locality.
- Ensuring availability of specialized vehicles with cold storage for certain medications.

INTERVIEWER: The user experience is crucial. How would you design the UI for the Pharmacy category to enhance user retention?

CANDIDATE:

- The UI design should focus on listing comprehensive product details such as quantity, medicine potency (mg), salt composition, and product ratings.
- Clear information on payment gateways and prescription uploading for prescription drugs.
- The design should be intuitive to facilitate ease of navigation and ensure more user retention.

INTERVIEWER: What is your strategy for onboarding suppliers to ensure a robust product offering?

CANDIDATE:

- Forming partnerships with local medical shops and hospital pharmacies.
- Collaborating with wholesale suppliers and distributors, although they would be a lower priority.
- Ensuring a diverse and reliable supplier base to maintain stock availability.

INTERVIEWER: Let's discuss your Sales, Marketing, and Pricing Strategy. How do you plan to position Amazon Pharmacy in the market?

CANDIDATE:

- Utilizing Amazon's brand value for umbrella branding and advertising on the Amazon website.
- Offering initial promotions and combining products with services to attract customers.
- Setting competitive pricing lower than competitors and retail outlets, with additional discounts to increase user base and supplier participation.
- Promoting cross-selling and upselling to maximize revenue.

INTERVIEWER: After the sale, customer satisfaction remains paramount. What steps would you take to ensure After-Sales Services are top-notch?

CANDIDATE:

- Verifying product quality and checking expiry dates before delivery.
- Providing personalized recommendations for regular customers via push notifications.
- Offering a 24/7 customer helpline for any product or service clarification.
- Ensuring stock availability and having sales representatives from reputed pharmacies available online for queries.

INTERVIEWER: Your strategy sounds comprehensive. Finally, how would you justify and ensure the rationale behind each decision in your strategy?

CANDIDATE:

- Each decision should be backed by thorough market research and data analysis.
- Constantly monitoring performance metrics and customer feedback to adapt and refine the strategy.
- Providing clear, evidence-based justifications for all strategic choices to stakeholders to ensure alignment and support.

INTERVIEWER: Thank you for sharing your detailed strategy. This concludes our interview.

CANDIDATE: Thank you for the opportunity. I look forward to potentially contributing to this exciting launch.

Case 2: Online Alcohol Category Launch for Amazon in Bangalore

INTERVIEWER: Thank you for joining us today. Could you start by providing an overview of the strategy you would design for launching Amazon's online alcohol category in Bangalore?

CANDIDATE: Certainly. The strategy for launching Amazon's online alcohol category in Bangalore involves several key steps: understanding and defining the category, conducting preliminary market research, evaluating business feasibility, developing a pricing and marketing strategy, addressing pain points, and calculating market potential. This structured approach ensures a comprehensive launch plan.

INTERVIEWER: Let's begin with understanding and defining the category. How would you approach this?

CANDIDATE:

- **Total Population:** Approximately 1 crore.
- **Consumption Contexts:** B2B/B2C, at home vs. out-of-home consumption.
- **Types of Alcohol:** Foreign Manufactured Foreign Liquor (FMFL), Indian Manufactured Foreign Liquor (IMFL), and country liquor.

- **Varieties:** Beers, spirits (whisky, rum, vodka, etc.), and wine.
- **Existing Market Players:** Offline retail stores, delivery through third-party apps, bars, and restaurants.

Initially, I would partner with retail stores and bars for delivery, transitioning to an inventory model once the category scales. The focus would be on FMFL and IMFL with no restrictions on varieties.

INTERVIEWER: How would you conduct the preliminary market research for this category?

CANDIDATE: I would segment the target market into different groups:

- **Regular Buyers:** Frequent purchasers.
- **Experimenters:** Those trying different types of alcohol.
- **Special Occasion Buyers:** Purchase for events or celebrations.
- **Nervous Buyers:** Those hesitant to buy alcohol offline due to social stigma.

Understanding these segments helps tailor marketing and service offerings to meet their specific needs.

INTERVIEWER: What steps would you take to evaluate the business feasibility?

CANDIDATE:

- **Value Chain Analysis:** Amazon will control only the last mile delivery.
- **Dedicated Delivery Associates:** Setup specialized delivery associates with packaging designed to safely transport alcohol.
- **Special Packaging:** Use Amazon-branded packaging to deliver alcohol discreetly.
- **Refrigeration:** Consider refrigeration for certain alcohol varieties.

These steps ensure that the logistics and delivery mechanisms are robust and efficient.

INTERVIEWER: Let's discuss your pricing and marketing strategy. How would you approach this?

CANDIDATE:

- **Pricing:** Start with below-cost pricing to attract customers. Implement promotional pricing and discounts on bulk or recurring purchases. Share margins with onboarded retail stores to boost incremental sales.
- **Marketing:** Use online channels and in-app exclusive promotions based on the user's PIN code. Focus on rural and urban segments with tailored campaigns.

This strategy aims to quickly capture market share and build a loyal customer base.

INTERVIEWER: What are the pain points you have identified, and how would you address them?

CANDIDATE: The key pain points include:

- **Selection and Availability:** Ensure a wide range of products is always available.
- **Quick Delivery:** Implement efficient delivery systems to meet customer expectations.
- **Quality Assurance:** Verify the quality and authenticity of products.
- **Payment Convenience:** Offer multiple payment options.
- **Social Stigma:** Provide discreet packaging to address shyness in purchasing offline.
- **Personalized Information and Recommendations:** Offer tailored suggestions based on buyer persona.
- **Cart Collaboration:** Allow customers to share and build orders for special occasions collaboratively.

Addressing these pain points enhances customer satisfaction and encourages repeat purchases.

INTERVIEWER: Could you elaborate on the calculation of market potential provided in your strategy?

CANDIDATE: Certainly. Here's a detailed breakdown:

- **Population Segmentation and Consumption Propensity:**
 - **Rural (50%):** 50 lakhs
 - **High Income (30%):** 15 lakhs, consuming 5 liters/month
 - **Medium Income (50%):** 25 lakhs, consuming 4 liters/month
 - **Low Income (20%):** 10 lakhs, consuming 2 liters/month
 - **Urban (50%):** 50 lakhs
 - **Young Adults (40%):** 20 lakhs, consuming 3 liters/month
 - **Adults (50%):** 25 lakhs, consuming 2 liters/month
 - **Seniors (10%):** 5 lakhs, consuming 1 liter/month

Total Consumption Calculation

- **Rural High Income:** The calculation is 15 lakhs of high-income individuals multiplied by a consumption rate of 5 liters/month, which equals 75 lakh liters/month.
- **Rural Medium Income:** The calculation is 25 lakhs of medium-income individuals multiplied by a consumption rate of 4 liters/month, which equals 100 lakh liters/month.

- **Rural Low Income:** The calculation is 10 lakhs of low-income individuals multiplied by a consumption rate of 2 liters/month, which equals 20 lakh liters/month.
- **Urban Young Adults:** The calculation is 20 lakhs of young adults multiplied by a consumption rate of 3 liters/month, which equals 60 lakh liters/month.
- **Urban Adults:** The calculation is 25 lakhs of adults multiplied by a consumption rate of 2 liters/month, which equals 50 lakh liters/month.
- **Urban Seniors:** The calculation is 5 lakhs of seniors multiplied by a consumption rate of 1 liter/month, which equals 5 lakh liters/month.

Total Monthly Consumption: The sum of the consumption for all segments (75 + 100 + 20 + 60 + 50 + 5) results in a total of **310 lakh liters/month**.

Average Pricing and Market Value

Total Market Size: The total monthly consumption of 310 lakh liters/month is multiplied by the average price per liter of ₹1000, resulting in a total market size of **₹3100 crores/month**.

Market Penetration and Sales Target

Sales Target: The total market size of ₹3100 crores/month is multiplied by the target market penetration range of 5% to 10%, which equals a sales target of **₹155 to ₹310 crores/month**.

INTERVIEWER: Could you summarize the overall strategy for us?

CANDIDATE:

- **Define the Category:** Focus on FMFL and IMFL, partnering initially with retail stores and bars.
- **Market Research:** Segment the market into regular buyers, experimenters, special occasion buyers, and nervous buyers.
- **Business Feasibility:** Control the last mile delivery, set up specialized packaging and delivery systems.
- **Pricing and Marketing:** Use aggressive pricing and targeted marketing to capture market share quickly.
- **Address Pain Points:** Ensure product availability, quick delivery, quality assurance, payment convenience, discreet packaging, personalized recommendations, and cart collaboration.
- This structured approach ensures a successful launch of Amazon's online alcohol category in Bangalore.

INTERVIEWER: Thank you for sharing your detailed strategy. This concludes our interview.



Case 3: Category Launch Strategy for Flipkart Grocery

INTERVIEWER: Suppose Flipkart wants to aggressively scale its **Grocery category** to compete with BigBasket, Blinkit, and JioMart. How would you design the launch strategy?

CANDIDATE: I'd structure the approach across a few phases: first, establishing the business feasibility and category scope, then setting up operations and logistics, designing the platform experience, onboarding suppliers and brands, building a strong marketing and pricing strategy, addressing user and partner pain points, and finally ensuring solid after-sales with clear success metrics.

INTERVIEWER: Let's start with the Business Feasibility.

CANDIDATE: Grocery is one of the largest e-commerce opportunities in India, valued at over \$600 billion offline and with online penetration growing rapidly. The category is tricky though because margins are thin and logistics are complex. For Flipkart, the strategy should be two-pronged. In metros, where instant commerce is heating up, the focus can be on 30-minute delivery through a hyperlocal dark store model. In Tier 2 and Tier 3 cities, where grocery adoption is still nascent, a scheduled same-day or next-day delivery model would make more sense.

Initially, Flipkart should position grocery under its existing "Supermart" branding and scale it city by city, starting with Bangalore, Delhi NCR and Mumbai.

INTERVIEWER: What about Operations & Logistics?

CANDIDATE: Operations are critical in grocery. Flipkart should set up dark stores and micro-warehouses in metro cities to enable express deliveries within 30–60 minutes. For smaller towns, it can rely on regional warehouses combined with local kirana partnerships for last-mile delivery. Cold chain logistics should be implemented for perishables like dairy, meat, and frozen foods. To ensure reliability, Flipkart must integrate real-time inventory systems so that out-of-stock situations are minimized. Delivery fleet optimization, that is, using a mix of two-wheelers for hyperlocal drops and vans for bulk orders will also be important to balance cost with speed.

INTERVIEWER: And how would you approach the UI/UX design?

CANDIDATE: The grocery experience should feel fast, personalized, and trustworthy. A dedicated grocery tab on the Flipkart app should highlight essentials, personalized recommendations based on past purchases, and quick re-order options for frequently bought items.

Time-sensitive slots like “Deliver in 30 minutes” versus “Deliver tomorrow morning” should be clearly visible. Product pages should focus on transparency: expiry dates, weight, nutritional info, and customer ratings. To increase stickiness, Flipkart could also introduce “Smart Baskets,” which auto-fill based on user behavior. Payment options like UPI, Wallet, Cash on Delivery, and even subscription models for milk or staples would improve convenience.

INTERVIEWER: Good. How do you see Supplier & Brand Onboarding?

CANDIDATE: Flipkart should start with partnerships with large FMCG companies like HUL, ITC, Nestlé, and P&G to ensure core essentials are always in stock. Alongside that, onboarding regional brands and local suppliers is critical to cater to India’s fragmented grocery market. For fruits and vegetables, Flipkart can source directly from farmer-producer organizations and mandis to improve freshness and margins. In Tier 2+ cities, onboarding local kirana stores through an aggregator model can expand coverage quickly without heavy investment. Suppliers can be incentivized by providing guaranteed demand visibility, better shelf placement on the app, and lower commission rates in the launch phase.

INTERVIEWER: And what’s your Sales, Marketing & Pricing strategy?

CANDIDATE: Pricing in grocery is highly competitive. Flipkart should start with below-cost pricing on essentials like rice, wheat, oil, and milk to draw users in, while cross-subsidizing with higher-margin categories like snacks and beverages. Subscription-based discounts like “Supermart Plus” could encourage repeat orders. On the marketing side, grocery should be aggressively promoted during Flipkart Big Billion Days with “Stock Up” campaigns. Daily app notifications for deals on essentials and personalized coupons for staples can improve retention. Cross-selling is also powerful, for example, nudging users who buy diapers on Flipkart to also try baby food and milk through grocery.

INTERVIEWER: What are the pain points you’d focus on solving?

CANDIDATE: For users, the common issues are out-of-stock items, delivery delays, poor quality of perishables, and lack of trust in online groceries. Flipkart should address these by ensuring real-time stock visibility, reliable delivery windows, and offering quality assurance with instant refunds or replacements if products don’t meet standards.

For suppliers, the pain points are visibility and demand consistency, which Flipkart can solve by offering data-driven demand forecasts and better digital tools to manage inventory.

INTERVIEWER: How would you handle After-Sales & Metrics?

CANDIDATE: After-sales service in grocery has to be extremely responsive. Flipkart should have quick refund mechanisms for damaged or missing items, 24/7 support for complaints, and hassle-free replacements for perishables. In terms of metrics, the **North Star Metric** would be the number of repeat monthly grocery orders per household. Supporting metrics would include order fill rate, delivery time adherence, percentage of out-of-stock cancellations, and customer satisfaction scores.

INTERVIEWER: That's quite comprehensive. Can you summarize your approach?

CANDIDATE: Sure. Flipkart should scale grocery with a dual strategy: hyperlocal express delivery in metros and scheduled delivery in Tier 2+ cities. The experience should be app-first, focusing on convenience, trust, and personalization. Supplier onboarding should balance FMCG giants with local brands and kiranas, ensuring breadth and freshness.

Pricing should be aggressive at launch, using essentials as entry points and subscriptions for retention. By solving stockouts, quality issues, and delivery delays, Flipkart can build trust and repeat usage. Success should ultimately be measured by repeat orders per household and delivery reliability.



Case 4: Category Launch Strategy for Furniture by Blinkit

INTERVIEWER: Suppose Blinkit, known for quick commerce, wants to expand into the Furniture category. How would you approach this launch strategy?

CANDIDATE: I'd approach the launch in phases similar to other category expansions. First, I'd validate the business feasibility and define the scope of furniture that Blinkit should sell. Then, I'd think through the operational model, since furniture is very different from groceries, and design the app experience. From there, supplier onboarding becomes critical, followed by crafting the pricing and marketing strategy. Finally, I'd solve for key userpain points and define the after-sales support structure with measurable success metrics.

INTERVIEWER: Let's begin with the Business Feasibility.

CANDIDATE: The Indian furniture market is worth around \$20–25 billion, but online penetration is still below 10%. Players like Pepperfry, Urban Ladder, and Ikea dominate the organized online segment, while local carpenters and offline stores cover the rest. Blinkit's strength is speed and convenience, but large furniture doesn't fit that model.

So the feasible entry point is small and quick-deliverable furniture—items like chairs, stools, study tables, bedside tables, shoe racks, foldable desks, and storage organizers. These can fit into a dark-store logistics model. Larger furniture like beds and wardrobes could be a phase two play, requiring a warehouse-to-home logistics setup. Blinkit can position itself around “instant furniture needs”- things customers don't want to wait weeks to receive.

INTERVIEWER: What about Operations & Logistics?

CANDIDATE: Blinkit would need a hybrid logistics model. For smaller furniture, items can be stocked in micro-warehouses or dark stores and delivered within hours. For bulkier furniture, Blinkit can adopt a scheduled delivery model with third-party logistics partners who handle assembly as well. Packaging will be critical—flat-pack furniture with easy self-assembly instructions ensures it fits into existing supply chains. Blinkit could also experiment with “DIY furniture kits” for quick-ship convenience. Over time, adding an installation service layer, either through Urban Company tie-ups or an in-house team, would be essential for larger products.

INTERVIEWER: And how should the UI/UX design be planned?

CANDIDATE: The furniture section should sit as a new category tab within the Blinkit app. Since furniture is a high-consideration purchase, listings need to be very visual with multiple images, AR-based room previews, and videos showing assembly and usage. Filters like size, material, price, and delivery time are critical. To build confidence, ratings and verified customer photos should be highlighted. A “Quick Ship” label for items available for instant delivery would align with Blinkit’s brand promise. Payment should support EMIs and Pay Later, since ticket sizes are higher than groceries. To enhance stickiness, Blinkit could allow users to build “room bundles”- like a study kit (desk + chair + lamp) that can be delivered together.

INTERVIEWER: Now, let’s discuss Supplier Onboarding.

CANDIDATE: For furniture, sourcing is as important as logistics. Blinkit should initially partner with local manufacturers and furniture startups who can supply flat-pack, easy-to-ship products. Established brands like Nilkamal, Godrej Interio, or Wakefit could be onboarded for credibility.

At the same time, Blinkit should create its own private-label line of compact furniture, just like it has done for groceries and daily essentials, because that improves margins and gives better control over quality. Incentives for suppliers could include guaranteed visibility in the app, faster payments, and co-branding on promotional campaigns.

INTERVIEWER: How would you approach the Sales, Marketing & Pricing strategy?

CANDIDATE: Blinkit should position this category as “Furniture at the speed of life.” Pricing should be competitive with online marketplaces but emphasize speed of delivery as the differentiator. Initial offers could include deep discounts on compact, high-demand items like foldable desks or storage units. Marketing campaigns should target young professionals, students, and renters in metros who frequently need affordable, quick furniture. In-app promotions could tie furniture to life moments: for example, “Moving to a new flat? Order your starter kit today.” Social media campaigns with AR try-outs would also work well. For loyalty, Blinkit could integrate furniture offers with Blinkit Gold subscriptions, bundling perks like free delivery or assembly discounts.

INTERVIEWER: What are the key pain points you'd solve for users?

CANDIDATE: The major issues with buying furniture online are long delivery timelines, high delivery costs, poor quality, and complicated assembly. Blinkit can solve these by offering instant or same-day delivery for essentials, transparent pricing with no hidden delivery charges, quality checks with verified ratings, and flat-pack products that require minimal or no assembly. Another common pain point is lack of trust, so Blinkit could offer a "7-day easy return" or "Blinkit guarantee" on furniture. For suppliers, the pain points are visibility and high return costs; Blinkit can solve those with upfront demand insights and dedicated logistics for bulky returns.

INTERVIEWER: And finally, how would you handle After-Sales & Metrics?

CANDIDATE: After-sales should focus on customer confidence. Blinkit should offer quick refunds or replacements for damaged items, provide optional assembly support for complex products, and keep a 24/7 support line for furniture queries. Metrics-wise, the North Star Metric would be the number of furniture orders per month.

Supporting metrics would include average delivery time, return rate, customer satisfaction scores, repeat purchase rates for furniture buyers, and supplier retention. Measuring average ticket size would also be important since furniture is a higher-value category.

INTERVIEWER: Good. Can you summarize the strategy?

CANDIDATE: In short, Blinkit should enter furniture through the quick-commerce angle - focusing first on compact, ready-to-ship essentials like chairs, desks, and storage units. Operations should leverage dark stores for small furniture and scheduled logistics for larger items. The app experience must be highly visual with AR try-ons, fast delivery filters, and easy payment options. Supplier onboarding should combine local manufacturers, established brands, and private labels. Marketing should target young professionals and renters with a promise of instant, affordable furniture. Pain points around delivery delays, assembly, and quality should be directly addressed through packaging, guarantees, and optional installation. Success should be tracked through monthly orders, delivery reliability, and customer retention.

Case 5: Category Launch Strategy for Instant Maids by Urban Company

INTERVIEWER: Suppose Urban Company wants to launch a new category called Instant Maids, where users can book domestic help on-demand for a few hours or for a day. How would you approach the launch strategy?

CANDIDATE: I'd structure the launch in stages, just like any other category rollout. First, I would validate business feasibility and market demand, then design the operations and supply side, think through the product design and customer journey, figure out the supplier (maid workforce) onboarding, plan the marketing and pricing strategy, solve for key pain points on both sides, and finally ensure after-sales service with robust metrics to measure success.

INTERVIEWER: Let's start with Business Feasibility.

CANDIDATE: Domestic help is a massive but largely unorganized market in India, particularly in urban centers. Many households rely on part-time maids, but issues of reliability, last-minute availability, and trust are common.

The opportunity for Urban Company lies in formalizing this market- providing vetted, trained, and on-demand maids for daily tasks like cleaning, cooking, or babysitting. Initially, the service could target metro cities like Bangalore, Delhi NCR, and Mumbai, where demand for flexible, trustworthy domestic help is highest. The model would be similar to ride-hailing - users pay for hours booked, while maids earn through flexible work shifts. Urban Company already has brand equity in home services, so this launch would feel like a natural extension.

INTERVIEWER: How would you think about Operations & Logistics?

CANDIDATE: The operational backbone would need to combine speed and reliability. For instant availability, Urban Company would maintain a pool of trained maids within each neighborhood, much like Uber drivers in a zone. Bookings could be served within 30–60 minutes of request, depending on proximity. Maids should be equipped with uniforms and ID badges to build trust and standardization. Training centers could ensure quality, hygiene and safety standards. Urban Company would also need to handle background verification, medical check-ups, and emergency support for both customers and workers.

A geo-based dispatch system would match demand with the nearest available maid.

INTERVIEWER: And what about the UI/UX design?

CANDIDATE: The app should make booking extremely simple. I'd imagine a new "Instant Maid" tab alongside Urban Company's other categories. Users could select the type of task (cleaning, utensils, babysitting, cooking), the number of hours needed, and preferred time slots. The app should then show available maids nearby along with ratings and expected arrival times. Payment should be integrated seamlessly with UPI, Wallet, or Pay Later. To reduce friction, repeat users should get one-click rebooking of the same maid. There should also be built-in SOS and live-tracking features for safety, especially since this involves in-home services.

INTERVIEWER: What's your approach to Workforce Onboarding?

CANDIDATE: Workforce onboarding is critical. Urban Company should recruit maids from informal domestic worker networks, NGOs, and training institutes. To attract them, UC can offer higher and more consistent earnings compared to traditional setups, along with benefits like accident insurance, medical checkups, and flexible work hours.

Training programs should cover not only cleaning and basic tasks but also etiquette, hygiene, and customer interaction. To ensure retention, UC could introduce a tiered reward system—maids with high ratings get priority bookings, bonuses, and reduced commission rates.

INTERVIEWER: Now, let's talk about the Sales, Marketing & Pricing strategy.

CANDIDATE: The pricing should be per-hour, competitive with local part-time help but with added value for reliability and safety. For example, if the local rate is ₹120–150/hour, UC could charge ₹180–200/hour but bundle in trust guarantees, training, and insurance. Marketing should emphasize safety, reliability, and flexibility "Your help, when you need it." UC could run launch campaigns in metros, targeting working professionals, nuclear families, and bachelors who struggle with daily chores. Cross-selling would also help users booking a deep-cleaning session could be nudged to try Instant Maids for regular upkeep. Promotions like first 2 hours free, referral discounts, or subscription packs for weekly maid hours could drive adoption.

INTERVIEWER: What are the key pain points for users and workers you'd look to solve?

CANDIDATE: For users, the biggest issues are unreliability of daily help, last-minute unavailability, lack of trust, and safety concerns. UC can address this through background-verified maids, instant availability, punctuality guarantees, and SOS features in the app. For workers, the pain points are inconsistent income, lack of dignity, and no benefits. UC can solve these by providing predictable demand, training, better pay, and health/insurance benefits. Essentially, the category should make the relationship more structured and equitable for both sides.

INTERVIEWER: And how would you manage After-Sales & Metrics?

CANDIDATE: After-sales service has to focus on feedback and trust. Customers should be able to rate every booking, and repeated complaints against a maid should trigger retraining or removal. Refunds or credits should be processed quickly if service quality is poor. On the metrics side, the North Star Metric would be the number of repeat hourly bookings per household. Supporting metrics could include maid utilization rate, average rating, time-to-fulfill booking, and worker retention. High repeat usage would validate both customer trust and worker satisfaction.

INTERVIEWER: How would you summarize the strategy?

CANDIDATE: In short, Urban Company should launch Instant Maids as an Uber-like marketplace for domestic help, starting in metro cities where reliability and trust gaps are most acute. Operations should be built around trained, verified neighborhood-based maids with quick dispatch systems. The app experience should prioritize safety, simplicity, and personalization, while pricing should be competitive but positioned around added value. By solving reliability issues for customers and dignity/income issues for workers, UC can formalize a massive unorganized sector. Success should be measured by repeat bookings, customer ratings, and workforce retention.

Puzzle 1

Problem Statement:

There are 100 closed doors. Nearby, there are 100 rats in a cage. Then the first Rat is released and opens all the doors. Then the second Rat is released and closes every second door (the 2nd, 4th, 6th, etc.). Then the third Rat is released and toggles every third door (the 3rd, 6th, 9th, etc.), closing doors that are open and opening doors that are closed. This pattern continues, with each subsequent Rat toggling doors that correspond to its number (the n th Rat toggles every n th door). After all, 100 Rats have taken their turns, how many doors remain open?

(Amazon)

Solution:

Consider door number 56. Rats will visit it for every divisor it has. For 56, the divisors are 1, 2, 4, 7, 8, 14, 28, and 56.

On pass 1, the first Rat will open the door.

On pass 2, the second Rat will close it.

On pass 4, the fourth Rat will open it.

On pass 7, the seventh Rat will close it.

On pass 8, the eighth Rat will open it.

On pass 14, the fourteenth Rat will close it.

On pass 28, the twenty-eighth Rat will open it.

On pass 56, the fifty-sixth Rat will close it.

For every pair of divisors, the door returns to its initial state. However, in cases where a divisor pair consists of the same number, such as door number 16, the scenario is different. The divisors of 16 are 1, 2, 4, 8, and 16, with 4 repeated because 16 is a perfect square. Therefore, door number 16 will be visited on passes 1, 2, 4, 8, and 16, leaving it open at the end.

In conclusion, only doors corresponding to perfect squares will remain open.

Puzzle 2

Problem Statement:

In your basement, there are three light switches, all in the OFF position. Each switch controls one of three light bulbs on the floor above. You can toggle any of the switches but can only go upstairs to check the bulbs once. Once you are upstairs, you cannot return to the switches. What is the minimum number of trips upstairs needed to identify which switch controls each bulb?

(Amazon)

Solution:

Turn light switch 1 to the ON position for a few minutes, then turn it off. Next, turn on switch 2 and go upstairs.

The light bulb that is off but warm is controlled by switch 1.

The light bulb that is on is controlled by switch 2.

The light bulb that is off and cool is controlled by switch 3.

Puzzle 3

Problem Statement:

You have 12 identical-looking balls. All of them have the same weight except for one defective ball, which could be either heavier or lighter.

Using a two-sided balance scale, determine the minimum number of weighings required to identify the defective ball.

(Microsoft)

Solution:

Divide the 12 balls into 3 groups of 4 balls each.

First Weighing:

Weigh 2 groups (4 balls each) against each other.

Case 1: Both sides are equal

This means the defective ball is in the remaining group of 4 balls.

Let's call these balls B1, B2, B3, and B4.

Second Weighing: Weigh B1 and B2 against each other.

If they are unequal, either B1 or B2 is defective. Compare B1 with one of the 8 balls known to be of equal weight.

If B1 is equal to the reference ball, B2 is defective.

If B1 is not equal to the reference ball, B1 is defective.

If B1 and B2 are equal, either B3 or B4 is defective. Compare B3 with one of the 8 balls known to be of equal weight.

If B3 is equal to the reference ball, B4 is defective.

If B3 is not equal to the reference ball, B3 is defective.

In both cases, it takes 3 weighings (1 initial + 1 second weighing + 1 comparison).

Case 2: One side is heavier or lighter

This means the defective ball is in one of these 8 balls (4 on the heavier side and 4 on the lighter side). Let's call the balls on the lighter side L1, L2, L3, and L4, and those on the heavier side M1, M2, M3, and M4.

Second Weighing: Weigh 2L and 1M against 2L and 1M, reserving 1L and 1M.

If side A is down (heavier) and side B is up (lighter):

One of the L balls on side A is lighter, or The M ball on side B is heavier.

If side B is down (heavier) and side A is up (lighter):

One of the L balls on side B is lighter, or

The M ball on side A is heavier.

If both sides are balanced:

The defective ball is one of the reserved 1L and 1M.

Third Weighing: For cases 1 and 2, where we have 3 suspect balls (2L and 1M):

Weigh the 2L balls against each other.

If they are equal, the M ball is defective (heavier).

If they are unequal, the lighter L ball is defective.

For case 3, where one of the reserved balls (1L and 1M) is defective:

Weigh the 2 reserved balls against each other.

The heavier one (if any) is defective.

In this scenario, it also takes 3 weighings in total.



Puzzle 4

Problem Statement:

Three ants are situated at the corners of a triangle, one ant per corner. At a specific moment, each ant randomly starts moving toward one of the other corners. What is the likelihood that none of the ants will collide with each other?

(Intuit)

Solution:

Consider three ants labeled as A, B, and C. They will avoid a collision in two scenarios: if all three move clockwise or if all three move counterclockwise. A collision will occur if any two ants move toward each other, while the third ant can move either clockwise or counterclockwise. For each pair of ants, there are two such collision scenarios. With three possible pairs (A, B), (B, C), and (C, A), there are a total of $3 * 2 = 6$ collision scenarios. Therefore, the probability that the ants will not collide is $2 / (2 + 6)$, which simplifies to $\frac{1}{4}$.

Puzzle 5

Problem Statement:

Imagine a two-player game taking place on a circular table of unknown size.

Both players have an unlimited number of coins and take turns placing a coin on the table. Each coin must lie entirely on the table and cannot overlap with any previously placed coins.

The winner is the player who places the last coin legally.

Which player, if any, has a winning strategy, given that the table's diameter is larger than that of the coins?

(Microsoft)

Solution:

Player A has a winning strategy. His initial move is to place a coin at the exact center of the table. Player B then places a coin anywhere on the table, except the center. To ensure a win, Player A can always place his next coin directly opposite Player B's last coin, with Player A's first coin being the midpoint between them. This pattern continues until the table is full, ensuring that Player A always has a valid move while Player B eventually runs out of options. Consequently, Player A wins. This strategy is effective even if the table is as small as a single coin.

Puzzle 6

Problem Statement:

You are blindfolded, and there are 10 coins placed on the table in front of you. You can touch the coins but cannot determine which side is facing up by feel alone. You know that 5 coins are heads up, and 5 are tails up, but you don't know which ones. Is it possible to create two piles of coins, each with the same number of heads up? You can flip the coins as many times as you like.

Solution:

Divide the 10 coins into two piles with an equal number of coins. Next, flip all the coins in one of the piles. This will ensure that each pile has the same number of heads-up coins.

Here's an example to illustrate how it works:

Initial Scenario:

There are 5 heads and 5 tails.

Case 1:

Pile 1: H H T T T

Pile 2: H H H T T

When you flip all the coins in Pile 1:

Pile 1: T T H H H

Now, Pile 1 has 3 heads, and Pile 2 also has 3 heads.

Case 2:

Pile 1: H T T T T

Pile 2: H H H H T

When you flip all the coins in Pile 1:

Pile 1: T H H H H

Now, Pile 1 has 4 heads, and Pile 2 also has 4 heads.

In both cases, flipping one pile ensures that both piles end up with the same number of heads.

Puzzle 7

Problem Statement:

Ram and Shyam are on separate islands. Shyam is sick, and Ram has the medicine. Anuj has a boat and a chest that can be locked. He is willing to transport objects between Ram and Shyam, but only in the chest. If the chest is unlocked, Anuj will steal whatever is inside. Both Ram and Shyam have a padlock and a key, and each key can only open their own lock.

Is it possible for Ram to send the medicine to Shyam? If so, how many minimum trips are required? Note that neither Ram nor Shyam can travel in the boat. The boat traveling from Ram to Shyam or Shyam to Ram counts as one trip.

Answer: 4

Solution:

Trip 1: Ram places the medicine in the chest and locks it with his padlock. Anuj takes the locked chest to Shyam.

Trip 2: Shyam receives the locked chest, adds his own padlock to it (so now the chest has two padlocks), and sends it back to Ram.

Trip 3: Ram removes his padlock and sends the chest (still locked with Shyam's padlock) back to Shyam.

Trip 4: Shyam unlocks the chest with his key and retrieves the medicine.

the hat, those wearing it must dunk themselves underwater exactly at midnight.

If there are n people and c hats, how long will it take for the men to remove the hats? The men cannot communicate in any way to indicate who has a hat.

Note: The genie does not reveal the number of hats placed. All the men are perfect logicians and are aware that everyone else is also a perfect logician.

Answer: C days

Solution:

Let's consider a few simple scenarios first:

$c = 1$: The person with the hat will see that no one else has a hat. Realizing this, he will understand that he must be the one with the hat.

$c = 2$: Each person with a hat will see one other person with a hat, while everyone else will see two hats. If there were only one hat, the person would have resolved it on the first day. Since no one goes on the first night, the person who sees one hat will realize that he too must have a hat. Thus, both will go underwater on the second night.

$c = 3$: Each person with a hat will see two hats, while everyone else will see three hats. If there were one or two hats, someone would have gone on the first or second night. When no one goes

Puzzle 8

Problem Statement:

A group of men find themselves on an island. A genie appears, gathers everyone, and places a magical hat on some of their heads (ensuring at least one person has a hat). This magical hat can be seen by others but not by the person wearing it. To remove the hat, those wearing it must dunk themselves underwater exactly at midnight.

If there are n people and c hats, how long will it take for the men to remove the hats? The men cannot communicate in any way to indicate who has a hat.

Note: The genie does not reveal the number of hats placed. All the men are perfect logicians and are aware that everyone else is also a perfect logician.

Solution:

Let's consider a few simple scenarios first:

$c = 1$: The person with the hat will see that no one else has a hat. Realizing this, he will understand that he must be the one with the hat.

$c = 2$: Each person with a hat will see one other person with a hat, while everyone else will see two hats. If there were only one hat, the person would have resolved it on the first day.

Since no one goes on the first night, the person who sees one hat will realize that he too must have a hat. Thus, both will go underwater on the second night.

$c = 3$: Each person with a hat will see two hats, while everyone else will see three hats. If there were one or two hats, someone would have gone on the first or second night. When no one goes by the third night, those who see two hats will understand they must also have a hat and will all go underwater on the third night. In this manner, we can see that it will take c days to remove all the hats.

Puzzle 9

Problem Statement:

You are on a spaceship equipped with a computer that has n processors. An alien laser beam strikes the spaceship, damaging some of the processors. You are aware that more than half of the processors remain functional. You can inquire with one processor about whether it considers another processor to be good or bad. A good processor will always tell the truth, whereas a bad one will always lie. A 'step' is defined as asking one processor if another processor is good or bad. Determine the minimum number of steps required to identify at least one good processor.

Solution:

Number the processors from 1 to n . Start by asking processor 1 if processor 2 is good. If processor 1 says "no," remove both processors from the list and start over, as you have removed one good and one bad processor, leaving more than half of the remaining processors still good. If processor 1 says processor 2 is good, proceed by asking processor 2 about processor 3, processor 3 about processor 4, and so on, until you receive a "no" answer.

Suppose processor j says processor $j+1$ is bad. Remove both processors j and $j+1$ from the list, as this removes one good and one bad processor, maintaining the majority of good processors in the remaining list. and continue by asking processor $j-1$ about processor j (the processor that was after $j+1$ before you removed it and the original processor j).

You are forming a "chain" of processors, where each processor believes the next one is good. This chain consists either entirely of good processors or bad ones. Once the chain includes more than half of the remaining processors, you can be certain they are all good. Alternatively, if only one or two processors are left after several removals, those remaining processors are good.

This process involves at most $n-2$ steps. Every processor is asked about at most once, and neither the first processor nor the last processor in the chain is asked, since the process stops once the chain length exceeds half of the list.

Puzzle 10

Problem Statement:

Imagine you have a 4 L jug and a 9 L bucket, both without measurement markings. How can you accurately measure out 6 L using only these containers and an unlimited water supply?
(Microsoft)

Solution:

Take two buckets: one with a capacity of 4 L and the other with 9L.

Fill the 9 L bucket completely: 0 L and 9 L.

Pour water into the 4 L bucket: 4 L and 5 L.

Empty the 4 L bucket: 0 L and 5 L.

Repeat steps 2 and 3 twice: 4 L and 1 L.

Now, the 9 L bucket will have 1 L of water remaining: 0 L and 1 L.

Pour this 1 L into the 4 L bucket: 1 L and 0 L.

Refill the 9 L bucket: 1 L and 9 L.

Pour water from the 9 L bucket into the 4 L bucket until it fills up: 4L and 6 L.

Now, you have exactly 6 L of water in the 9 L bucket.

Interview Experiences



Candidate 1

Process: CV Shortlisting, GD Round (2-3 days before SIP week), 2 Interview Rounds (Day 1)

I. GD Round: Problem Statement:

The global demand for food is increasing due to population growth, stressing agricultural systems. Climate change further affects food production, disrupting supply chains. How can a tech company promote sustainable agriculture, ensure food security, and reduce environmental impact while supporting farmers?

II. Interview Round 1 (30–35 mins):

TMAY Case: “Design a photo app for specially-abled people” (10–12 mins to solve, then present) Cross-questioned on solutions

III. Interview Round 2 (30–35 mins):

TMAY CV Questions – only about Workex

Case: Design a government app to improve K–12 education outcomes No cross-questioning on the case; returned to CV discussion

Candidate 2

I. GD Round: Problem Statement:

Topic: The fitness industry is saturated with products, yet users fail to achieve lasting results. Design a product that improves user fitness significantly compared to current solutions.

Interview Round 1:

Started with favorite products – mentioned two, then Discord

Discussed:

1. What is Discord?
2. Key Features
3. Target Audience
4. Differentiation from other platforms

Explored Discord bots, low latency (interviewer explained about server optimization) Why students prefer Discord over WhatsApp

Case: Improve WhatsApp to capture market from Discord

Metrics to evaluate the changes Conversational and unstructured – many follow-ups based on responses Interview

Round 2:

Focused on CV: Workex, responsibilities, independent projects Since candidate worked at AMEX, finance questions dominated

Question: Key financial challenges in rural vs. urban areas
Focused on farmers → interviewer asked to expand beyond farmers
Case (approximate): Design a finance app for community-based pooled investment.

Candidate 3

Round 2 (50 mins):

Taken by an alumnus – explained the structure
TMAY

Hooks picked from TMAY, CV Grilling – ZS Associates, Bajaj Auto, Research Paper

International Trade: Challenges and bottlenecks in India
Mentioned Vizhinjam hub, China's advantage since liberalization

Tip: Use interesting examples for your interests

Why Product Management after consulting workex – required 3 levels of explanation. Tip: Prepare L1, L2, L3 for all HR questions

Q: Do you use ChatGPT? Yes

Case: Design ChatGPT for student engagement. Tips: Align with company vision, ask clarifying questions, think aloud, use personas, define metrics. Tip: Mention tech terms (e.g., LLM, hallucination, RAG); if unsure, general tech is fine

Q: Do you have a question? Asked about MS carbon neutrality initiative.

Asked to show the working sheet at the end

Round 3 (40 mins):

Taken by a Partner Director TMAY

CV Questions: Research paper, ZS, Bajaj, Regatta, Alumni Cell

Case: PM at Amazon – increase purchases via Alexa Creativity-focused
Worked on least-prioritized personas. Tip: Use examples and storytelling in case discussions.

Q: Tell me about a time when your idea wasn't accepted by the team
Needed to justify the example when interviewer wasn't convinced
Q: Do you have a question? Asked about MS carbon neutrality again.

Candidate 1

Process: CV Shortlist, Case Submission, Buddy Round (eliminative, before placement week), Final Round (Day 1)

Case Submission: 2-part case Product Design + New Revenue Stream Buddy

Round 1: Very evaluative – taken by a PM TMAY No fluff – be direct Assumptions must be logical

Focus on metrics

Guesstimate: Revenue generated by the tool between Delhi and Ghaziabad

RCA: 10% decline in Amazon orders – focus on approach, not just solution

Dashboarding: Top 5 metrics for Uber CEO

Technical round: Cookies and cache

Candidate was asked to pose a question

Final Round:

TMAY Stress interview

Guesstimate: Number of elevators needed for a 100-storey building

Product Design: Design an aircraft for pets as a PM at

Airbus Metrics: Top 5 metrics for the CPO of Airbnb Stick to a structure Question for the interviewer.

Candidate 2

Round 1

(Online - Pre-SIP) - 2 panelists : Senior Manager (observer only) and recent hire

TMAY, Workex related questions, grilled on workex, discussion on free open source softwares, contributions of tech community in the same as well, again workex (how you can correlate the same to workex)

Dashboarding case: Give the KPI metrics a PM would look at for a newly launched fintech app

1. Asked in detail about North Star Metric and the understanding of metrics (not the formula),

2. What is the difference b/w an Active User and an Engaged User.

3. Define Engagement and Activity - Explain the same related to Fintech industry

4. What is churn rate? What do you mean by customers “leaving”?

Last 3 questions pertaining to metrics and its understanding - grilled for 25 minutes

Candidate 1

CV Shortlist, 1 round of interview (start at 6 am - Day 1) 10 shortlisted - 3 interviewed

The interview experience was unique due to the unique CV - UPSC 3 years gap, APM experience. The role is focused towards Product Marketing (customer research).

Taken by a single Alumn - need to be precise and to the point

- What brings you here
- Don't you want to explore - why Product Management
- About education (why UG college, what other options, why this stream)
- Asked a question from the stream
 - Can deflect that - be truthful
- Favourite subject - how that subject applied in customer research
- UPSC motivation - why give up on it
 - Grilling discussion about that
- Why you chose FMS
- CV grilling - Internships, FSC, POR
- 3 number series
 - Others were also asked a guesstimate

- Do you know how to code
 - You can answer no, or yes - the next questions depend on your choice
- More questions on customer research
- Asked about the design experience, dashboards
- Best feature of Figma (it was in CV)
- One weakness you are working on
- How do you stay motivated in your life
- Biggest learning from your APM work
- Do you have a question for me

Candidate 2

- TMAY
- Why Adobe?
- CV based grilling
- Puzzles
 - N prisoners in a line, each sees only the hats ahead (red/blue), must guess their own color
 - Heaviest ball via weighings
- DSA questions
 - What are queues
 - Some basic algorithms like binary search on sorted arrays and palindrome in a string
- One failure in life and what do you think about it now?

Candidate 1

Round 1

- Tmay
- Case: how many Starbucks accessories are sold in 1 day in cybercity
- Favourite app – Discord. Why doesn't Meta go into this space? Revenue model of Discord
- What is unique about it?

Round 2

- How would you advertise yourself?
- Why prodman?
- Abt workex.
- Case: Favourite FMCG product. Market size of men's face wash in Mumbai.
- Favourite show on OTT (Answered TBBT)
- Its production house?
- How much should HBO bid for TBBT's next season (aiming for profitability)
- Other revenue streams for HBO

Round 3 (HR)

- Why MMT
- 3 things you don't like about MMT

Candidate 2

Pre-interview: Shortlisting basis one-pager submission on GL concepts and CV evaluation

Round 1

- Feedback on MMT app/features led to a discussion on Homestays feature, KPIs, user flow, and hotel cannibalization
- Case on feature improvement. Multiple suggestions evaluated on customer funnel, technical feasibility, and implementation
- Guesstimate - Ticketing volumes for Kashi Vishwanath temple and pricing strategy

Round 2

- This round was highly grilling, testing detailed knowledge of MMT's business lines and app functionalities
- Proposing new features, differentiating them from existing ones, mapping user flows, and detailing technical requirements from UI to backend
- Guesstimate - whether a truck carrying an oil tanker filled with its own fuel could circle the globe. No other data was provided.

Round 3 (HR)

- Brief and light conversation, with limited questioning.

Candidate 3

Round 1

- Tmay
- Follow ups on TMAY (What is pet sitting – as it was mentioned as a hobby)
- Case – Design a pet sitting services app. Guesstimate the market size
- If could choose one pain point to solve for, what would it be?

Round 2

- TMAY
- How has FMS been?
- What is your favourite part about FMS?
- Was asked about the closed airport to hometown – mentioned Bangalore Airport
- Guesstimate - Number of people going to Bangalore airport in a day

Round 3 (HR)

- TMAY
- Why MMT
- What did you do in sustainability in Alstom (CV based question)
- Why MMT
- If you are a sustainability consultant for MMT, what 3 changes would you do?
- Any questions for me?

Candidate 1

Process: Shortlist, GD, Personality test, PI. Comes for 2 roles - Product Management and Product Marketing

Personal Interview

- 4 in panel
- What is the meaning of your name?
- Are you interested in Product Management or Product Marketing
- TMAY, About her family, noted all points
- Choose an industry and note the pain points of the industry - have to give solutions - need to be software solutions, similar to what NewGen does. Grilled on that
- Grilled on CV, especially on the FSC project (grilling), asked about time in college
- The candidate is supposed to ask the question
- Stress interview

Candidate 2

Process: Shortlist, GD, Personality test, PI. Comes for 2 roles - Product Management and Product Marketing

Personal Interview:

- Can you give me a brief introduction of yourself?
- So you were preparing for UPSC, why MBA now?
- What do you think is the most hardest thing you ever did?
- Are you aware about our products?
- Case – Where do you think we can innovate?
- Give me specific industries where we can enter?
- Assume I am MD of A hospital chain, how will you pitch me your product?
- Do you have anything that you want to build ?
- Do you have any questions for us?

Candidate 1

Process: Shortlist, 2 rounds of Interview

1st Round - HR round - very conversational

- Let's talk about stakeholder management
- Tell me about a time when you had to convince a difficult stakeholder
- An instance when you bounced backed from a difficult situation

Candidate 2

Process: Shortlist, 2 rounds of Interview

1st Round

- Conducted in a rapid-fire style; interviewer asked quick questions and wanted straightforward and concise answers.
- Covered all CV-based topics including college projects, PORs and resume highlights.
- Included HR questions
 - Time you overcame a problem
 - When you demonstrated leadership
 - Where do you see yourself in 10 years

2nd Round

- Interviewer set expectation: 3 abstract questions.

- How would you pitch an idea as an intern to senior management?
- How do you define creativity? (Expected answer around problem-solving and not art/design.)

- Extended discussion on blockchain applications at Car Dekho:

- Brainstormed ideas
- Evaluated feasibility of blockchain solutions
- Interviewer shared why Car Dekho was not pursuing blockchain actively

Candidate 1

Round 1

Q. TMAY

Q. What did you do at work, expand on it?

Q. You worked only 6 months, what are all these numbers, did you work on all of them?

Q. What is this automatic cheque verification project(A project I did in UG, explained about it)

Q. Where did you get the training dataset from?

Q. What other models did you compare with(I remembered only 1 of them)

Q. What do you know about Amex(answered about how they are a closed loop system unlike competition and how they make revenue)

Q. How does closed loop make a difference(answered about how they would have all the data and can utilize for various purposes)

Q. Suppose you have a group of prospects who are looking for a credit card, how would you measure the credit risk or worthiness of them, what are few metrics that you would look at. (Was able to give 6-7 metrics/data points)

Q. Now you have to decide what credit limit can be given to

them, how would you predict the limit you can give(Didn't get any good ideas, reused 3 metrics I stated for above question and framed something)

Q. Any questions for me?

Round 2

Different from typical case interviews.

Why Amex, What do you know about Amex, let us start with a case.

Case: You are a company which sells furniture online like Pepperfry, Urban Ladder. The logistics cost associated with selling furniture is very high as unlike Amazon and Flipkart you would be dealing with large packages. Your company wants to launch a expedited delivery feature which provides 1 day delivery like Amazon. But instead of charging a fixed fee like Amazon, the company wants to offer an optional fee along with a purchase, now since the logistics cost of furniture delivery is very high, you want to offer this option to people who are less likely to return a product. What are few metrics/data points on which you would predict the likelihood of someone returning a product, give rationale behind all the metrics (Had a 25-30 min discussion on which data can be used, how we would get it and what it means).

Q. Do you play cricket or are you a passive watcher(I had Cricket as an interest on my CV, answered I play and follow it regularly and am very passionate about bowling)

Q. Who's your favorite cricketer(answered Pat Cummins right now and Dale Steyn is my all time favorite)

Q. How would you compare Pat Cummins and Jasprit Bumrah, which stats would you use to compare them, what would they mean?

A. Started with normal stats but told these are common ones, we can use different or less known statistics and gave 5-6 measures. He cross-questioned on how would these be measured, answered that.

Q. What is it with these EV projects, is it because of your interest or because of the hype around EV these days(I had a UG project and an FSC project on EVs)

Q. 1 very small question on FSC EV project

Q. Any questions for me?

Candidate 2

Round 1

- Grilling Round
- TMAY
- Why Amex
- Difference between Amex , Mastercard and Visa
- Revenue model of Amex

- SQL queries and logic associated with it. Eg: write a query that gives 3 letters after the second A in American Express
- Excel functions and logic
- What is standard deviation? Make me understand as if I am a 5 year old
- Guesstimate: How much revenue Starbucks makes in a day in Delhi?
- What data points or columns should one look for in a database of customers to define who is a premium customer?

Round 2:

- TMAY
- Why Amex?
- What do you like about the work culture of Amex?
- Can you give an example from your daily life wherein you generally use your calculative and analytical personality?
- Case: Pepperfry is going to launch a feature called quick delivery but since their package size is large, they don't want to make this feature available for every customer because of logistics cost. How to analyse and suggest which customers should be shown this feature? Give answer from both business and data sense(probability of a customer not returning the order and params associated with it)
- Pushed me a lot on data params. Wanted me to give 25 params. Any questions?

Candidate 3

Round 1

- Focus Areas:
- Basic probability questions, puzzles, and case studies.
- Example Question:

Light Bulb Puzzle: “There are 100 bulbs initially turned off. A child toggles all the bulbs, then every second bulb, then every third, and so on up to 100 times. Which bulbs will remain on?”

- Case Study:
- Scenario: “You are shifting to another city. Walk through the scenario, focusing on where data comes in, how to consider all scenarios, and your overall thought process.”
- Project Discussion:
- In-depth discussion of projects involving AI/ML, algorithms, and technical problem-solving.

Round 2

- About the Company:
- Questions to assess your understanding of American Express and its operations.
- Case Study:
 - Product Case: Example focused on predicting product return likelihood for a platform like Urban Ladder.

- Key factors: product price, category, dimensions; user behavior (cart dwell time, return history); transaction details (discounts, payment mode); logistics (delivery time).
- Importance of feature engineering (e.g., cart dwell time, price-category interactions) and predictive modeling (e.g., Logistic Regression, Random Forest).
- Emphasis on approach rather than just the solution, focusing on identifying high-risk returns and actionable insights.

This experience highlights the importance of a structured approach to problem-solving, technical depth and clear communication during the interviews.



Thank You

